Designing Experiments for Policies Beyond the Experiment

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Abstract

We study how to use pilot data to design experiments when the policy of interest cannot be directly implemented in the experimental setting. For instance, there may be noncompliance in the experiment, while the policy may treat the population as a whole. Accordingly, the payoff of the policy is only set-identified by the experimental data, so it is no longer clear what makes a good design. We answer this question from the perspective of ultimately implementing the policy. This leads to a two-wave statistical decision problem. We characterise the asymptotically optimal rules for choosing the experimental design and the subsequent policy. Evaluating a design requires integrating over the possible second-wave data and over the posterior of all the take-up rates and mean outcomes in the experimental arms. To facilitate implementation of optimal rules, we introduce regret-directed dimension reduction (RDDR), which we show reduces the dimension of integration while preserving the first-order asymptotic performance. In a numerical exercise calibrated to a Supplemental Nutrition Assistance Program (SNAP) outreach experiment, choosing the design with our rule reduces regret by 27% relative to the allocation in the original study's main analysis sample. RDDR retains nearly all of this improvement at about one-third of the computation time.

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1 Introduction

Experiments are often run to inform policy decisions, such as whether to adopt a welfare programme, launch a product feature, or change an eligibility rule. The experimental evidence is valuable to the extent that it improves the decision making that follows. The goal of experimental design is therefore to allocate limited resources so that the subsequent policy decision problem can be solved as well as possible.

What an experiment can randomise is often not the policy that must ultimately be chosen. Many experiments rely on voluntary take-up of a treatment. An electricity pricing experiment, for example, may require households to opt in to a new tariff (George et al., 2018). A government agency running a benefit programme can send letters to randomly selected eligible households, encouraging them to apply (Bhargava and Manoli, 2015; Heard et al., 2017; Domurat, Menashe, and Yin, 2021; Linos et al., 2022). The policy of interest, however, may make the new tariff the default or enrol eligible households automatically using administrative records. Such policies can require legal authority or administrative systems that are unavailable during the experiment. Some individuals would take up the treatment under such a policy but under none of the experimental arms. Their outcomes under treatment are never observed in the experiment, so the experiment can only set-identify the policy's payoff.

This paper studies a *two-wave* statistical decision problem for experimental design and policy choice with set-identified payoffs. The *decision procedure* consists of two rules. The *design rule* uses the pilot sample to select a follow-up experiment from a list of candidate designs. The *policy rule* uses the combined evidence from both waves to select a terminal policy. We propose an approach for choosing asymptotically optimal decision procedures for this two-wave problem.

The Pennsylvania Supplemental Nutrition Assistance Program (SNAP) outreach experiment provides our running example. Finkelstein and Notowidigdo (2019) randomised status quo, information, and information plus application assistance among older adults who were likely eligible but not enrolled. These interventions increased enrolment but require households to apply by themselves. In New York, however, the state runs a Combined Application Project that removes this step for qualifying Supplemental Security Income recipients classified as living alone and enrols them automatically using administrative records.[1] Unlike information letters or application assistance, this approach requires a waiver of federal SNAP rules from the US Department of Agriculture and coordination across agencies and

[1] See New York State OTDA, "New York State Combined Application Project (NYSCAP)" (April 22, 2020).

administrative systems (Negoita, Levin, and Paprocki, 2023), so it cannot be offered as an experimental arm of an outreach experiment.[2] The policy question is whether Pennsylvania should automatically enrol eligible older adults using administrative records the same way New York does. Such a proactive enrolment policy could also reach households that would remain unenrolled under every experimental arm. So how should the experiment be designed to inform this policy decision?

Standard guidance on experimental design does not speak to this situation. It often asks which allocation delivers the *most precise estimate* of a chosen treatment effect (Neyman, 1934; Hahn, Hirano, and Karlan, 2011; Kasy, 2016; Bai, 2022; Tabord-Meehan, 2023; Armstrong, 2026). Targeting precision is sensible when the choice is between two experimental arms, since the decision then turns on a single welfare contrast that the experiment identifies, and a more precise estimate of that contrast generally means a better-informed choice.[3] When the policy lies beyond the experimental menu, the decision depends on the bounds on the policy's set-identified payoff, and no single point-identified treatment effect determines which policy to choose, even when only two policies are compared.[4] Thus an experimental design can be very precise about some treatment effect and of little use for the decision. In the SNAP calibrated numerical exercise below, a rule that minimises the variance of a local average treatment effect leaves the eventual policy choice 79% worse than an even split across experimental arms.

To design the experiment for the decision, we first specify how the policymaker chooses among terminal policies when some of their payoffs are only set-identified. The bounds of the identified set depend on the experiment through three moments in each covariate stratum x and experimental arm z, namely the take-up rate $\mathbb{E}[D \mid X = x, Z = z]$ and the outcome moments $\mathbb{E}[YD \mid X = x, Z = z]$ and $\mathbb{E}[Y(1-D) \mid X = x, Z = z]$, where X, Z, D and Y denote the covariate stratum, the assigned arm, take-up and the outcome. We call the vector of these moments the *reduced form*. Following Manski (2000, 2007, 2011), we compare

[2] See USDA Food and Nutrition Service, "Supplemental Nutrition Assistance Program: Demonstration Projects", *Federal Register* (April 8, 2024).

[3] Hirano and Porter (2009) establish conditions under which efficient estimation of welfare contrasts leads to asymptotically optimal treatment rules in smooth, point-identified settings. Russo (2020), Kasy and Sautmann (2021) and Higbee (2025) study experimental design for choosing among more than two policies with point-identified payoffs.

[4] Experimental design under noncompliance is also studied by Bugni and Gao (2023, Section 6), who use a pilot experiment to choose the design that minimises the asymptotic variance of an estimated local average treatment effect, Chandak et al. (2024), who target the mean-squared error of an estimated counterfactual response function under an additive outcome model, and Oprescu, Cho, and Kallus (2025), who derive variance-optimal instrument assignment for ATE estimation under an additional unconfounded-compliance assumption that delivers point identification. All three papers target the precision of a quantity that their assumptions point-identify. We impose no assumption that point-identifies the payoff of the policy of interest and evaluate designs by the policy choice they lead to.

the terminal policies by their *maximum regret*. A policy's regret is the payoff lost by not choosing the best policy. The maximum regret of a policy is its largest regret over all payoffs consistent with the reduced form and the maintained assumptions. When the reduced form is known, the policymaker picks the policy with the smallest maximum regret.[5] Thus the policy is robust to the ambiguity about payoffs that the experimental data cannot resolve.

In practice, the reduced form must be estimated, so policy choice must also account for sampling uncertainty. Finite-sample analyses of policy choice with set-identified payoffs (see, e.g., Stoye, 2012; Ishihara and Kitagawa, 2026; Yata, 2025; Montiel Olea, Qiu, and Stoye, 2026a) derive minimax-regret rules assuming that the data follow a specified distribution. We instead follow the approach of Christensen, Moon, and Schorfheide (2026), hereafter CMS. We use their criterion, which combines minimax reasoning about payoff ambiguity with averaging over nearby values of the reduced form. This combination follows the generalised Bayes–minimax principle of Hurwicz (1951). The minimax part enters through the *excess regret* of the terminal policy selected by a decision procedure, which is the difference between that policy's maximum regret and the smallest maximum regret achievable when the reduced form is known. The averaging part enters through *local alternatives*, which are nearby values of the reduced form whose differences shrink with sample size at the same rate as estimation error. They remain statistically difficult to distinguish and may favour different policies, so the large-sample problem retains the uncertainty that the policymaker faces in a finite sample. We compare decision procedures through the large-sample behaviour of their expected excess regret averaged over local alternatives, which we call the *local-average excess regret*. The approach then uses the *limits-of-experiments* framework of Le Cam (1972) to represent the data in large samples by Gaussian observations, which makes the large-sample problem solvable without specifying the finite-sample distribution of the data.[6]

Hirano and Porter (2009) pioneer the use of the limits-of-experiments framework to characterise asymptotically optimal treatment rules under point identification. Hirano and Porter (2025) extend the asymptotic representation approach to sequential experiments, where earlier decisions affect what is observed later. Building on this sequential approach, Higbee (2025) derives asymptotically optimal two-wave experimental designs under a Bayes welfare criterion in a parametric model with point-identified payoffs and smooth welfare function. CMS use this framework to study a general static decision problem with set-identified pay-

[5] Other criteria include maximin welfare, which chooses the policy with the highest worst-case payoff, and maximisation of subjective expected welfare, which averages payoffs under a specified probability distribution over feasible payoff vectors. See Manski (2011) for a discussion of these criteria.

[6] In this paper, an experiment is the randomised experiment that the policymaker designs. In Le Cam's terminology, an experiment is more generally a family of probability distributions for the data, indexed by unknown parameters. The data from the two-wave randomised experiment give such a family.

offs.[7] Neither line of work covers our problem. The designs of Higbee (2025) inform a policy choice that depends on a smooth, point-identified welfare contrast, and CMS take the data as given. When a follow-up design must inform a policy choice under set-identified payoffs, the design problem itself changes.

To see how, consider a policymaker who has observed pilot data, which could also be reduced-form evidence from earlier studies, and must decide how to design a follow-up experiment with a view to implementing a policy. The policymaker ranks the candidate designs by the expected excess regret of the policy choice that each design would lead to. A design is then valuable to the extent that its second wave is expected to settle which policy has the smallest maximum regret. With point-identified payoffs and smooth welfare, only a scalar welfare contrast matters for the choice between two close policies, and a good design estimates the contrast as precisely as possible. With set-identified payoffs, the maximum regret can be nonsmooth in the reduced form. A bound can switch from one formula to another as the reduced form crosses a threshold, so the combination of moments (components of the reduced form) that the maximum regret depends on changes from one region of the reduced form to another. When the pilot leaves uncertain which region contains the true reduced form, several combinations can matter even for the choice between just two policies. A good design therefore uses the pilot to balance information across them, and its value cannot be summarised by the variance of any single estimate.

To find such a design, we apply the limits-of-experiments framework to each wave of data, so that in the limit the policymaker observes two Gaussian random variables in sequence, choosing a design after observing the first and a terminal policy after observing the second. We call this limit problem the *sequential Gaussian decision problem*. Unlike in a static problem, the design selected after the pilot determines how informative the second wave is, so the precision of the second Gaussian observation depends on that design. Because the design must anticipate a terminal policy that is chosen under set-identified payoffs, we solve this problem by backward induction. Given both Gaussian observations, the optimal policy rule chooses the policy with the smallest expected excess regret. Given only the pilot observation, we evaluate each candidate design by its *Gaussian design loss*, which is the expected excess regret of this policy choice, averaged over the second-wave observations that the design could produce. The design loss thus values a design by how much its second wave is expected to improve the policy choice, and the optimal design rule selects the candidate with the lowest design loss. We then link the sequential Gaussian decision problem back

[7] CMS focus on a discrete decision problem. Xu (2026) develops related asymptotic theory for continuous decisions under general losses. Montiel Olea, Qiu, and Stoye (2026b) study a local asymptotic regime in which sampling uncertainty and the degree of partial identification both diminish.

to the two-wave problem in large samples. We show that no decision procedure in the class we consider has a limiting local-average excess regret smaller than the optimal value of the sequential Gaussian decision problem. We also establish conditions under which some procedure attains this bound and is therefore asymptotically optimal in the two-wave problem.

The optimal policy and design rules depend on how the criterion weights the local alternatives, and this choice of weights is arbitrary. Following CMS, we let the weights become flat. The optimal rules then converge to a limiting procedure that does not depend on the weights. In this limit, the policymaker's posterior for the reduced form after each wave is centred at an efficient estimate from the data observed so far, with that estimate's asymptotic covariance matrix. For implementation, we therefore construct a finite-sample quasi-Bayesian decision procedure that uses an efficient estimator and its estimated covariance matrix to form a *quasi-posterior distribution* for the reduced form (Doksum and Lo, 1990; Kim, 2002; Müller, 2013). We show that the quasi-Bayesian procedure makes the same choices of experiment and terminal policy as the limiting procedure in large samples, so that it is asymptotically optimal as the weights become flat.

While the theory characterises the optimal procedure, computing it in practice is challenging. The quasi-Bayesian procedure ranks experiments by a finite-sample counterpart of the Gaussian design loss, which we call the *full design loss*. Computing it requires two nested integrations. The inner integration averages each policy's excess regret over the updated quasi-posterior, and the outer integration averages over the second-wave evidence that the chosen design could produce. Both integrations are over vectors with as many components as the reduced form, which has three moments in each covariate stratum and experimental arm. The cost therefore grows quickly with the number of arms and strata.

We therefore propose *regret-directed dimension reduction (henceforth RDDR)*, which performs both integrations only along the linear combinations of the reduced form that matter for excess regret. In Higbee (2025)'s point-identification setting, a quadratic approximation to smooth welfare identifies the relevant combinations through marginal welfare effects. This approximation does not apply to our problem, because the maximum regret can be nonsmooth in the reduced form as noted above. We instead use the fact that the first-order approximation to excess regret is piecewise linear. The gradients of these linear pieces, for the policies with the smallest maximum regret, span the directions that matter, and RDDR estimates them from the pilot. We show that any increase in the full design loss from using RDDR is of higher order, so that, asymptotically, decisions based on RDDR are equivalent to those based on the full design loss. In the SNAP-calibrated exercise without covariates, the reduced form contains 9 moments, and the baseline RDDR rule integrates over about

1.5 dimensions on average.

We illustrate the methods in a numerical exercise calibrated to the SNAP outreach experiment (Finkelstein and Notowidigdo, 2019). Comparing allocations of the same pilot and second-wave sample sizes, our design rule reduces expected excess regret by 27 per cent relative to a benchmark allocation based on the original study. RDDR retains nearly all of this gain at about one-third of the computation time. A design targeting the precision of a local average treatment effect has 79 per cent higher expected excess regret than equal allocation, and it would need about twice as many second-wave observations to match the baseline RDDR rule. This illustrates the importance of matching the design criterion to the policy decision.

To our knowledge, this paper is the first to design experiments for a terminal policy choice when the payoffs of the policies of interest are only partially identified. Besides the literature discussed above, much of the work on adaptive experimentation targets the rewards earned during the experiment. Bandit methods, including Thompson sampling, balance information acquisition against these rewards (Russo et al., 2018; Adusumilli, 2025). Cesa-Bianchi, Colomboni, and Kasy (2025) study adaptive maximisation of cumulative social welfare that must be inferred from observed behaviour. We instead evaluate a design only by the policy chosen after the experiment. Russo (2020) and Kasy and Sautmann (2021) share this focus, but their designs inform a choice among the experimental treatments themselves, whose payoffs the experiment point-identifies.

The rest of the paper proceeds as follows. Section 2 formulates the two-wave decision problem, characterises the identified set for policy payoffs, and defines maximum regret. Section 3 constructs the sequential Gaussian decision problem and characterises asymptotically optimal decision procedures. Section 4 develops a finite-sample quasi-Bayesian decision procedure and regret-directed dimension reduction, which simplifies the computation of design losses. Section 5 reports the SNAP-calibrated design exercises.

Notation. For a finite set $\mathcal{X}$, $\Delta(\mathcal{X})$ denotes its probability simplex. For a probability law Q, $\Pr_Q$ and $\mathbb{E}_Q$ denote probability and expectation under Q, with subscripts omitted when clear. We write $\mathbf{1}\{A\}$ for the indicator of an event A and $[a]_+ := \max\{a, 0\}$ for the positive part of $a \in \mathbb{R}$. The notation $N(m, \Sigma)$ denotes a Gaussian law with mean m and covariance Σ, allowing singular Σ. We use $\|\cdot\|$ for the Euclidean norm of vectors and its induced operator norm for matrices. The matrix Id_d is the $d \times d$ identity matrix. We write $\mathrm{blockdiag}_j(A_j)$ for the matrix with diagonal blocks A_j and all off-diagonal blocks equal to zero. For symmetric matrices, $A \succeq B$ ($A \succ B$) means that $A - B$ is positive semidefinite (positive definite). For $A \succeq 0$, $A^{1/2}$ denotes its unique symmetric positive-semidefinite square root. We write $\Rightarrow$ and $\xrightarrow{p}$ for convergence in distribution and in probability.

2 Policy Problem, Identification, and Robust Terminal Choice

The policy of interest may not be directly implementable in the experiment. Besides the SNAP running example, three further examples illustrate this point.

Example: EITC Expansion. The U.S. Earned Income Tax Credit (EITC) is a benefit programme that gives low-income workers a cash payment when they file their tax returns. Under the current EITC scheme, workers without dependent children receive much less than workers with children. The research organisation MDRC ran the Paycheck Plus experiment in New York City and Atlanta to study the effects of an expanded EITC for these workers (Miller, Katz, and Isen, 2022). The researchers randomly offered some low-income adults without dependent children a bonus designed to mimic the additional credit under such an expansion. The experiment could not change the existing EITC scheme, which is set by federal law. The bonus was therefore paid separately from the tax refund. Filing taxes did not give participants the bonus automatically. They also had to claim it through a participating free tax-preparation site. In the first year, 54 per cent of those offered the bonus were eligible and filed their taxes, but only 42 per cent claimed the bonus. If the expanded EITC were implemented, the eligible filers who did not claim the bonus in the experiment would receive the credit when they file their taxes. Their outcomes with the extra payment, such as employment or earnings, are never observed in the experiment, so the experiment can only set-identify the payoff of the expanded EITC.

Example: Time-of-use electricity pricing. In 2016 and 2017, California's three largest investor-owned utilities ran opt-in experiments with time-of-use (TOU) electricity tariffs, which charge more during peak hours. Households first agreed to participate and were then randomly assigned to a TOU tariff or to the existing tiered tariff. The California Public Utilities Commission, which regulates these utilities, required the experiments to guide its later decision on default TOU pricing, but legal restrictions prohibited default enrolment before January 2018 (George et al., 2018, p. 1, footnote 1, and p. 2). An experiment to inform whether to make a given TOU tariff the default could therefore only randomise invitations or administrative help to enrol in that tariff, using meter data to observe electricity use among participants and nonparticipants. The default policy would allow households to opt out. Some households would decline

every experimental invitation but remain enrolled under the default. Their peak-period electricity use under the new tariff is never observed in the experiment, so the experiment can only set-identify the payoff of default enrolment. This gap matters. In Sacramento, where a municipal utility did randomise default enrolment, households enrolled only by default reduced peak use about half as much as households that opted in (Fowlie et al., 2021).

Example: AI tutoring. Under the EU AI Act, an AI tutor that evaluates students' work and adapts their lessons is a high-risk AI system (European Union, 2024). Before such a tutor passes a conformity assessment against the Act's requirements, it can be tested in classrooms only with parents' consent. Afterwards, it can be used with every student without asking for consent. For a policymaker who has such a tutor developed for its schools, the policy of interest is to complete the assessment and then use the AI tutor in every class. The decision whether to adopt it is made before the assessment, using an experiment that can only randomly invite some families to take part. Students whose parents would not consent under any invitation never use the AI tutor in the experiment. Such parents may be many. In a 2025 US poll, nearly seven in ten parents did not support giving AI software access to students' grades and assessment results (Prothero, 2025). Their children are unlikely to be typical, because school studies that need parents' written consent tend to leave out students who are more at risk, such as those with lower grades (Unger et al., 2004; Liu et al., 2017). Under the policy, these students would use the AI tutor in class. How much they would learn with it is never observed in the experiment, so the experiment can only set-identify the payoff of this policy.

In all of these examples and our SNAP running example, experimental evidence can help compare policies without point identifying all of their payoffs. We now formalise a two-wave decision problem in which the policymaker uses pilot data to choose a follow-up experiment, then uses the combined evidence to choose a policy.

2.1 Individuals, policies, and payoffs

A policymaker must choose a policy for a target population from a finite menu. We let $\mathcal{Z}^T$ denote the set of terminal policies. Before making this choice, the policymaker can run a two-wave experiment to gather information. The experimental arms available in the pilot and the follow-up experiment form a finite menu, which we denote $\mathcal{Z}^E$. These menus

need not coincide. Some policies under consideration may be infeasible to implement in the experiment, while some experimental arms may not be available as terminal policies. Write $\mathcal{Z} := \mathcal{Z}^E \cup \mathcal{Z}^T$ for the union of the two menus. We assume that $|\mathcal{Z}^E| \geq 2$ and $|\mathcal{Z}^T| \geq 2$. The status-quo policy, which we denote by 0, is included in both menus.

> **Running example.** In the SNAP example introduced above, the experimental and terminal menus are $\mathcal{Z}^E = \{0, 1, 2\}$ and $\mathcal{Z}^T = \{0, 1, z^\star\}$. Here 0 denotes the status quo, 1 information outreach, 2 information plus application assistance, and $z^\star$ proactive enrolment. We assume that application assistance can be provided in the experiment but cannot be offered at the scale required for the terminal policy. Proactive enrolment is a terminal policy that cannot be implemented in the outreach experiment. ◁

Each individual is characterised by a random vector $(X, V, Y(0), Y(1))$. The observed covariates $X \in \mathcal{X}$ indicate the individual's stratum, where $\mathcal{X}$ is finite.[8] The random variables $Y(0)$ and $Y(1)$ are the individual's potential outcomes under non-take-up and take-up, respectively. We assume that both potential outcomes are bounded and normalise them to lie in $[0, 1]$. For each policy $z \in \mathcal{Z}$, let $D(z) \in \{0, 1\}$ denote the individual's potential take-up under z. The response type $V := (D(z))_{z \in \mathcal{Z}} \in \mathcal{V} := \{0, 1\}^{|\mathcal{Z}|}$ collects these potential take-up decisions.

> **Running example.** In the SNAP example, the response type is $V = (D(0), D(1), D(2), D(z^\star))$. Suppose take-up is monotone across the four policies: $D(0) \leq D(1) \leq D(2) \leq D(z^\star)$. Then only five response types are possible:
>
> | | $D(0)$ | $D(1)$ | $D(2)$ | $D(z^\star)$ |
> |---|---|---|---|---|
> | v_0 | 0 | 0 | 0 | 0 |
> | v_1 | 0 | 0 | 0 | 1 |
> | v_2 | 0 | 0 | 1 | 1 |
> | v_3 | 0 | 1 | 1 | 1 |
> | v_4 | 1 | 1 | 1 | 1 |
>
> (1)
>
> For example, type v_1 consists of households that enrol only under policy $z^\star$. ◁

This formulation connects the model to noncompliance and instrumental-variables models (Angrist, Imbens, and Rubin, 1996; Vytlacil, 2002). We assume that policies affect outcomes only through take-up, so the outcome under z is $Y(D(z)) = D(z)Y(1) + (1 - D(z))Y(0)$.

[8] Assuming a finite $\mathcal{X}$ is not very restrictive in practice, because experiments typically stratify on a few discrete covariates, and continuous covariates can be grouped into cells.

Let P specify the conditional distribution of $(V, Y(0), Y(1))$ given X. Probabilities $\Pr_P(\cdot \mid X = x)$ and expectations $\mathbb{E}_P[\cdot \mid X = x]$ refer to this conditional law. The target population has known stratum shares $\omega = (\omega_x)_{x \in \mathcal{X}}$, with $\omega_x > 0$ and $\sum_{x \in \mathcal{X}} \omega_x = 1$.[9]

The payoff θ_z of a terminal policy $z \in \mathcal{Z}^T$ is its improvement in the average outcome relative to the status quo, net of costs. For each $z \in \mathcal{Z}^T$, let $c_z(x) \geq 0$ be the cost per additional treatment recipient relative to the status quo, and let $k_z(x) \geq 0$ be the implementation cost per member of stratum x, incurred regardless of take-up. Both costs are known and expressed in outcome units, with $c_0(x) = k_0(x) = 0$. Under P, the payoff of policy z is

$$\theta_z(P) := \sum_{x \in \mathcal{X}} \omega_x \mathbb{E}_P \Big[Y(D(z)) - Y(D(0)) \\ - c_z(x)(D(z) - D(0)) - k_z(x) \;\Big|\; X = x \Big], \tag{2}$$

with $\theta_0(P) = 0$. We collect the terminal payoffs given P in $\theta_{\mathcal{Z}^T}(P) := (\theta_z(P))_{z \in \mathcal{Z}^T}$.

Remark 1 (Utility and multiple outcomes). By redefining Y, we can think of θ_z as $\mathbb{E}[u(Y(D(z)))] - \mathbb{E}[u(Y(D(0)))]$, net of costs, for a known bounded utility function u. If the policymaker cares about several outcomes $Y_1, \ldots, Y_m$, u can take all of them as arguments. For example, u could be a weighted sum $\sum_{k=1}^m \lambda_k Y_k$ with known weights λ_k. The analysis below then applies to the redefined outcome. ◁

Remark 2 (Policies that differ across strata). We can also allow the policymaker to choose a policy $z_x \in \mathcal{Z}^T$ for each stratum x, for example proactive enrolment for one stratum and information outreach for the others. The payoff of the resulting policy is given by (2) with z replaced by z_x in the term for stratum x. ◁

2.2 Experimental data and decision procedures

The policymaker observes a pilot sample $S_1 = \{W_i^O\}_{i=1}^n$, where $W_i^O = (Y_i, D_i, Z_i, X_i)$ and $Z_i \in \mathcal{Z}^E$. Subscripts n below refer to the pilot sample size. Here Z_i records the experimental arm, D_i actual take-up, and Y_i the realised outcome. The pilot sampling and assignment mechanism is taken as given. After observing S_1, the policymaker chooses a follow-up experiment.

The design of the follow-up experiment is a vector $g = (\rho, q, \pi)$, where ρ is the sample-size ratio between the follow-up and pilot samples, $q = (q_x)_{x \in \mathcal{X}}$ gives the sampling probabilities

[9] We can think of ω as coming from administrative or census records on the target population. If it is estimated from a sample, we assume that the sample size is much larger than the experiment, so its estimation error is negligible relative to the sampling error of the experiment. When $|\mathcal{X}| = 1$, conditioning on X and the weight $\omega_x = 1$ are suppressed.

across strata, and $\pi = (\pi_x)_{x\in\mathcal{X}}$, with $\pi_x = (\pi_{zx})_{z\in\mathcal{Z}^E}$, gives the assignment probabilities within each stratum. For a fixed $0 < \bar{\rho} < \infty$, the design space is $\mathcal{G} := (0, \bar{\rho}] \times \Delta(\mathcal{X}) \times \Delta(\mathcal{Z}^E)^{|\mathcal{X}|}$. We write $\rho(g)$, $q_x(g)$, and $\pi_{zx}(g)$ for the coordinates of g. The follow-up sample size is $n_2(g) = \lfloor \rho(g)n \rfloor$.[10] A *stratum–arm group* (x, z) consists of units with $X = x$ and $Z = z$ and has sampling probability $q_x(g)\pi_{zx}(g)$.

After observing the pilot $S_1 = s_1$, the policymaker chooses an experiment from a finite *candidate list* $\mathcal{G}_{n,J}(s_1) = \{g_{1,n}(s_1), \ldots, g_{J,n}(s_1)\} \subseteq \mathcal{G}$.

The candidate list collects the designs that are feasible for the second wave, for example those that satisfy the experimental budget. The policymaker fixes the list before the pilot, as fixed designs or as prespecified measurable maps $g_{j,n}$ from pilot data to designs, and chooses one candidate after observing the pilot. The number of candidates J is finite and does not grow with n. When every design in $\mathcal{G}$ is feasible, a fine grid over $\mathcal{G}$ can serve as the list.

Running example. With one stratum and a fixed sample-size ratio ρ, we suppress ρ and the stratum sampling probability $q = 1$, and identify a SNAP design with its allocation vector $g = (\pi_0, \pi_1, \pi_2) \in \Delta(\mathcal{Z}^E)$. For example, a candidate list could be

$$\mathcal{G}_{n,2}(s_1) = \{(1/3, 1/3, 1/3),\ g_{2,n}(s_1)\},$$

where $g_{2,n}(s_1)$ is a feasible allocation calculated from pilot estimates of arm-specific outcome variances. The first candidate is fixed in advance; the second can vary across pilot realisations. ◁

Under design g, the follow-up experiment generates the sample $S_2(g) = \{W_i^E\}_{i=1}^{n_2(g)}$, where $W_i^E = (Y_i, D_i, Z_i, X_i)$ and $Z_i \in \mathcal{Z}^E$. Let Q_P^O denote the marginal distribution of each pilot observation W_i^O, and let $Q_{P,g}^E$ denote the marginal distribution of each second-wave observation W_i^E under design g. Q_P^O is determined by the conditional law P and the pilot experimental design. $Q_{P,g}^E$ is determined by the same P and the experimental design g. Their marginal distributions of X may differ from each other and from the target-population shares ω.

Definition 1 (Decision procedure)**.** A decision procedure is a pair (γ_n, δ_n) of measurable maps. The design rule γ_n assigns to each pilot sample s_1 a design $\gamma_n(s_1) \in \mathcal{G}_{n,J}(s_1)$. The

[10] In the asymptotic analysis of Section 3, $\rho(g)$ is fixed, so the two waves grow at the same rate. In any finite sample both waves are finite, and letting them grow together keeps the asymptotic analysis representative of the finite-sample problem.

policy rule δ_n assigns a terminal policy $\delta_n(s_1, s_2) \in \mathcal{Z}^T$ to each pilot sample s_1 and second-wave sample s_2.

We write $G_n := \gamma_n(S_1)$ for the selected design, so the procedure chooses the terminal policy $\delta_n(S_1, S_2(G_n))$.[11]

To make the rules below well defined when several policies or designs tie, fix a total order $\preceq_T$ on $\mathcal{Z}^T$. Terminal ties are broken by the $\preceq_T$-smallest minimiser, and design ties by the smallest candidate index.

The following assumptions specify the sampling structure, link the observed data to the potential outcomes, and restrict the model. Let $\mathcal{P}$ denote the set of conditional laws P of $(V, Y(0), Y(1))$ given X that satisfy these restrictions.

Assumption 1. *(i) Potential take-up and outcomes are well-defined at the unit level and do not depend on other units' assignments.*[12] *If an individual is observed in arm $Z = z$, then $D = D(z)$ and $Y = Y(D)$; thus assignment affects the outcome only through take-up.*

(ii) For every $P \in \mathcal{P}$, the pilot observations are i.i.d. from Q_P^O, assignment satisfies $Z \perp (V, Y(0), Y(1)) \mid X$, and $\Pr_{Q_P^O}(X = x, Z = z) > 0$ for every $x \in \mathcal{X}$ and $z \in \mathcal{Z}^E$.

(iii) For every $P \in \mathcal{P}$ and every design rule γ_n, conditional on $S_1 = s_1$, the second-wave sample consists of $n_2(\gamma_n(s_1))$ i.i.d. draws from $Q^E_{P,\gamma_n(s_1)}$. Under every design, assignment satisfies $Z \perp (V, Y(0), Y(1)) \mid X$.

(iv) For every $P \in \mathcal{P}$, $\big(\Pr_P(V = v \mid X = x)\big)_{x \in \mathcal{X}, v \in \mathcal{V}} \in \mathcal{L}_V$, where $\mathcal{L}_V \subseteq (\Delta(\mathcal{V}))^{|\mathcal{X}|}$ is known, nonempty, and polyhedral.

Part (i) imposes the stable unit treatment value assumption (SUTVA) (Rubin, 1980) that rules out interference between units. Parts (ii) and (iii) require that, in both waves, units are randomly assigned to arms within each stratum. The pilot must also include every arm in every stratum, so that every experimental moment can be estimated from it. For the follow-up experiment, a design may skip some arms and spend its sample on those that matter most for the decision. The response-type restrictions in part (iv) can include monotonicity and bounds on take-up probabilities under terminal policies, as illustrated below. Their polyhedral form allows these restrictions to be expressed as finitely many linear equalities

[11] Randomised rules can be formulated by adding an independent random variable at each decision. Theorem 2(i) shows that randomised rules do not lower regret.

[12] A related gap between experimental evidence and policy payoffs can arise under marketplace interference; see Holtz et al. (2024) for an Airbnb pricing experiment. Extending the analysis to such settings would require an identified set for marketwide payoffs and a sampling model that accommodates interference.

and inequalities. This leads to a linear programming approach to identification, following Balke and Pearl (1997).

Running example. In the SNAP example, monotonicity sets the probabilities of all response types outside (1) to zero. Bounds on take-up probabilities under proactive enrolment can further restrict the probabilities of the remaining types. One possibility is a bound on the take-up probability in the target population:

$$\underline{p}^{D}_{z^\star} \le \sum_{x \in \mathcal{X}} \omega_x \Pr_P\{D(z^\star) = 1 \mid X = x\} \le \overline{p}^{D}_{z^\star}, \qquad P \in \mathcal{P}.$$

Bounds may also be specified separately for each stratum:

$$\underline{p}^{D}_{x,z^\star} \le \Pr_P\{D(z^\star) = 1 \mid X = x\} \le \overline{p}^{D}_{x,z^\star}, \qquad x \in \mathcal{X}, \quad P \in \mathcal{P}.$$

Setting the two bounds equal fixes the take-up probability and setting both to one describes a policy that enrols everyone automatically, as when a platform switches permanently to a new user interface or when a government imposes a new compulsory policy. All of these bounds are linear in the response-type probabilities and can be included in $\mathcal{L}_V$. ◁

Remark 3 (Sources of pilot evidence). The pilot may be a small first wave of a newly planned study or an earlier experiment that was not originally intended as a pilot. Observational data are also allowed if they satisfy part (ii) of Assumption 1. In each case, the earlier sample and the second-wave sample must share the conditional law P specified above. ◁

2.3 What the experiments identify

The experiments provide information about take-up and outcomes in the experimental arms. For each $z \in \mathcal{Z}$ and $x \in \mathcal{X}$, define the population moments

$$\begin{aligned}
\mu^{D}_{xz}(P) &:= \mathbb{E}_P[D(z) \mid X = x], \\
\mu^{Y1}_{xz}(P) &:= \mathbb{E}_P[Y(1)D(z) \mid X = x], \\
\mu^{Y0}_{xz}(P) &:= \mathbb{E}_P[Y(0)(1 - D(z)) \mid X = x].
\end{aligned}$$

For an experimental arm $z \in \mathcal{Z}^E$, Assumption 1 implies that these three moments equal the observable conditional moments $\mathbb{E}[D \mid X = x, Z = z]$, $\mathbb{E}[YD \mid X = x, Z = z]$, and

$\mathbb{E}[Y(1-D) \mid X = x, Z = z]$, respectively. We call them the moments of stratum–arm group (x, z). The first is the take-up probability and the other two are the contributions of takers and non-takers to the arm's mean outcome. We stack these three moments across all stratum–arm groups $(x, z) \in \mathcal{X} \times \mathcal{Z}^E$ into a vector, which we call the *reduced form*:

$$\mu(P) := \big(\mu^D_{xz}(P), \mu^{Y1}_{xz}(P), \mu^{Y0}_{xz}(P)\big)_{x\in\mathcal{X},\, z\in\mathcal{Z}^E}.$$

We first consider what can be learned about policy payoffs when the reduced form is known. Write μ for a generic reduced-form value and $\mathcal{M} := \{\mu(P) : P \in \mathcal{P}\}$ for the set of values allowed by the model. For a terminal policy that is also an experimental arm, $z \in \mathcal{Z}^T \cap \mathcal{Z}^E$, the expectations in (2) are determined by μ. Replacing them with the corresponding experimental moments gives

$$\begin{aligned}\theta_z(\mu) := \sum_{x\in\mathcal{X}} \omega_x \Big[&(\mu^{Y1}_{xz} + \mu^{Y0}_{xz}) - (\mu^{Y1}_{x0} + \mu^{Y0}_{x0}) \\ &- c_z(x)(\mu^D_{xz} - \mu^D_{x0}) - k_z(x)\Big].\end{aligned} \tag{3}$$

Thus $\theta_z(\mu(P)) = \theta_z(P)$ for these policies. For a terminal-only policy, $z \in \mathcal{Z}^T \setminus \mathcal{Z}^E$, the corresponding moments are not observed. The response-type restrictions can bound its payoff but need not point identify it.

> **Running example.** In the SNAP example, proactive enrolment brings in types v_1, v_2, and v_3, none of whom enrol under the status quo. Differences between adjacent arms identify the contributions of v_2 and v_3 to its payoff. Type v_1 behaves like v_0 in every experimental arm and differs only under proactive enrolment. Within each stratum x, the experiments therefore reveal only the combined share of these two types, $1 - \mu^D_{x2}$, and their combined outcome mass without enrolment, μ^{Y0}_{x2}. They never reveal $Y(1)$ for v_1. A bound on take-up under proactive enrolment limits the share of v_1, and $Y(1) \in [0, 1]$ limits its outcome mass under enrolment. Even when the reduced form is known, the payoff of proactive enrolment is therefore bounded but generally not point-identified. ◁

To describe this payoff ambiguity, let $\mathcal{P}(\mu) := \{P \in \mathcal{P} : \mu(P) = \mu\}$ be the set of conditional laws consistent with μ. The identified set for terminal payoffs collects all terminal payoff vectors that these laws can generate.

Definition 2 (Identified set for terminal payoffs)**.** For $\mu \in \mathcal{M}$, define

$$\begin{aligned} \Theta_{\mathcal{Z}^T}(\mu) &:= \{\theta_{\mathcal{Z}^T}(P) : P \in \mathcal{P}(\mu)\}, \\ \underline{\theta}_z(\mu) &:= \inf_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} \theta_z, \qquad & z \in \mathcal{Z}^T, \\ \overline{\theta}_z(\mu) &:= \sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} \theta_z, \qquad & z \in \mathcal{Z}^T. \end{aligned} \tag{4}$$

For terminal policies in the experimental menu, both endpoints equal $\theta_z(\mu)$. For terminal-only policies, the endpoints may differ.

2.4 Linear programming representation

We compute the identified set for terminal payoffs by linear programming. The next definition introduces the variables of the programme and specifies which values are consistent with the experimental moments and the model assumptions. We then use these variables to calculate terminal payoffs.

For $v \in \mathcal{V}$ and $z \in \mathcal{Z}$, let $d_z(v)$ denote the coordinate of v corresponding to z.

Definition 3 (LP feasible set)**.** For $P \in \mathcal{P}$, let $\alpha(P)$ collect

$$\alpha^V_{xv}(P) := \Pr_P(V = v \mid X = x), \qquad \alpha^{Ya}_{xv}(P) := \mathbb{E}_P[Y(a)\mathbf{1}\{V = v\} \mid X = x], \quad a \in \{0,1\}.$$

For a generic array $\alpha = (\alpha^V, \alpha^{Y1}, \alpha^{Y0})$, define

$$\mathcal{F}(\mu) := \left\{ \alpha \;\middle|\; \begin{array}{ll} \alpha^V \in \mathcal{L}_V, & \\ 0 \le \alpha^{Ya}_{xv} \le \alpha^V_{xv} & (x \in \mathcal{X},\ v \in \mathcal{V},\ a \in \{0,1\}), \\ \displaystyle\sum_{v \in \mathcal{V}} \alpha^V_{xv} d_z(v) = \mu^D_{xz} & (x \in \mathcal{X},\ z \in \mathcal{Z}^E), \\ \displaystyle\sum_{v \in \mathcal{V}} \alpha^{Y1}_{xv} d_z(v) = \mu^{Y1}_{xz} & (x \in \mathcal{X},\ z \in \mathcal{Z}^E), \\ \displaystyle\sum_{v \in \mathcal{V}} \alpha^{Y0}_{xv} (1 - d_z(v)) = \mu^{Y0}_{xz} & (x \in \mathcal{X},\ z \in \mathcal{Z}^E) \end{array} \right\}.$$

The bound $0 \le \alpha^{Ya}_{xv} \le \alpha^V_{xv}$ follows from $Y(a) \in [0,1]$. The remaining equalities require the latent type shares and outcome contributions to match the observed moments in every experimental arm.

We now rewrite the same payoff using the variables in Definition 3. Since

$$Y(D(z)) - Y(D(0)) = (D(z) - D(0))(Y(1) - Y(0)),$$

substituting this identity into (2) and summing over response types gives, for every $z \in \mathcal{Z}^T$ and $\alpha \in \mathcal{F}(\mu)$,

$$\theta_z(\alpha) = \sum_{x \in \mathcal{X}} \omega_x \left[\sum_{v \in \mathcal{V}} (d_z(v) - d_0(v)) \left(\alpha_{xv}^{Y1} - \alpha_{xv}^{Y0} - c_z(x) \alpha_{xv}^{V} \right) - k_z(x) \right]. \tag{5}$$

Thus $\theta_z(\alpha(P)) = \theta_z(P)$ for every terminal policy. For $z \in \mathcal{Z}^T \cap \mathcal{Z}^E$, the moment constraints in Definition 3 also imply $\theta_z(\alpha) = \theta_z(\mu)$ for every $\alpha \in \mathcal{F}(\mu)$, with $\theta_z(\mu)$ given by (3).

The constraints in Definition 3 hold for every law in $\mathcal{P}(\mu)$, so the linear programme never excludes a possible value of a payoff vector in $\Theta_{\mathcal{Z}^T}(\mu)$. Theorem 1 shows that the LP characterisation is also sharp. Every feasible α is generated by some law in $\mathcal{P}(\mu)$, so the linear programme returns only payoff vectors that can actually occur. The bounds therefore cannot be narrowed without strengthening the assumptions. Because the set is sharp for the whole payoff vector and is compact, the worst case in maximum regret, defined below, is attained. For each policy, some law consistent with the data and the assumptions makes its regret equal to its maximum regret.

Theorem 1 (LP characterisation of the identified set for terminal payoffs)**.** *Under parts (i) and (iv) of Assumption 1, for every $\mu \in \mathcal{M}$,*

$$\Theta_{\mathcal{Z}^T}(\mu) = \{(\theta_z(\alpha))_{z \in \mathcal{Z}^T} : \alpha \in \mathcal{F}(\mu)\}.$$

The identified set for terminal payoffs $\Theta_{\mathcal{Z}^T}(\mu)$ is nonempty, compact, and convex. Consequently, for every $r \in \mathbb{R}^{|\mathcal{Z}^T|}$,

$$\sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} r^\top \theta = \max_{\alpha \in \mathcal{F}(\mu)} \sum_{z \in \mathcal{Z}^T} r_z \theta_z(\alpha).$$

The theorem replaces optimisation over conditional laws by a finite-dimensional linear programme, and the vector r selects the combination of payoffs to bound. With e_z the unit vector for policy z, the directions $r = e_z$ and $r = -e_z$ give the upper and lower bounds on θ_z. Since $\Theta_{\mathcal{Z}^T}(\mu)$ is compact and convex, its projection onto coordinate z, the identified set for θ_z, is the whole interval $[\underline{\theta}_z(\mu), \overline{\theta}_z(\mu)]$. The direction $r = e_{z'} - e_z$ gives the largest payoff difference between z' and z, which is the building block of maximum regret we will define in Subsection 2.5.

2.5 Maximum regret and policy choice

Given a particular payoff vector of all policies, regret would be the payoff lost by not choosing the best policy. When the reduced form leaves several payoff vectors possible, we evaluate a policy by its largest regret over the identified set for them.

Definition 4 (Maximum regret)**.** Define the maximum regret of policy $z \in \mathcal{Z}^T$ at reduced form μ by

$$R(z,\mu) = \sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} \left\{ \max_{z' \in \mathcal{Z}^T} \theta_{z'} - \theta_z \right\}.$$

When μ is known, the oracle decision minimises $R(z,\mu)$. The corresponding minimax-regret value is $R^*(\mu) := \min_{z \in \mathcal{Z}^T} R(z,\mu)$.

By Theorem 1, maximum regret can be computed using one LP for each alternative policy $z' \in \mathcal{Z}^T$:

$$R(z,\mu) = \max_{z' \in \mathcal{Z}^T} \sup_{\alpha \in \mathcal{F}(\mu)} \{\theta_{z'}(\alpha) - \theta_z(\alpha)\}. \tag{6}$$

The oracle value $R^*(\mu)$ can be strictly positive since knowing the reduced form need not resolve payoff ambiguity. Define the *excess regret* as $R(z,\mu) - R^*(\mu)$. Because excess regret is built from the linear programmes in (6), it is piecewise affine in μ, with kinks where the binding constraints change or two policies tie (Proposition 1). These kinks shape the local analysis in Section 3.

Running example. In the SNAP example, information outreach has point-identified payoff $\theta_1(\mu)$, whereas the identified interval for the payoff of proactive enrolment is $[\underline{\theta}_{z^\star}(\mu), \overline{\theta}_{z^\star}(\mu)]$. With a single stratum, we suppress X, set $c_1 = c_{z^\star} = 0$, and allow implementation costs per person k_1 and $k_{z^\star}$. Impose the bound on the take-up probability $\underline{p}^D_{z^\star} \leq \mathbb{E}_P[D(z^\star)] \leq \overline{p}^D_{z^\star}$. Equation (3) gives

$$\theta_1(\mu) = \mu_1^{Y1} + \mu_1^{Y0} - \mu_0^{Y1} - \mu_0^{Y0} - k_1.$$

Appendix S2.2 derives the endpoints for proactive enrolment:

$$\begin{aligned} \underline{\theta}_{z^\star}(\mu) &= \mu_2^{Y1} - \mu_0^{Y1} - \mu_0^{Y0} - k_{z^\star} \\ &\quad + \left[\mu_2^{Y0} + \mu_2^D - \overline{p}^D_{z^\star}\right]_+, \\ \overline{\theta}_{z^\star}(\mu) &= \mu_2^{Y1} - \mu_0^{Y1} - \mu_0^{Y0} + 1 - \mu_2^D - k_{z^\star} \\ &\quad - \left[1 - \overline{p}^D_{z^\star} - \mu_2^{Y0}\right]_+. \end{aligned} \tag{7}$$

With zero per-recipient costs, both endpoints can be attained at $\mathbb{E}_P[D(z^\star)] = \overline{p}^D_{z^\star}$, so only the upper bound on the take-up probability enters these expressions.

The positive parts make the bounds nonsmooth. The lower bound has a kink when μ_2^{Y0}, the outcome mass among non-enrollees in arm 2, equals $\overline{p}^D_{z^\star} - \mu_2^D$, the largest feasible mass induced to enrol only by proactive enrolment. The upper bound has a kink when μ_2^{Y0} equals $1 - \overline{p}^D_{z^\star}$, the smallest feasible mass that remains unenrolled under proactive enrolment.

The maximum regrets are

$$R(0, \mu) = \max\{[\theta_1(\mu)]_+, \overline{\theta}_{z^\star}(\mu)\}, \tag{8}$$

$$R(1, \mu) = \max\{[\theta_1(\mu)]_+, \overline{\theta}_{z^\star}(\mu)\} - \theta_1(\mu), \tag{9}$$

$$R(z^\star, \mu) = \left[[\theta_1(\mu)]_+ - \underline{\theta}_{z^\star}(\mu)\right]_+. \tag{10}$$

The moments of the assistance arm enter both payoff bounds. Learning these moments can therefore improve the terminal policy decision even though assistance is not a terminal option. ◁

In practice, μ is unknown and the terminal policy depends on the data. We evaluate a decision procedure (γ_n, δ_n) at P by its expected excess regret,

$$\mathbb{E}\left[R(\delta_n(S_1, S_2(G_n)), \mu(P)) - R^*(\mu(P))\right],$$

where the expectation is over both waves under P and the design rule γ_n. As in CMS (2026), the worst case over payoffs is taken at the true reduced form, before averaging over sampling variation.[13] The design rule affects expected excess regret through the second-wave evidence available to the terminal decision. Because the criterion depends on the unknown P, a procedure that does well at one P can do poorly at another, and typically no procedure minimises it at every P. Procedures must therefore be compared through an aggregate of their expected excess regret over P. Section 3 does so in a local asymptotic framework.

[13] With $r(z, \theta) := \max_{z' \in \mathcal{Z}^T} \theta_{z'} - \theta_z$, we have $R(z, \mu) = \sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} r(z, \theta)$, so the criterion averages the worst case $\sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} r(\delta(S), \theta)$ over the data S. This generally differs from the worst case of the average, $\sup_{\theta \in \Theta_{\mathcal{Z}^T}(\mu)} \mathbb{E}[r(\delta(S), \theta)]$. Finite-sample minimax expected regret instead takes a worst case over the full parameter space, including the reduced form, after the sampling expectation; see Stoye (2012); Yata (2025); Montiel Olea, Qiu, and Stoye (2026a).

3 Optimal Decision Procedures in Large Samples

This section characterises decision procedures that are optimal in large samples. At a fixed law P, many procedures do equally well as the sample grows. Whenever one policy has strictly smaller maximum regret than the others, any procedure based on a consistent estimate of the reduced form selects it with probability approaching one, so the expected excess regret of each such procedure vanishes. Following Hirano and Porter (2009) and CMS, we therefore compare procedures under local alternatives, whose reduced forms lie within $O(n^{-1/2})$ of a reference point. The reduced forms of these alternatives differ from the reference point by amounts of the same order as sampling error, so the alternatives remain hard to tell apart in large samples. When several policies tie in maximum regret at the reference point, these alternatives favour different policies, and procedures can choose a worse policy with probability that does not vanish. Comparing procedures under local alternatives therefore retains the uncertainty about the reduced form that the policymaker faces in finite samples. Using the limits-of-experiments framework of Le Cam (1972), we show that the two waves are asymptotically equivalent to two Gaussian observations, in the sense that whatever a sequence of procedures achieves in the limit can also be achieved by a procedure that sees only these two observations. We call the problem of choosing the design and the policy from these two observations the *sequential Gaussian decision problem*. Solving it therefore tells us how well any procedure can do in large samples, and the problem can be solved exactly by backward induction.

We begin by defining the criterion for comparing procedures in Subsection 3.1. This criterion, *local-average excess regret*, averages the expected excess regret of a procedure, scaled by $\sqrt{n}$, over local alternatives. Subsection 3.2 derives the Gaussian representation of the two waves. Subsection 3.3 solves the sequential Gaussian decision problem by backward induction, which gives its optimal procedure. It then links this problem back to the two-wave problem in large samples. No procedure for the two-wave problem in the class we consider has a limiting local-average excess regret smaller than the optimal value of the sequential Gaussian decision problem, and the subsection gives conditions under which some procedure attains this bound and is therefore asymptotically optimal in the two-wave problem. Finally, Subsection 3.4 lets the weights on local alternatives become increasingly spread out and shows that the optimal procedure for each choice of weights converges to a limiting procedure that does not depend on the weights, which we call the *diffuse Gaussian decision procedure*. Section 4 then constructs a finite-sample quasi-Bayesian procedure that makes the same choices of experiment and terminal policy as the diffuse Gaussian decision procedure in large samples.

3.1 Local alternatives and local-average excess regret

To study procedures near a reference law $P_0 \in \mathcal{P}$, we perturb its reduced form $\mu_0 := \mu(P_0)$. We allow the model $\mathcal{P}$ to be semiparametric. The reduced form $\mu \in \mathbb{R}^{d_\mu}$ of Subsection 2.3, with $d_\mu := 3|\mathcal{X}||\mathcal{Z}^E|$, is finite-dimensional, but the conditional law P of $(V, Y(0), Y(1))$ given X may contain infinite-dimensional nuisance parameters, such as the distribution of outcomes within each response type when Y is continuous. Following CMS (2026, Section 5.2), we therefore perturb the reduced form within a least favourable parametric submodel of $\mathcal{P}$, namely a path $t \mapsto P_t$ through P_0 with $\mu(P_t) = t$.[14] For now, we simply assume that a path with $\mu(P_t) = t$ exists.

Assumption 2 (Common path)**.** *Fix $P_0 \in \mathcal{P}$ and set $\mu_0 := \mu(P_0)$. There exist a neighbourhood $\mathcal{N}_0 \subset \mathbb{R}^{d_\mu}$ of μ_0 and a path $t \mapsto P_t \in \mathcal{P}$ satisfying $P_0 = P_{\mu_0}$ and $\mu(P_t) = t$ for every $t \in \mathcal{N}_0$.*

Because the laws of both waves are determined by the same P, the path is common in the sense that the same law P_t induces the laws of the pilot data and the second-wave data under every design. The path is also chosen to be least favourable, meaning that, among all parametric submodels of $\mathcal{P}$ through P_0, it locally carries the least information about the reduced form. We will explain the requirements in detail in Subsection 3.2. The local alternative in direction $h \in \mathbb{R}^{d_\mu}$ is $P_{n,h} := P_{\mu_0 + h/\sqrt{n}}$, defined whenever $\mu_0 + h/\sqrt{n} \in \mathcal{N}_0$. Let $\mathcal{H}_n$ denote the set of such directions. The corresponding reduced form is $\mu(P_{n,h}) = \mu_0 + h/\sqrt{n}$.

Let $\mathcal{Z}_0 := \arg\min_{z \in \mathcal{Z}^T} R(z, \mu_0)$ denote the terminal policies with the smallest maximum regret at the reference point. If $\mathcal{Z}_0$ contains a single policy, that policy remains the best under every local alternative once n is large, and any procedure based on a consistent estimate of the reduced form selects it with probability approaching one. The choice is hard only when several policies tie at the reference point. Nearby reduced forms can then favour different policies in $\mathcal{Z}_0$ and hence call for different experiments. Under local alternatives, the excess regret of each policy in $\mathcal{Z}_0$ is of order $n^{-1/2}$. We scale excess regret by $\sqrt{n}$ so that its large-sample limit is not degenerate, and define

$$\ell_n(z, h) := \sqrt{n}\left[R(z, \mu(P_{n,h})) - R^*(\mu(P_{n,h}))\right], \qquad (z, h) \in \mathcal{Z}^T \times \mathcal{H}_n.$$

To define the criterion, let $Q_{n,h}^{\gamma_n}$ denote the joint law of both waves under $P_{n,h}$ and the design rule γ_n. We weight local alternatives by a Gaussian density. Let φ_σ denote the $N(0, \sigma^2 \mathrm{Id}_{d_\mu})$ density and let $\varphi_{n,\sigma}$ be its restriction to $\mathcal{H}_n$, defined by

[14] An alternative is to use a tangent space and a Gaussian sequence formulation (Hirano and Porter, 2009; Xu, 2026).

$\varphi_{n,\sigma}(h) := \frac{\varphi_\sigma(h)\,\mathbf{1}\{h\in\mathcal{H}_n\}}{\int_{\mathcal{H}_n}\varphi_\sigma(u)\,du}$.[15]

Definition 5 (Local-average excess regret)**.** For $\sigma > 0$, the *local-average excess regret* of (γ_n, δ_n) at P_0 is

$$\mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) = \int_{\mathcal{H}_n} \mathbb{E}_{Q^{\gamma_n}_{n,h}}\left[\ell_n\left(\delta_n(S_1, S_2(G_n)), h\right)\right] \varphi_{n,\sigma}(h)\, dh.$$

As in CMS, this criterion is a hybrid of minimax and Bayes in the spirit of Hurwicz (1951). The minimax part is inside ℓ_n. At each local alternative, ℓ_n is built from the maximum regret of the chosen policy, which takes the worst case over all payoff vectors consistent with the reduced form. The Bayes part is outside. The criterion averages ℓ_n over the data from both waves and over local alternatives.

The weights $\varphi_{n,\sigma}$ can be read in two ways. From a frequentist perspective, they average the expected scaled excess regret across local alternatives. A procedure can do well at some local alternatives and poorly at others, so we rank procedures by this average. From a Bayesian perspective, they form a prior on the local parameter, and the criterion is the Bayes risk of the procedure under this prior. The scale σ is arbitrary, and there is no a priori reason to give one local alternative more weight than another. As in CMS (2026, Section 4.4), we first let $n \to \infty$ with σ fixed, which keeps the criterion finite, and then let $\sigma \to \infty$ in Subsection 3.4, which makes the weights flat.[16]

We next derive the limit of ℓ_n. For the policies in $\mathcal{Z}_0$, excess regret is zero at μ_0, so $\ell_n(z, h)$ is determined by how $R(z, \cdot)$ and $R^*(\cdot)$ change near μ_0. The next proposition gives two properties that we use repeatedly.

Proposition 1 (Properties of excess regret)**.** *Under Assumption 2, the functions $R(z, \cdot)$, $R^*(\cdot)$, and $R(z, \cdot) - R^*(\cdot)$ are continuous and piecewise affine on $\mathcal{M}$ for every $z \in \mathcal{Z}^T$. Moreover, the following conclusions hold.*

(i) For every $z \in \mathcal{Z}_0$, $R(z, \cdot)$ is Hadamard directionally differentiable at μ_0, with continuous, positively homogeneous derivative $\dot{R}_z : \mathbb{R}^{d_\mu} \to \mathbb{R}$.

(ii) For some $C < \infty$, $0 \le \ell_n(z, h) \le C(1 + \|h\|)$ for every n, $z \in \mathcal{Z}_0$, and $h \in \mathcal{H}_n$.

[15] Since $\mathcal{N}_0$ is a neighbourhood of μ_0, $\mathcal{H}_n$ contains an expanding ball. Hence every compact subset of $\mathbb{R}^{d_\mu}$ is contained in $\mathcal{H}_n$ for all sufficiently large n. For each fixed $\sigma > 0$, the denominator therefore tends to one and $\int_{\mathbb{R}^{d_\mu}} |\varphi_{n,\sigma}(h) - \varphi_\sigma(h)|\, dh \to 0$. Thus the two weighting densities become arbitrarily close as the sample size grows.

[16] With flat weights from the start, the criterion would integrate the expected excess regret over $\mathbb{R}^{d_\mu}$ with respect to Lebesgue measure. As CMS (2026, Section 4.4) note, this integral can be infinite for every procedure when $d_\mu > 1$, and then it cannot rank procedures. This happens, for example, when some directions of h do not affect the regret of any policy.

By part (i), for each $z \in \mathcal{Z}_0$ and each fixed h, the scaled excess regret $\ell_n(z,h)$ converges to

$$\ell(z,h) := \dot{R}_z(h) - \min_{z' \in \mathcal{Z}_0} \dot{R}_{z'}(h), \qquad z \in \mathcal{Z}_0.$$

Convergence for each fixed h does not by itself imply that the integral over h in the criterion converges. Part (ii) supplies a bound that holds for every n and is integrable under Gaussian weights, so dominated convergence gives, for every fixed $\sigma > 0$,

$$\int_{\mathcal{H}_n} \max_{z \in \mathcal{Z}_0} |\ell_n(z,h) - \ell(z,h)| \, \varphi_{n,\sigma}(h)\, dh \longrightarrow 0.$$

Hence, in large samples, the local-average excess regret of any procedure whose terminal choices lie in $\mathcal{Z}_0$ can be computed with ℓ, which does not depend on n, in place of ℓ_n.

Running example. To see what ℓ looks like, return to the one-stratum specialisation in (7). We choose a reference point at which two policies tie in maximum regret and the lower bound $\underline{\theta}_{z^\star}$ on the payoff of proactive enrolment is at a kink. Specifically, fix a reference point μ_0 with $\mu_2^{Y0} + \mu_2^D = \overline{p}_{z^\star}^D$, which puts $\underline{\theta}_{z^\star}$ at its kink, and $\mu_2^{Y0} > 1 - \overline{p}_{z^\star}^D$, which keeps the upper bound $\overline{\theta}_{z^\star}$ away from its kink. We also take $\theta_1(\mu_0) \neq 0$, so that one observed policy is strictly better, and place its payoff $[\theta_1(\mu_0)]_+$ at the midpoint of $[\underline{\theta}_{z^\star}(\mu_0), \overline{\theta}_{z^\star}(\mu_0)]$.

Let $z_b = 1$ if $\theta_1(\mu_0) > 0$ and $z_b = 0$ otherwise, so z_b is the locally better of policies 0 and 1. By (8)–(10),

$$R(z_b, \mu_0) = R(z^\star, \mu_0) = \frac{1}{2}\big\{\overline{\theta}_{z^\star}(\mu_0) - \underline{\theta}_{z^\star}(\mu_0)\big\},$$
$$R(1 - z_b, \mu_0) = R(z_b, \mu_0) + |\theta_1(\mu_0)|.$$

Hence $\mathcal{Z}_0 = \{z_b, z^\star\}$. Near μ_0, (8)–(10) give $R(z_b,\mu) - R(z^\star,\mu) = \overline{\theta}_{z^\star}(\mu) + \underline{\theta}_{z^\star}(\mu) - 2\theta_{z_b}(\mu)$.

Write $h = (h_z^D, h_z^{Y1}, h_z^{Y0})_{z \in \{0,1,2\}}$ for the local parameter. Differentiating this difference at μ_0 gives $\dot{R}_{z_b}(h) - \dot{R}_{z^\star}(h) = 2(h_2^{Y1} - h_{z_b}^{Y1} - h_{z_b}^{Y0}) - h_2^D + [h_2^{Y0} + h_2^D]_+$. The local comparison thus depends on h only through two linear combinations, which we collect in the linear map $\Lambda_{\mathrm{SNAP}}(z_b) : \mathbb{R}^9 \to \mathbb{R}^2$,

$$\Lambda_{\mathrm{SNAP}}(z_b)h := \begin{pmatrix} 2(h_2^{Y1} - h_{z_b}^{Y1} - h_{z_b}^{Y0}) - h_2^D \\ h_2^{Y0} + h_2^D \end{pmatrix}. \tag{11}$$

The second coordinate enters only through its positive part, which comes from

the kink in $\underline{\theta}_{z^\star}$. If $[\Lambda_{\text{SNAP}}(z_b)h]_2 \leq 0$, the comparison depends on h only through the first coordinate, and if $[\Lambda_{\text{SNAP}}(z_b)h]_2 > 0$, it depends on the sum of the two. Which combination of moments decides the comparison thus depends on the side of the kink on which the local alternative lies. The definition of ℓ gives

$$\begin{aligned} \ell(z_b, h) &= \Big[[\Lambda_{\text{SNAP}}(z_b)h]_1 + \big[[\Lambda_{\text{SNAP}}(z_b)h]_2\big]_+\Big]_+, \\ \ell(z^\star, h) &= \Big[-[\Lambda_{\text{SNAP}}(z_b)h]_1 - \big[[\Lambda_{\text{SNAP}}(z_b)h]_2\big]_+\Big]_+. \end{aligned} \tag{12}$$

◁

The approximation of ℓ_n by ℓ covers only the policies in $\mathcal{Z}_0$. Every policy outside $\mathcal{Z}_0$ has strictly positive excess regret at μ_0, and maximum regret is bounded, so $\ell_n(z, h)$ is of exact order $\sqrt{n}$ for such a policy. A procedure that chooses such a policy even with a small probability can therefore have local-average excess regret that diverges as n grows. These choices are easy to avoid, because the data distinguish such policies from those in $\mathcal{Z}_0$ with probability approaching one. We therefore restrict attention to sequences of procedures whose probability of choosing a policy outside $\mathcal{Z}_0$, averaged over local alternatives, vanishes faster than $n^{-1/2}$. The contribution of these choices to local-average excess regret then vanishes.

Definition 6 (Admissible local sequence)**.** Let $\mathbb{D}(P_0)$ denote the class of sequences of decision procedures $\{(\gamma_n, \delta_n)\}$ as in Definition 1 satisfying the following condition for every fixed $\sigma > 0$:

$$\sqrt{n} \int_{\mathcal{H}_n} \Pr_{Q_{n,h}^{\gamma_n}} \{\delta_n(S_1, S_2(G_n)) \notin \mathcal{Z}_0\}\, \varphi_{n,\sigma}(h)\, dh \longrightarrow 0.$$

The definition adapts the comparison class of CMS (2026, Definition 1) to our criterion. We now define asymptotic optimality within the class.

Definition 7 (Asymptotic optimality)**.** Fix P_0 and $\sigma > 0$. We say that $\{(\gamma_n, \delta_n)\} \in \mathbb{D}(P_0)$ is *asymptotically optimal* if its local-average excess regret converges and

$$\lim_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) = \inf_{\{(\gamma_n', \delta_n')\} \in \mathbb{D}(P_0)} \liminf_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_n', \delta_n'; P_0).$$

This definition evaluates the design rule and the policy rule jointly. A sequence of procedures is asymptotically optimal if no sequence in the class has smaller local-average excess regret in the limit.

3.2 Gaussian representation of the two waves

To find asymptotically optimal procedures, we first show that in large samples, the data from the two waves are equivalent to two Gaussian observations. The pilot behaves like a single observation $\mathsf{T}_1 \sim N(I_1 h, I_1)$. Under design g, the second wave behaves like a second observation $\mathsf{T}_{2,g} \sim N(I_2(g)h, I_2(g))$. The matrices I_1 and $I_2(g)$ measure how much the pilot and the second wave reveal about the reduced form, so the experimental design determines how precise the second observation is. In the limit, the policymaker observes T_1, chooses a design G, observes $\mathsf{T}_{2,G}$, and chooses a policy.

We begin by explaining how the local alternatives perturb both waves together. We then show that the pilot behaves like T_1 and that, for each design g fixed in advance, the second wave behaves like $\mathsf{T}_{2,g}$. Next, we show that when the design G is chosen after seeing the pilot, the second wave behaves like $\mathsf{T}_{2,G}$. Finally, we show that whatever a sequence of procedures achieves in the limit can also be achieved in a sequential Gaussian decision problem, in which the design is chosen from T_1 and the policy from T_1 and $\mathsf{T}_{2,G}$.

The local alternatives lie on the path of Assumption 2. This path perturbs the law P instead of the laws of the observed data, because arbitrary perturbations of the observed laws may not be compatible with the response-type restrictions in Assumption 1(iv). In the SNAP example, monotonicity implies $\mu^D_{x2} - \mu^D_{x1} = \Pr_P(V = v_2 \mid X = x) \geq 0$, so no law in $\mathcal{P}$ makes take-up in arm 2 lower than in arm 1 within a stratum. For each t, the same P_t must generate the pilot law and the second-wave law under every design. This is what we mean by a *common path*.[17]

The common path induces a *pilot submodel* $t \mapsto Q^O_t := Q^O_{P_t}$ of the pilot model $\{Q^O_P : P \in \mathcal{P}\}$ and, for each design g, a *second-wave submodel* $t \mapsto Q^E_{t,g} := Q^E_{P_t,g}$ of the second-wave model $\{Q^E_{P,g} : P \in \mathcal{P}\}$. These are parametric models for the law of a single pilot observation and of a single second-wave observation, indexed by the reduced form t. For the local alternative $P_{n,h}$, let $Q^O_{n,h} := (Q^O_{P_{n,h}})^{\otimes n}$ and $Q^E_{n,h,g} := (Q^E_{P_{n,h},g})^{\otimes n_2(g)}$ denote the corresponding laws of the full pilot sample S_1 and of the full second-wave sample S_2 under a fixed design g. By parts (ii) and (iii) of Assumption 1, $Q^{\gamma_n}_{n,h}(ds_1, ds_2) = Q^O_{n,h}(ds_1)\, Q^E_{n,h,\gamma_n(s_1)}(ds_2)$.

To show that each wave behaves like a Gaussian observation, we also need the likelihood to be smooth in t along these submodels. The standard condition is differentiability in quadratic mean. A submodel $t \mapsto Q_t$ with densities q_t with respect to a dominating measure

[17] One can instead specify the local pilot and second-wave laws first and then check whether they are induced by a common path $P_t \in \mathcal{P}$ through P_0 with $\mu(P_t) = t$. In the parametric models of Higbee (2025, Sections 2.2 and 4.1) and Hirano and Porter (2025, Section 4.2), a finite-dimensional parameter indexes the laws of all stages and designs, so local alternatives of the form $\theta_0 + h/\sqrt{n}$ already link these laws.

ν is *differentiable in quadratic mean* (DQM) at t_0, with score $\dot{\ell}$, if

$$\int \left[\sqrt{q_t} - \sqrt{q_{t_0}} - \tfrac{1}{2}(t-t_0)^\top \dot{\ell}\sqrt{q_{t_0}}\right]^2 d\nu = o\big(\|t-t_0\|^2\big) \qquad \text{as } t \to t_0.$$

DQM holds for smooth parametric families, such as the multinomial model for (D, Y) within each stratum–arm group when Y is binary (van der Vaart, 1998, Chapter 7). Its key consequence is an expansion of the log-likelihood ratio. For n i.i.d. observations from a DQM submodel with score $\dot{\ell}$ and information $I := \mathbb{E}[\dot{\ell}\dot{\ell}^\top]$, the log-likelihood ratio of the local alternative $t_0 + h/\sqrt{n}$ against t_0 satisfies

$$\sum_{i=1}^{n} \log \frac{q_{t_0+h/\sqrt{n}}(W_i)}{q_{t_0}(W_i)} = h^\top \mathsf{T}_n - \frac{1}{2} h^\top I h + o_p(1), \qquad \mathsf{T}_n := \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \dot{\ell}(W_i) \Rightarrow N(0, I),$$

under t_0 (van der Vaart, 1998, Theorem 7.2). This expansion is called *local asymptotic normality* (LAN). For a single Gaussian observation $\mathsf{T} \sim N(Ih, I)$, the log-likelihood ratio of h against $h = 0$ is exactly $h^\top \mathsf{T} - \frac{1}{2} h^\top I h$. The expansion above has the same form with T_n in place of T, and T_n converges to $N(0, I)$ which is the law of T at $h = 0$. The log-likelihood ratio of the sample therefore converges in distribution to that of the single observation. As a result, whatever a decision rule based on the sample achieves in the limit can also be achieved by a rule that sees only this one Gaussian observation (van der Vaart, 1998, Theorem 7.10).

The information I of this Gaussian observation depends on the path. Because $\mathcal{P}$ is allowed to be semiparametric, many paths through P_0 satisfy $\mu(P_t) = t$, and they can move the nuisance parameters in different ways, so they can carry different information about μ. We use a *least favourable* path, meaning that, of all parametric submodels, it is the one along which the data carry the least information about μ, as in CMS (2026, Section 5.2). Its score is then the efficient score for μ, and the information is the efficient information for μ, the inverse of the semiparametric efficiency bound.[18] Estimating μ along this path is therefore asymptotically as hard as estimating it in $\mathcal{P}$.

Since one path serves the pilot and every candidate design, it must be least favourable for all of these experiments. This is less demanding than it may appear. Within stratum–arm group (x, z), the stratum and the arm are fixed, so an observation varies only through its take-up D and outcome Y. In both waves, individuals come from the same population, and within each stratum x they are randomly assigned to the experimental arms (Assumption 1). The conditional law of (D, Y) in each group is therefore the same in the pilot and under every design. A design only determines the number of second-wave observations in each group.

[18] For a related use of least favourable submodels to derive efficiency bounds across experimental designs, see Armstrong (2026, Section 4).

Suppose that, near P_0, the model allows each within-group law to be perturbed slightly in any direction without changing the law in other groups. Then the efficient score for the moments of a group does not depend on how the sample is spread across groups, so it is the same in every experiment that samples the group. A path that is least favourable for the pilot is then least favourable for every design. In the SNAP model, the condition holds, for example, when the inequalities that monotonicity implies across arms hold strictly at P_0 (see Appendix S3.5 for sufficient conditions).

The next assumption collects the smoothness and efficiency requirements.

Assumption 3 (DQM and score regularity)**.** *(i) There exists a neighbourhood $\mathcal{N}_1 \subseteq \mathcal{N}_0$ of μ_0 such that Q_t^O is absolutely continuous with respect to $Q_{P_0}^O$ for all $t \in \mathcal{N}_1$. The pilot submodel $t \mapsto Q_t^O$ is DQM at μ_0, and its score $\dot{\ell}^O$ is the efficient score for μ. For some $\delta > 0$, $\mathbb{E}_{Q_{P_0}^O} \|\dot{\ell}^O\|^{2+\delta} < \infty$, and $I_1 := \mathbb{E}_{Q_{P_0}^O}[\dot{\ell}^O \dot{\ell}^{O\top}] \succ 0$.*

(ii) For each $g \in \mathcal{G}$, the score $\dot{\ell}_g^E$ of the second-wave submodel $t \mapsto Q_{t,g}^E$ at μ_0 is the efficient score for μ.

The information matrix of the second wave is $I_2(g) := \rho(g)\mathbb{E}_{Q_{P_0,g}^E}[\dot{\ell}_g^E \dot{\ell}_g^{E\top}]$.

By part (i), the pilot is locally asymptotically normal along the common path, and I_1^{-1} is the asymptotic variance of an efficient estimator of the reduced form based on the pilot. The condition $I_1 \succ 0$ means that the pilot is informative about every coordinate of the reduced form. For this, the pilot must at least sample every arm in every stratum, which Assumption 1(ii) ensures.

Part (i) also implies that the second-wave submodels are DQM. As explained above, the law of (D, Y) within each group is the same in both waves and under every design, and the law of a single second-wave observation depends on the design only through the probabilities $q_x(g)\pi_{zx}(g)$ of the groups. Along the path, the law of a second-wave observation is therefore the pilot law reweighted across groups. As we show in Appendix S3.2, every second-wave submodel is then DQM at μ_0, uniformly over designs, with the same score $\dot{\ell}^O$ as the pilot, so part (ii) only needs to add that this score is efficient under every design. As discussed before the assumption, this holds when each within-group law can be perturbed without changing the law in other groups.

The matrix $I_2(g)$ may be singular, for example when g does not sample some group. Because I_1 is positive definite, the two waves together remain informative about every coordinate of the reduced form, that is, $I_1 + I_2(g)$ is positive definite.

Running example. In the SNAP model, suppose that the inequalities that monotonicity implies across arms hold strictly at P_0. The moments of group

(x, z) are then estimated efficiently by the sample mean of $(D, YD, Y(1-D))$ in that group. With m observations in the group, this sample mean has variance Σ_{xz}/m, where Σ_{xz} is the covariance matrix of $(D, YD, Y(1-D))$ in the group under P_0, and the information block for the group is the inverse of n times this variance, $(m/n)\Sigma_{xz}^{-1}$. In the pilot, m/n is the share p_{xz}^O of pilot observations in the group, and in the second wave it is $\rho(g)q_x(g)\pi_{zx}(g)$, so

$$\begin{aligned} I_1 &= \text{blockdiag}_{x\in\mathcal{X},\, z\in\mathcal{Z}^E}\left(p_{xz}^O \Sigma_{xz}^{-1}\right), \\ I_2(g) &= \rho(g)\,\text{blockdiag}_{x\in\mathcal{X},\, z\in\mathcal{Z}^E}\left(q_x(g)\pi_{zx}(g)\Sigma_{xz}^{-1}\right). \end{aligned} \tag{13}$$

◁

Under Assumption 3, the log-likelihood ratios of the two waves depend on the data only through the normalised sums of efficient scores up to negligible terms[19]

$$\begin{aligned} \mathsf{T}_{1,n} &:= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \dot{\ell}^O(W_i^O), \\ \mathsf{T}_{2,n}(g) &:= \frac{1}{\sqrt{n}} \sum_{i=1}^{n_2(g)} \dot{\ell}_g^E(W_i^E). \end{aligned} \tag{14}$$

For a fixed design g, the central limit theorem gives $\mathsf{T}_{1,n} \Rightarrow \mathsf{T}_1 \sim N(0, I_1)$ under $Q_{n,0}^O$ and $\mathsf{T}_{2,n}(g) \Rightarrow N(0, I_2(g))$ under $Q_{n,0,g}^E$. Under the local alternative h, Le Cam's third lemma shifts these limits to $N(I_1 h, I_1)$ and $N(I_2(g)h, I_2(g))$ (van der Vaart, 1998, Example 6.7). Each wave thus behaves like a Gaussian observation of h.

The limits above hold for a fixed design. Because the design rule selects $G_n = \gamma_n(S_1)$ from the candidate list after observing the pilot, the limit of the second wave must also be derived for a design that depends on the pilot data. We need to impose the following additional requirement about the candidate design list.

Assumption 4 (Limits of the candidate list)**.** *For fixed J, there exist measurable maps $g_j : \mathbb{R}^{d_\mu} \to \mathcal{G}$, $j = 1, \ldots, J$, such that, under $Q_{n,0}^O$,*

$$\left(\mathsf{T}_{1,n}, g_{1,n}(S_1), \ldots, g_{J,n}(S_1)\right) \Rightarrow \left(\mathsf{T}_1, g_1(\mathsf{T}_1), \ldots, g_J(\mathsf{T}_1)\right).$$

The assumption holds trivially for a list fixed in advance, because each g_j is then constant and $\mathsf{T}_{1,n}$ satisfies the central limit theorem. A list computed from the pilot must converge jointly with $\mathsf{T}_{1,n}$. A common case is a candidate computed from consistent pilot estimates,

[19] In Le Cam's terminology, $\mathsf{T}_{1,n}$ and $\mathsf{T}_{2,n}(g)$ are the central sequences of the two LAN expansions.

such as the allocation based on estimated outcome variances in the SNAP candidate list. The variance estimates converge to the population variances, so this candidate converges in probability to the allocation computed from the population variances.

We can now show that the two waves behave like the two Gaussian observations when the design is chosen after seeing the pilot. The selected design $G_n = \gamma_n(S_1)$ is a known function of the pilot data, so the likelihood ratio of the two waves is the pilot likelihood ratio times the second-wave likelihood ratio at G_n. Because the second-wave expansion holds uniformly over designs, it also holds at the data-dependent design G_n. By Assumption 4, the candidates from which G_n is chosen converge jointly with $\mathsf{T}_{1,n}$. In large samples, the two waves together therefore have the same likelihood ratios as the two Gaussian observations described at the start of this subsection. The next definition states formally how a procedure uses these observations. Like a decision procedure in Definition 1, a Gaussian decision procedure consists of a design rule and a policy rule. The design rule chooses a candidate design from the first Gaussian observation, and the policy rule chooses a terminal policy from both observations.

Definition 8 (Gaussian decision procedure)**.** A *Gaussian decision procedure* $(\gamma_\infty, \delta_\infty)$ consists of a measurable design rule $\gamma_\infty(t_1, u_1) \in \{g_1(t_1), \ldots, g_J(t_1)\}$ and a measurable policy rule $\delta_\infty(t_1, t_2, g, u_2) \in \mathcal{Z}^T$. At local parameter $h \in \mathbb{R}^{d_\mu}$, the procedure generates observations and choices in the order

$$\begin{aligned} \mathsf{T}_1 &= I_1 h + I_1^{1/2}\zeta_1, \\ G &= \gamma_\infty(\mathsf{T}_1, U_1), \\ \mathsf{T}_{2,G} &= I_2(G)h + I_2(G)^{1/2}\zeta_2, \\ Z_\infty &= \delta_\infty(\mathsf{T}_1, \mathsf{T}_{2,G}, G, U_2), \end{aligned} \tag{15}$$

where ζ_1 and ζ_2 are standard d_μ-dimensional Gaussian vectors, U_1 and U_2 are uniform on $[0, 1]$, and all four are mutually independent.

The uniform variables U_1 and U_2 allow the choices to be random given the observations.[20] The next proposition shows that the limits of $\mathsf{T}_{1,n}$, $\mathsf{T}_{2,n}(G_n)$, the selected design, and the chosen policy can all be generated by a Gaussian decision procedure.

Proposition 2 (Gaussian representation)**.** *Suppose Assumptions 2, 3, and 4 hold, and let* (γ_n, δ_n) *be a sequence of decision procedures as in Definition 1. Every subsequence has a further subsequence, indexed again by* n*, and a Gaussian decision procedure* $(\gamma_\infty, \delta_\infty)$ *such*

[20]The limiting design or policy of a deterministic finite-sample procedure can remain random given the Gaussian observations, for example when the procedure uses a statistic that is asymptotically independent of $\mathsf{T}_{1,n}$ and $\mathsf{T}_{2,n}(G_n)$. The variables U_1 and U_2 reproduce this randomness in the limit.

that, for every fixed h, under $Q_{n,h}^{\gamma_n}$,

$$\begin{aligned}\big(\mathsf{T}_{1,n}, G_n, \mathsf{T}_{2,n}(G_n), \delta_n(S_1, S_2(G_n))\big)\\ \Rightarrow \big(\mathsf{T}_1, G, \mathsf{T}_{2,G}, Z_\infty\big),\end{aligned} \tag{16}$$

where the limit is generated by $(\gamma_\infty, \delta_\infty)$ at h as in (15).

Proposition 2 is the formal version of the statement at the start of this subsection that, in large samples, the data from the two waves are equivalent to two Gaussian observations. In other words, whatever a sequence of procedures achieves in the limit can be achieved by a Gaussian decision procedure.[21]

Running example. In the one-stratum SNAP example, the information matrices in (13) and the statistics in (14) become

$$\begin{aligned} I_1 &= \mathrm{blockdiag}_z\big(p_z^O \Sigma_z^{-1}\big), & \mathsf{T}_{1,n} &= I_1 \sqrt{n}(\widehat{\mu}_1 - \mu_0) + o_p(1),\\ I_2(g) &= \rho(g)\,\mathrm{blockdiag}_z\big(\pi_z(g)\Sigma_z^{-1}\big), & \mathsf{T}_{2,n}(g) &= I_2(g)\sqrt{n}(\widehat{\mu}_{2,g} - \mu_0) + o_p(1),\end{aligned}$$

where $\widehat{\mu}_1$ and $\widehat{\mu}_{2,g}$ stack the sample means of $(D, YD, Y(1-D))$ in the three arms in the pilot and in the second wave under g. Under the local alternative h, these sample means are approximately normal around $\mu_0 + h/\sqrt{n}$, so $\mathsf{T}_{1,n}$ and $\mathsf{T}_{2,n}(G_n)$ are approximately the two Gaussian observations $\mathsf{T}_1 = I_1 h + I_1^{1/2}\zeta_1$ and $\mathsf{T}_{2,G} = I_2(G)h + I_2(G)^{1/2}\zeta_2$, where the allocation G is chosen from the candidate list after T_1 is observed. By Proposition 2, a procedure that chooses the allocation using the pilot sample means and the policy using the sample means of both waves behaves, in large samples, like one that chooses G using T_1 and the policy using T_1 and $\mathsf{T}_{2,G}$. ◁

3.3 The sequential Gaussian decision problem

We now find the optimal procedure in large samples. By the Gaussian representation in Proposition 2, it suffices to look among Gaussian decision procedures, which see sequentially only the two Gaussian observations. The resulting problem is the sequential Gaussian

[21] Hirano and Porter (2025, Theorem 2) give a similar representation for sequential procedures whose choices converge in distribution under every local alternative. Proposition 2 does not require this convergence and passes to subsequences instead, because the admissible class $\mathbb{D}(P_0)$ also contains procedures whose choices keep oscillating as n grows. Higbee (2025, Lemma 4.5) assumes joint convergence of the selected design and of the terminal-policy process over all designs, together with stochastic equicontinuity of that process.

decision problem, in which the policymaker chooses the design after the first Gaussian observation and the policy after the second. This problem is easier to solve, and we later show how to solve it exactly by backward induction. To set up this problem, we need a criterion for Gaussian decision procedures. Each Gaussian decision procedure represents the sequences of procedures that make the same choices as it does in large samples. We evaluate it by the local-average excess regret that these sequences attain in the limit, which we derive next. The best Gaussian decision procedure then gives the optimal procedure in large samples.

Consider a sequence of procedures (γ_n, δ_n) whose terminal choices lie in $\mathcal{Z}_0$, the policies for which ℓ_n has a finite limit. Suppose that it makes the same choices as a Gaussian decision procedure $(\gamma_\infty, \delta_\infty)$ in large samples, meaning that under each local alternative its terminal choices converge in distribution to those of $(\gamma_\infty, \delta_\infty)$. By Subsection 3.1, we can replace ℓ_n by its limit ℓ in the local-average excess regret of this sequence. At each h, the expectation of ℓ at the chosen policy is the sum over $\mathcal{Z}_0$ of $\ell(z, h)$ times the probability of choosing z. Hence

$$\mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) = \int_{\mathcal{H}_n} \sum_{z \in \mathcal{Z}_0} \ell(z, h)\ \Pr_{Q^{\gamma_n}_{n,h}}\big\{\delta_n(S_1, S_2(G_n)) = z\big\}\, \varphi_{n,\sigma}(h)\, dh + o(1).$$

As n grows, the probability of choosing z approaches $\Pr_h\{Z_\infty = z\}$, and the weights $\varphi_{n,\sigma}$ approach φ_σ. Here $\Pr_h$ and $\mathbb{E}_h$ denote probability and expectation over $(\zeta_1, \zeta_2, U_1, U_2)$ when the observations and choices of $(\gamma_\infty, \delta_\infty)$ are generated by (15) at the local parameter h. Local-average excess regret therefore converges to

$$\int_{\mathbb{R}^{d_\mu}} \sum_{z \in \mathcal{Z}_0} \ell(z, h) \Pr_h\{Z_\infty = z\}\, \varphi_\sigma(h)\, dh = \int_{\mathbb{R}^{d_\mu}} \mathbb{E}_h\big[\ell(Z_\infty, h)\big]\, \varphi_\sigma(h)\, dh,$$

which depends on the sequence only through $(\gamma_\infty, \delta_\infty)$. It is the criterion in the next definition.

Definition 9 (Sequential Gaussian decision problem)**.** Fix $\sigma > 0$. The *Gaussian excess regret* of a Gaussian decision procedure $(\gamma_\infty, \delta_\infty)$ whose policy rule takes values in $\mathcal{Z}_0$ is

$$\mathfrak{R}_{\infty,\sigma}(\gamma_\infty, \delta_\infty; P_0) := \int_{\mathbb{R}^{d_\mu}} \mathbb{E}_h\big[\ell(Z_\infty, h)\big]\, \varphi_\sigma(h)\, dh.$$

The *sequential Gaussian decision problem* is to minimise $\mathfrak{R}_{\infty,\sigma}$ over all such procedures, with value

$$\mathfrak{R}^*_{\infty,\sigma}(P_0) := \inf_{(\gamma_\infty, \delta_\infty)} \mathfrak{R}_{\infty,\sigma}(\gamma_\infty, \delta_\infty; P_0).$$

The policy rule is restricted to $\mathcal{Z}_0$, where $\ell(z, \cdot)$ is defined. This is without loss for

sequences in $\mathbb{D}(P_0)$, whose choices outside $\mathcal{Z}_0$ do not affect local-average excess regret in the limit.

Running example. In the SNAP example, $\mathcal{Z}_0 = \{z_b, z^\star\}$ by Subsection 3.1. The sequential Gaussian decision problem is to choose an allocation G from the candidate list after observing T_1, and then a policy in $\{z_b, z^\star\}$ after observing $\mathsf{T}_{2,G}$, so as to minimise $\int \mathbb{E}_h[\ell(Z_\infty, h)]\, \varphi_\sigma(h)\, dh$. The allocation matters through $I_2(G)$ in (13), which determines how precisely $\mathsf{T}_{2,G}$ measures the moments of each arm. ◁

To solve the sequential Gaussian decision problem, we rewrite it as a Bayesian problem. The Gaussian excess regret averages $\mathbb{E}_h[\ell(Z_\infty, h)]$ over h. Draw H from φ_σ, independently of $(\zeta_1, \zeta_2, U_1, U_2)$, and generate the observations and choices by (15) at $h = H$. Write $\Pr_\sigma$ for the resulting joint law of $(H, \mathsf{T}_1, G, \mathsf{T}_{2,G}, Z_\infty)$ and $\mathbb{E}_\sigma$ for expectation under it. Because H is independent of $(\zeta_1, \zeta_2, U_1, U_2)$, the conditional law of the observations and choices given $H = h$ is the law under which $\mathbb{E}_h$ is computed. Hence $\mathbb{E}_\sigma[\ell(Z_\infty, H) \mid H = h] = \mathbb{E}_h[\ell(Z_\infty, h)]$, and averaging over H, whose density is φ_σ, gives

$$\mathbb{E}_\sigma\big[\ell(Z_\infty, H)\big] = \int_{\mathbb{R}^{d_\mu}} \mathbb{E}_h\big[\ell(Z_\infty, h)\big]\, \varphi_\sigma(h)\, dh = \mathfrak{R}_{\infty,\sigma}(\gamma_\infty, \delta_\infty; P_0).$$

In this joint law, φ_σ is a prior on the local parameter. Since $\ell \geq 0$, Tonelli's theorem lets us compute $\mathbb{E}_\sigma[\ell(Z_\infty, H)]$ by integrating first over h given the data and then over the data. This order matches the policymaker's viewpoint. The policymaker never observes h, so at each decision it averages over what remains unknown given the data observed so far. The sequential Gaussian decision problem is therefore a Bayesian sequential decision problem with prior φ_σ, and we solve it by backward induction. We first find the best policy for any data from both waves, and then the best design, anticipating that the policy will be chosen optimally.

Backward induction uses the posterior of H after each wave. Write $\pi_{\sigma,1}(\cdot \mid t_1)$ for the conditional law of H given $\mathsf{T}_1 = t_1$, and $\pi_{\sigma,+}(\cdot \mid t_1, t_2, g)$ for its conditional law after the second-wave observation t_2 under design g. With a Gaussian prior and Gaussian observations, these posteriors are Gaussian:

$$\begin{aligned}
\pi_{\sigma,1}(\cdot \mid t_1) &= N\left(V_{\sigma,1} t_1, V_{\sigma,1}\right), \\
\pi_{\sigma,+}(\cdot \mid t_1, t_2, g) &= N\left(V_{\sigma,+}(g)(t_1 + t_2), V_{\sigma,+}(g)\right),
\end{aligned}$$

where $V_{\sigma,1} := (\sigma^{-2}\mathrm{Id}_{d_\mu} + I_1)^{-1}$ and $V_{\sigma,+}(g) := (\sigma^{-2}\mathrm{Id}_{d_\mu} + I_1 + I_2(g))^{-1}$.

At the last stage, the policymaker has observed (t_1, t_2, g), so only h is unknown, and the optimal policy minimises the posterior expectation of $\ell(z, H)$:

$$\delta^*_{\infty,\sigma}(t_1, t_2, g) = \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\sigma,+}(dh \mid t_1, t_2, g).$$

Here and below, $\min_{\preceq_T} A$ denotes the element of a set $A \subseteq \mathcal{Z}^T$ that comes first in the order $\preceq_T$ fixed in Subsection 2.2.

Running example. At the SNAP reference point of Subsection 3.1, $\ell(z_b, \cdot)$ and $\ell(z^\star, \cdot)$ depend on h only through $\Lambda_{\mathrm{SNAP}}(z_b)h$, so the policy choice depends on the data only through the posterior of $\Lambda_{\mathrm{SNAP}}(z_b)H$. Given (t_1, t_2, g), let m_1, m_2 be the posterior means of its two coordinates, and let s_2 be the posterior standard deviation of the second coordinate,

$$\begin{aligned} m_j &= [\Lambda_{\mathrm{SNAP}}(z_b) V_{\sigma,+}(g)(t_1 + t_2)]_j, \qquad j = 1, 2, \\ s_2^2 &= [\Lambda_{\mathrm{SNAP}}(z_b) V_{\sigma,+}(g) \Lambda_{\mathrm{SNAP}}(z_b)^\top]_{22}. \end{aligned}$$

Write $\Delta_\sigma(t_1, t_2, g)$ for the posterior expectation of $\ell(z_b, H) - \ell(z^\star, H)$ given (t_1, t_2, g). By (12), this difference equals $[\Lambda_{\mathrm{SNAP}}(z_b)H]_1 + \big[[\Lambda_{\mathrm{SNAP}}(z_b)H]_2\big]_+$, and the expectation of the positive part of a Gaussian variable gives

$$\Delta_\sigma(t_1, t_2, g) = m_1 + s_2 \phi(m_2/s_2) + m_2 \Phi(m_2/s_2), \tag{17}$$

where ϕ and Φ are the standard normal density and distribution function. The optimal policy chooses proactive enrolment $z^\star$ when $\Delta_\sigma > 0$ and z_b when $\Delta_\sigma < 0$.

By Jensen's inequality, $\Delta_\sigma \geq m_1 + [m_2]_+$, the value of $\ell(z_b, h) - \ell(z^\star, h)$ at the posterior mean of H. Uncertainty about which side of the kink in $\underline{\theta}_{z^\star}$ the local alternative lies on therefore tilts the policy choice towards $z^\star$. This is why the policy rule averages excess regret over the posterior instead of plugging in the posterior mean. ◁

At the first stage, when the design is chosen, the policymaker knows only t_1 and anticipates that the policy will then be chosen by $\delta^*_{\infty,\sigma}$. Given $\mathsf{T}_1 = t_1$, the second-wave observation under design g has the predictive law

$$\pi^{\mathrm{pred}}_{\sigma,g}(\cdot \mid t_1) := N(I_2(g) V_{\sigma,1} t_1,\; I_2(g) + I_2(g) V_{\sigma,1} I_2(g))\,.$$

The *Gaussian design loss* of design g is the smallest posterior expectation of $\ell(z, H)$ after both waves, averaged over this predictive law:

$$C_\sigma(t_1, g) = \int \left[\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\sigma,+}(dh \mid t_1, t_2, g) \right] \pi^{\text{pred}}_{\sigma,g}(dt_2 \mid t_1).$$

Let $\iota^*_\sigma(t_1)$ be the smallest index $j \in \{1, \dots, J\}$ minimising $C_\sigma\big(t_1, g_j(t_1)\big)$. The optimal design rule is $\gamma^*_{\infty,\sigma}(t_1) = g_{\iota^*_\sigma(t_1)}(t_1)$, and we call $(\gamma^*_{\infty,\sigma}, \delta^*_{\infty,\sigma})$ the *optimal Gaussian decision procedure*.

The design loss values a design by how much its second wave is expected to improve the policy choice. Designs differ in how much they are expected to move the posterior in the directions that matter for comparing the policies in $\mathcal{Z}_0$.

Running example. Return to the SNAP comparison between z_b and $z^\star$. Applying $\min\{a, b\} = (a + b - |a - b|)/2$ to the posterior expectations of $\ell(z_b, H)$ and $\ell(z^\star, H)$ and averaging over the predictive law gives

$$\begin{aligned} C_\sigma(t_1, g) = \frac{1}{2} \int \{\ell(z_b, h) + \ell(z^\star, h)\}\, \pi_{\sigma,1}(dh \mid t_1) \\ - \frac{1}{2} \int |\Delta_\sigma(t_1, t_2, g)|\, \pi^{\text{pred}}_{\sigma,g}(dt_2 \mid t_1). \end{aligned}$$

The first term does not depend on the design, because averaging a posterior expectation over the predictive law returns the corresponding pilot expectation. The optimal design therefore maximises the predictive mean of $|\Delta_\sigma|$, the expected size of the difference between the posterior expectations of $\ell(z_b, H)$ and $\ell(z^\star, H)$ after the second wave. A good design is thus one whose second wave is expected to show clearly which of z_b and $z^\star$ is better.

The design affects C_σ only through $\Lambda_{\text{SNAP}}(z_b) V_{\sigma,+}(g) \Lambda_{\text{SNAP}}(z_b)^\top$, the posterior covariance of $\Lambda_{\text{SNAP}}(z_b) H$ after both waves. To see this, note that Δ_σ in (17) depends only on m_1, m_2, and s_2, and s_2^2 is a diagonal entry of this matrix. Before the second wave is observed, (m_1, m_2) is Gaussian with mean $\Lambda_{\text{SNAP}}(z_b) V_{\sigma,1} t_1$, which does not depend on g, and covariance matrix

$$\Lambda_{\text{SNAP}}(z_b) V_{\sigma,1} \Lambda_{\text{SNAP}}(z_b)^\top - \Lambda_{\text{SNAP}}(z_b) V_{\sigma,+}(g) \Lambda_{\text{SNAP}}(z_b)^\top,$$

the posterior covariance of $\Lambda_{\text{SNAP}}(z_b) H$ after the pilot minus that after both waves. A design that leaves less posterior uncertainty about $\Lambda_{\text{SNAP}}(z_b) H$ therefore spreads the posterior means more widely and tends to make $|\Delta_\sigma|$ larger. Because $V_{\sigma,+}(g)$ is block diagonal across arms by (13) and $\Lambda_{\text{SNAP}}(z_b)$ involves

only arms 2 and z_b, second-wave observations allocated to the remaining arm do not lower the design loss. Arm 2 is valuable to the design even though it is not a terminal policy, because its moments determine both bounds $\underline{\theta}_{z^\star}$ and $\overline{\theta}_{z^\star}$ on the payoff of proactive enrolment. Conversely, arm $1 - z_b$ has no value even though $1 - z_b$ is a terminal policy, because at this reference point the choice is between z_b and $z^\star$, and their comparison does not depend on its moments. Subsection 4.3 generalises this reduction. ◁

The next theorem shows that the optimal Gaussian decision procedure solves the sequential Gaussian decision problem and that its value is an asymptotic lower bound on the local-average excess regret of every admissible sequence. The bound can be attained under the following condition on the optimal Gaussian decision procedure.

Definition 10 (No first-order ties)**.** Fix $\sigma > 0$, and let $\Pr_\sigma$ be as defined above, under the optimal Gaussian decision procedure. We say that there are *no first-order ties in the policy choice* at σ if, $\Pr_\sigma$-almost surely, the minimiser of $z \mapsto \int \ell(z, h)\, \pi_{\sigma,+}(dh \mid \mathsf{T}_1, \mathsf{T}_{2,G}, G)$ over $\mathcal{Z}_0$ is unique. We say that there are *no first-order ties in the design choice* at σ if, $\Pr_\sigma$-almost surely, the minimiser of $g \mapsto C_\sigma(\mathsf{T}_1, g)$ over $\{g_1(\mathsf{T}_1), \ldots, g_J(\mathsf{T}_1)\}$ is unique. We say that there are *no first-order ties* at σ if both hold.

The condition extends the no-first-order-ties condition in CMS (2026, Section 4.2) to the two choices in the sequential problem. No first-order ties in the policy choice is the same condition as in CMS (2026), applied after both observations. As in CMS, policies may tie in maximum regret at μ_0, which is why $\mathcal{Z}_0$ can contain several policies, but they must be separated at first order. That is, almost surely one of them has a strictly smaller posterior expectation of $\dot{R}_z(H)$ than the others. Ties in $\dot{R}_z(h)$ at a particular value of h are allowed. In the SNAP running example, the two policies in $\mathcal{Z}_0 = \{z_b, z^\star\}$ tie only when $\Delta_\sigma = m_1 + s_2\phi(m_2/s_2) + m_2\Phi(m_2/s_2)$ in (17) is zero. Here Δ_σ is the posterior expectation of $\dot{R}_{z_b}(H) - \dot{R}_{z^\star}(H)$. Recall that m_1 and m_2 are the posterior means of the two coordinates of $\Lambda_{\mathrm{SNAP}}(z_b)H$, and s_2 is the posterior standard deviation of the second. Because Δ_σ is strictly increasing in m_1 and (m_1, m_2) has a continuous distribution, $\Delta_\sigma = 0$ has probability zero.

No-first-order ties in the design choice concerns the design chosen after the first observation. The optimal design may switch between candidates as T_1 varies, and the condition requires that the values of T_1 at which two candidates both attain $\min_j C_\sigma(\mathsf{T}_1, g_j(\mathsf{T}_1))$ have probability zero. It fails if two candidates $g \neq g'$ satisfy $C_\sigma(t_1, g) = C_\sigma(t_1, g')$ for every t_1. In the SNAP running example, this happens when $\Lambda_{\mathrm{SNAP}}(z_b)V_{\sigma,+}(g)\Lambda_{\mathrm{SNAP}}(z_b)^\top$, the posterior covariance of the two combinations in (11), is the same for g and g'. Choosing either of two such candidates gives the same Gaussian excess regret.

Theorem 2 (Gaussian backward-induction optimality)**.** *Suppose Assumptions 2, 3, and 4 hold, and fix $\sigma > 0$. Then:*

(i) The optimal Gaussian decision procedure attains the value of the sequential Gaussian decision problem, and

$$\mathfrak{R}^*_{\infty,\sigma}(P_0) = \mathfrak{R}_{\infty,\sigma}(\gamma^*_{\infty,\sigma}, \delta^*_{\infty,\sigma}; P_0) = \mathbb{E}_\sigma \left[\min_{1 \le j \le J} C_\sigma\big(\mathsf{T}_1, g_j(\mathsf{T}_1)\big) \right].$$

(ii) Every admissible sequence $(\gamma_n, \delta_n) \in \mathbb{D}(P_0)$ satisfies

$$\liminf_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) \ge \mathfrak{R}^*_{\infty,\sigma}(P_0).$$

(iii) If there are no first-order ties at σ, there exists an admissible sequence $(\gamma_{n,\sigma}, \delta_{n,\sigma}) \in \mathbb{D}(P_0)$ such that

$$\lim_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_{n,\sigma}, \delta_{n,\sigma}; P_0) = \mathfrak{R}^*_{\infty,\sigma}(P_0).$$

Part (i) says that the optimal Gaussian decision procedure derived by backward induction solves the sequential Gaussian decision problem. Parts (ii) and (iii) link this problem back to the two-wave problem. Part (ii) says that no admissible sequence of procedures does better in the limit, so the value of the sequential Gaussian decision problem is a lower bound. Part (ii) follows from Proposition 2. Along a suitable subsequence, any admissible sequence behaves in the limit like some Gaussian decision procedure, so its local-average excess regret converges to the Gaussian excess regret of that procedure. By part (i), no Gaussian decision procedure does better than the optimal one. Because the proposition holds along subsequences, the bound also covers sequences whose choices do not converge. Part (iii) says that the bound can be attained. When there are no first-order ties, the sequence in part (iii) is therefore asymptotically optimal in the sense of Definition 7, with optimal value $\mathfrak{R}^*_{\infty,\sigma}(P_0)$.

Before turning to how to implement asymptotically optimal procedures in Section 4, we discuss the role of σ, the scale of the weights on local alternatives.

3.4 Diffuse Gaussian decision procedure

The weights φ_σ keep the criterion finite, but the choice of σ is arbitrary. Following CMS (2026, Section 4.4), we let $\sigma \to \infty$, which makes the weights flat. In the Bayesian interpretation of the criterion, this yields a flat, improper prior on the local parameter. The limiting procedure does not depend on σ, and we later show that the finite-sample procedure of Section 4 makes the same choices as this procedure in large samples.

Write $I_+(g) := I_1 + I_2(g)$. As $\sigma \to \infty$, $V_{\sigma,1} \to I_1^{-1}$ and $V_{\sigma,+}(g) \to I_+(g)^{-1}$, so the posterior and predictive laws converge to

$$\begin{aligned}
\pi_{\infty,1}(\cdot \mid t_1) &= N\left(I_1^{-1} t_1, I_1^{-1}\right), \\
\pi_{\infty,+}(\cdot \mid t_1, t_2, g) &= N\left(I_+(g)^{-1}(t_1 + t_2), I_+(g)^{-1}\right), \\
\pi^{\mathrm{pred}}_{\infty,g}(\cdot \mid t_1) &= N\big(I_2(g) I_1^{-1} t_1,\ I_2(g) + I_2(g) I_1^{-1} I_2(g)\big).
\end{aligned}$$

Under the flat prior, the posterior covariance $I_+(g)^{-1}$ is the inverse of the total information $I_1 + I_2(g)$ of the two observations, and the posterior mean $I_+(g)^{-1}(t_1 + t_2)$ is their information-weighted combination. Under the predictive law, this mean is distributed as $N(I_1^{-1} t_1, I_1^{-1} - I_+(g)^{-1})$, whose covariance is the reduction in posterior covariance that the second wave achieves.

Flat weights also remove the reference point from the posterior. For finite σ, the posterior mean $V_{\sigma,1} t_1$ shrinks the local parameter toward zero, that is, the reduced form toward the unknown reference point μ_0. Under the flat prior, $I_1^{-1} t_1$ corresponds to $\sqrt{n}(\widehat{\mu}_1 - \mu_0)$ for an efficient pilot estimator $\widehat{\mu}_1$, so the posterior for the reduced form $\mu_0 + h/\sqrt{n}$ is centred at $\widehat{\mu}_1$ and μ_0 drops out. The same holds after the second wave. The diffuse procedure can therefore be approximated from data.

Applying the same backward-induction recursion gives the diffuse policy rule

$$\delta^*_\infty(t_1, t_2, g) = \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\infty,+}(dh \mid t_1, t_2, g)$$

and the diffuse Gaussian design loss

$$C_\infty(t_1, g) = \int \left[\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\infty,+}(dh \mid t_1, t_2, g) \right] \pi^{\mathrm{pred}}_{\infty,g}(dt_2 \mid t_1).$$

Let $\iota^*_\infty(t_1)$ be the smallest index $j \in \{1, \ldots, J\}$ minimising $C_\infty\big(t_1, g_j(t_1)\big)$. The corresponding design rule is $\gamma^*_\infty(t_1) = g_{\iota^*_\infty(t_1)}(t_1)$. We call the pair $(\gamma^*_\infty, \delta^*_\infty)$ the *diffuse Gaussian decision procedure*. We define no first-order ties in the policy choice and in the design choice at $\sigma = \infty$ as in Definition 10.[22] The diffuse Gaussian decision procedure is the limit of the optimal Gaussian decision procedures in the following sense.

Proposition 3 (Convergence to the diffuse Gaussian decision procedure)**.** *As $\sigma \to \infty$, the following hold.*

[22] That is, with C_∞, $\pi_{\infty,+}$, and the diffuse Gaussian decision procedure in place of C_σ, $\pi_{\sigma,+}$, and the optimal Gaussian decision procedure, and with "$\Pr_\sigma$-almost surely" replaced by "almost everywhere with respect to $\int \Pr_h\, dh$", which uses Lebesgue measure on h in place of φ_σ.

(i) For every $t_1 \in \mathbb{R}^{d_\mu}$,

$$\max_{1 \le j \le J} \left| C_\sigma\big(t_1, g_j(t_1)\big) - C_\infty\big(t_1, g_j(t_1)\big) \right| \longrightarrow 0.$$

If there are no first-order ties in the design choice at $\sigma = \infty$*, then, almost surely,* $\gamma^*_{\infty,\sigma}(\mathsf{T}_1) = \gamma^*_\infty(\mathsf{T}_1)$ *for all sufficiently large* σ*.*

(ii) If there are no first-order ties in the policy choice at $\sigma = \infty$*, then, almost surely,* $\delta^*_{\infty,\sigma}(\mathsf{T}_1, \mathsf{T}_{2,G}, G) = \delta^*_\infty(\mathsf{T}_1, \mathsf{T}_{2,G}, G)$ *for all sufficiently large* σ*, where* $G = \gamma^*_\infty(\mathsf{T}_1)$*.*

Together, the two parts say that the optimal Gaussian decision procedure makes the same choices as the diffuse Gaussian decision procedure once σ is large. By part (i), it almost surely selects the same design for all sufficiently large σ, so the two procedures see the same second observation $\mathsf{T}_{2,G}$. By part (ii), it then also selects the same terminal policy. We can therefore work with the diffuse Gaussian decision procedure directly, without choosing a particular σ.

4 Implementation and Dimension Reduction

This section constructs a feasible design rule γ_n and policy rule δ_n that make the same choices as the diffuse Gaussian decision procedure of Subsection 3.4 in large samples, so that they lead to asymptotically optimal decisions. The construction is simple. The diffuse Gaussian decision procedure chooses the policy using the posterior of the local parameter after both waves, and it ranks designs by averaging over the distribution of this posterior before the second wave is observed. In finite samples, we replace these Gaussian posteriors by Gaussian approximations built from an efficient estimator of the reduced form and its estimated covariance matrix. Subsection 4.1 defines the resulting quasi-Bayesian procedure and explains how to compute it. Subsection 4.2 gives conditions under which it makes the same choices as the diffuse Gaussian decision procedure in large samples. Computing the design loss of this procedure is challenging in practice. Subsection 4.3 therefore introduces regret-directed dimension reduction (RDDR), which makes the choice of design computationally tractable and leads to decisions that are asymptotically equivalent to those based on the full design loss.

4.1 A quasi-Bayesian decision procedure

The procedure has four inputs. Let $\widehat{\mu}_1$ be an efficient estimator of μ computed from the pilot, let $\widehat{\Sigma}_1$ be a positive-definite, consistent estimator of the asymptotic covariance matrix of $\sqrt{n}(\widehat{\mu}_1 - \mu)$, and write $\widehat{I}_1 := \widehat{\Sigma}_1^{-1}$. For each candidate design g, let $\widehat{I}_2(g; S_1)$ be a positive

semidefinite estimate of the second-wave information $I_2(g)$ computed from the pilot. After the second wave under design g, let $\widehat{\mu}_{2,g}$ be the estimate of μ computed from the second-wave sample.

Following CMS and Doksum and Lo (1990); Kim (2002); Müller (2013), we treat the Gaussian approximation to the sampling distribution of an efficient estimator as a likelihood and combine it with a flat prior on $\mathbb{R}^{d_\mu}$. After the pilot, this gives the quasi-posterior $N(\widehat{\mu}_1, \widehat{\Sigma}_1/n)$.[23] The second-wave estimate is treated in the same way, with its sampling distribution approximated by $N(\mu, \widehat{I}_2(g; S_1)^{-1}/n)$.[24] Updating the pilot quasi-posterior with this approximation gives the quasi-posterior after the second wave,

$$N\left(\widehat{\mu}_{+,g}, \widehat{\Sigma}_{+,g}/n\right), \qquad \widehat{\Sigma}_{+,g} := \{\widehat{I}_1 + \widehat{I}_2(g; S_1)\}^{-1},$$

whose mean

$$\widehat{\mu}_{+,g} := \widehat{\Sigma}_{+,g}\{\widehat{I}_1\widehat{\mu}_1 + \widehat{I}_2(g; S_1)\widehat{\mu}_{2,g}\}$$

is the information-weighted average of the two estimates. The policy rule chooses the policy with the smallest expected excess regret under this quasi-posterior,[25]

$$\delta^{\Pi}_{n,g}(S_1, \widehat{\mu}_{2,g}) := \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} \int \{R(z, \mu) - R^*(\mu)\}\, N\left(\widehat{\mu}_{+,g}, \widehat{\Sigma}_{+,g}/n\right)(d\mu).$$

To choose the design, each candidate g is evaluated before its second wave is observed. For each g, the covariance $\widehat{\Sigma}_{+,g}/n$ of the quasi-posterior after the second wave is already known, and only its mean is uncertain. The diffuse Gaussian design loss C_∞ of Subsection 3.4 averages over the distribution of the corresponding posterior mean, $N(I_1^{-1}t_1, I_1^{-1} - I_+(g)^{-1})$. We replace this distribution by $N(\widehat{\mu}_1, (\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})/n)$, the posterior $\pi_{\infty,+}$ by $N(\overline{\mu}, \widehat{\Sigma}_{+,g}/n)$, where $\overline{\mu}$ is the posterior mean, and $\ell(z, h)$ by $R(z, \mu) - R^*(\mu)$. This gives the *full design loss* of candidate g,

$$C_n(S_1, g) := \int \left[\min_{z \in \mathcal{Z}^T} \int \{R(z, \mu) - R^*(\mu)\}\, N\left(\overline{\mu}, \widehat{\Sigma}_{+,g}/n\right)(d\mu)\right] N\left(\widehat{\mu}_1, (\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})/n\right)(d\overline{\mu}). \tag{18}$$

[23] A construction based on the bootstrap distribution of the pilot estimator leads to the same asymptotic results (Appendix S4.2).

[24] If $\widehat{I}_2(g; S_1)$ is singular, this approximation is restricted to the range of $\widehat{I}_2(g; S_1)$, with $\widehat{I}_2(g; S_1)^{-1}$ replaced by its Moore–Penrose inverse. The formulas below are unchanged.

[25] The quasi-posterior laws may assign probability to values outside $\mathcal{M}$, where $R(z, \cdot)$ is not originally defined. We therefore extend it to $\mathbb{R}^{d_\mu}$ for these calculations, keeping its original values on $\mathcal{M}$ and its continuous, piecewise-affine form with finitely many pieces. These extensions are globally Lipschitz and have at most linear growth.

It averages the smallest expected excess regret among the policies over an approximation to the distribution of the quasi-posterior after the second wave. Let $\iota_n^C(s_1)$ be the smallest index $j \in \{1, \ldots, J\}$ minimising $C_n\big(s_1, g_{j,n}(s_1)\big)$, and define $\gamma_n^C(s_1) := g_{\iota_n^C(s_1),n}(s_1)$. The corresponding policy rule is $\delta_n^C(s_1, s_2) := \delta_{n,G}^{\Pi}(s_1, \widehat{\mu}_{2,G})$, where $G = \gamma_n^C(s_1)$ and $\widehat{\mu}_{2,G}$ is computed from s_2. Together, (γ_n^C, δ_n^C) is our finite-sample decision procedure.

For each candidate g, the full design loss can be computed by simulation:

(1) Draw $\overline{\mu}^{(1)}, \ldots, \overline{\mu}^{(B)}$ from $N(\widehat{\mu}_1, (\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})/n)$.

(2) For each b, draw $\mu^{(b,k)}$, $k = 1, \ldots, K$, from $N(\overline{\mu}^{(b)}, \widehat{\Sigma}_{+,g}/n)$, and approximate each policy's expected excess regret by $K^{-1} \sum_{k=1}^{K} \{R(z, \mu^{(b,k)}) - R^*(\mu^{(b,k)})\}$.

(3) Take the minimum over policies for each b, and average over b.

Each evaluation of $R(z, \mu)$ solves the linear programmes in (6). The term R^* does not affect either choice, because its average is the same under every candidate design, and it can be omitted in computation. Both $\overline{\mu}^{(b)}$ and $\mu^{(b,k)}$ are d_μ-dimensional, and d_μ grows with the numbers of strata and arms. Approximating the two nested integrals accurately may then require many draws, which can be computationally too expensive in practice. Subsection 4.3 introduces a method to address this problem.

Running example. Continue the one-stratum specialisation. The estimates $\widehat{\mu}_1$ and $\widehat{\mu}_{2,g}$ stack the sample means of $(D, YD, Y(1-D))$ in the three arms, in the pilot and in the second wave under g. Let n_{1z} be the number of pilot observations in arm z, and let $\widehat{\Sigma}_z$ be the sample covariance matrix of $(D, YD, Y(1-D))$ among them. The estimated information matrices replace Σ_z, p_z^O and $\rho(g)$ in (13) by $\widehat{\Sigma}_z$, n_{1z}/n and $n_2(g)/n$,

$$\widehat{I}_1 = \text{blockdiag}_{z \in \{0,1,2\}} \left(\frac{n_{1z}}{n} \widehat{\Sigma}_z^{-1} \right),$$
$$\widehat{I}_2(g; S_1) = \text{blockdiag}_{z \in \{0,1,2\}} \left(\frac{n_2(g)\pi_z(g)}{n} \widehat{\Sigma}_z^{-1} \right).$$

Hence

$$\widehat{\Sigma}_{+,g} = \text{blockdiag}_{z \in \{0,1,2\}} \left(\frac{n \widehat{\Sigma}_z}{n_{1z} + n_2(g)\pi_z(g)} \right),$$

and $\widehat{\mu}_{+,g}$ averages the pilot and second-wave sample means of each arm with weights n_{1z} and $n_2(g)\pi_z(g)$. Together with $\widehat{\mu}_1$ and $\widehat{\Sigma}_1 = \widehat{I}_1^{-1}$, these give the Gaussian laws in $\delta_{n,g}^{\Pi}$ and (18). With the SNAP regret expressions (8)–(10), we can then compute $\delta_{n,g}^{\Pi}$ and $C_n(S_1, g)$. ◁

4.2 Asymptotic optimality of the quasi-Bayesian procedure

We now show that the quasi-Bayesian procedure (γ_n^C, δ_n^C) makes the same choices as the diffuse Gaussian decision procedure $(\gamma_\infty^*, \delta_\infty^*)$ in large samples. Its quasi-posteriors have the same form as the diffuse posteriors of Subsection 3.4. The estimates $\widehat{\mu}_1$ and $\widehat{\mu}_{2,g}$ take the role of the Gaussian observations T_1 and $\mathsf{T}_{2,g}$, and $\widehat{I}_1$ and $\widehat{I}_2(g; S_1)$ take the role of I_1 and $I_2(g)$. The two procedures therefore make the same choices in large samples if the estimates are efficient and the information matrices are estimated consistently. The next assumption states these requirements.

Assumption 5.

(i) Under $Q_{n,0}^O$, $\sqrt{n}(\widehat{\mu}_1 - \mu_0) = I_1^{-1}\mathsf{T}_{1,n} + o_p(1)$, $\left\|\widehat{I}_1 - I_1\right\| = o_p(1)$, *and* $\max_{1\le j\le J}\left\|\widehat{I}_2(g_{j,n}(S_1); S_1) - I_2(g_{j,n}(S_1))\right\| = o_p(1)$.

(ii) Under $Q_{n,0}^{\gamma_n^C}$, *the selected design* $G_n = \gamma_n^C(S_1)$ *satisfies* $\widehat{I}_2(G_n; S_1)\sqrt{n}(\widehat{\mu}_{2,G_n} - \mu_0) = \mathsf{T}_{2,n}(G_n) + o_p(1)$.

Part (i) requires the pilot estimator to be a regular efficient estimator, and the information matrices of the pilot and of every candidate design to be estimated consistently. Part (ii) requires the second-wave estimate under the design selected from the pilot to be efficient in the same sense. Designs are ranked using the pilot alone, so the design choice requires only part (i). Part (ii) is needed for the policy choice.

Theorem 3 (Finite-sample implementation of the diffuse Gaussian decision procedure)**.** *Consider the quasi-Bayesian procedure above, and suppose Assumptions 2–4 and Assumption 5*(i) *hold.*

(i) Under $Q_{n,0}^O$,

$$\max_{1\le j,k\le J}\Big|\sqrt{n}\left[C_n(S_1, g_{j,n}(S_1)) - C_n(S_1, g_{k,n}(S_1))\right] - \left[C_\infty(\mathsf{T}_{1,n}, g_{j,n}(S_1)) - C_\infty(\mathsf{T}_{1,n}, g_{k,n}(S_1))\right]\Big| \xrightarrow{p} 0.$$

(ii) Suppose additionally that Assumption 5(ii) *holds. Fix* h, *generate the observations and choices at* h *by the diffuse Gaussian decision procedure, and write* $G_\infty := \gamma_\infty^*(\mathsf{T}_1)$. *If there are no first-order ties at* $\sigma = \infty$, *then, under* $Q_{n,h}^{\gamma_n^C}$,

$$\left(G_n, \delta_n^C(S_1, S_2(G_n))\right) \Rightarrow \left(G_\infty, \delta_\infty^*(\mathsf{T}_1, \mathsf{T}_{2,G_\infty}, G_\infty)\right).$$

Part (i) says that, after scaling by $\sqrt{n}$, the differences between the full design losses of the candidates behave asymptotically like the differences between their diffuse Gaussian design losses. Minimising C_n therefore attains the minimum diffuse Gaussian design loss up to $o_p(1)$. The same holds for any pilot-measurable rule whose design loss is within $o_p(n^{-1/2})$ of the minimum of C_n, so the design losses need only be computed accurately relative to the scale $n^{-1/2}$ on which candidates differ. Part (ii) says that the design and the policy chosen by the quasi-Bayesian procedure converge jointly to those chosen by the diffuse Gaussian decision procedure. The no first-order ties condition makes these choices stable to small perturbations of the design losses. Together with Proposition 3, this shows that, in large samples, the quasi-Bayesian procedure behaves like the limit of the optimal procedures as the weights on local alternatives become flat.

The optimality results of Section 3 compare procedures within the class $\mathbb{D}(P_0)$ of Definition 6, which rules out choosing a policy outside $\mathcal{Z}_0$ too often. The quasi-Bayesian procedure should therefore also belong to this class. In the SNAP running example, the procedure uses the stratum–arm sample means of $(D, YD, Y(1-D))$ and their sample covariances, and these sample means are efficient. Proposition 7 in Appendix S4.3 shows that Assumption 5 then holds and that the procedure belongs to $\mathbb{D}(P_0)$.

4.3 Regret-directed dimension reduction (RDDR)

The two nested integrations in C_n are over all $d_\mu = 3|\mathcal{X}||\mathcal{Z}^E|$ coordinates of the reduced form, that is, 9 coordinates with three arms and no covariates and 18 with a binary covariate. The SNAP running example suggests why far fewer dimensions are often needed. At the reference point μ_0 of the running example, $\ell(z_b, \cdot)$ and $\ell(z^\star, \cdot)$ depend on the local parameter only through the two combinations $\Lambda_{\mathrm{SNAP}}(z_b)h$, so the design loss depends on the design only through the posterior covariance of these two combinations (Subsection 3.3). It could therefore be computed with two-dimensional integrals instead of nine-dimensional ones. RDDR generalises this idea and finds the relevant combinations in general.

We first define the relevant directions when the reference point μ_0 is known. By Proposition 1, the excess-regret functions $R(z, \cdot) - R^*(\cdot)$, $z \in \mathcal{Z}^T$, are continuous and piecewise affine on $\mathcal{M}$. Let $\mathscr{A}$ be a finite polyhedral partition of $\mathcal{M}$ into closed, full-dimensional regions $\mathcal{A} \in \mathscr{A}$ with disjoint interiors, such that every excess-regret function is affine on each region. Each excess-regret function therefore has a single gradient on each region. The gradients of the policies in $\mathcal{Z}_0$ on the regions that contain μ_0 span a subspace, which we define next.

Definition 11 (Local regret-directed subspace)**.** The local regret-directed subspace at P_0 is

defined by

$$\mathcal{S}_\ell(P_0) := \operatorname{span}\left\{ \nabla_\mu \big[R(z,\mu) - R^*(\mu)\big]\big|_{\mu\in\operatorname{int}(\mathcal{A})} : z \in \mathcal{Z}_0,\ \mathcal{A} \in \mathscr{A},\ \mu_0 \in \mathcal{A} \right\}.$$

Let $r_\ell := \dim\big(\mathcal{S}_\ell(P_0)\big)$, and let $\Psi \in \mathbb{R}^{r_\ell \times d_\mu}$ be any matrix with orthonormal rows spanning $\mathcal{S}_\ell(P_0)$.

For every $z \in \mathcal{Z}_0$, $\ell(z,h)$ is built from these gradients, so it depends on h only through Ψh. Changes in h orthogonal to $\mathcal{S}_\ell(P_0)$ leave ℓ unchanged. In the SNAP running example, $r_\ell = 2$ at the reference point, whereas $d_\mu = 9$. The next theorem treats the subspace as known and shows that the Gaussian design losses can then be computed in r_ℓ dimensions.

Theorem 4 (Regret-directed dimension reduction of Gaussian design losses)**.** *For every t_1, $C_\sigma(t_1, g)$ can be evaluated using r_ℓ-dimensional Gaussian laws and depends on g only through $\Sigma_{\ell,\sigma}(g) := \Psi V_{\sigma,+}(g)\Psi^\top$. Moreover, if $\Sigma_{\ell,\sigma}(g_2) - \Sigma_{\ell,\sigma}(g_1) \succeq 0$, then $C_\sigma(t_1, g_1) \le C_\sigma(t_1, g_2)$. The same conclusions hold for C_∞, with $\Sigma_{\ell,\infty}(g) := \Psi I_+(g)^{-1}\Psi^\top$.*

The matrix $\Sigma_{\ell,\sigma}(g)$ is the posterior covariance of ΨH after both waves. When $r_\ell = 1$, $\Sigma_{\ell,\sigma}(g)$ is the posterior variance of a single linear combination of H, so the design with the smallest variance minimises $C_\sigma(t_1, \cdot)$. Choosing the experiment then amounts to estimating this combination as precisely as possible, as the standard experimental design literature does for a chosen treatment effect. This case arises, for example, when exactly two policies minimise maximum regret at the reference point and the difference between their maximum regrets, in which shared kinks cancel, is differentiable there.

When $r_\ell > 1$, a design that leaves less posterior uncertainty about Ψh, in the positive semidefinite order, still has a weakly smaller design loss. A design g_1 can also leave less posterior uncertainty than g_2 about one combination in Ψh and more about another. Then $\Sigma_{\ell,\sigma}(g_1)$ and $\Sigma_{\ell,\sigma}(g_2)$ cannot be ranked in the positive semidefinite order, so $C_\sigma(t_1, g_1)$ and $C_\sigma(t_1, g_2)$ must be computed to tell which design is better. In the SNAP running example, two policies tie in maximum regret at the reference point, and the reference point lies on the kink of the lower bound $\underline{\theta}_{z^\star}$ on the payoff of proactive enrolment. The difference between the two maximum regrets has a different gradient on each side of this kink, so $r_\ell = 2$. If the two policies tied at a reference point away from this kink, the difference would have a single gradient there, and $r_\ell = 1$.

In practice, the subspace must be estimated, because it depends on the unknown reference point. RDDR replaces the reference point by a neighbourhood of the pilot estimate that is large relative to the estimation error, so that it contains the reference point with probability approaching one, but small enough so that it eventually excludes regions and policies that

are not relevant at the reference point. Distances are measured in estimated standard errors, using the pilot covariance estimate $\widehat{\Sigma}_1$. More specifically, we choose a threshold $\tau_n \to \infty$ with $\tau_n/\sqrt{n} \to 0$, and define

$$\widehat{\mathcal{S}}_{\ell,n} := \operatorname{span}\left\{ \nabla_\mu \big[R(z,\mu) - R^*(\mu)\big]\big|_{\mu \in \operatorname{int}(\mathcal{A})} : \begin{array}{l} z \in \mathcal{Z}^T,\ \mathcal{A} \in \mathscr{A},\ \text{and} \\ \text{there exists } \mu \in \mathcal{A} \text{ such that} \\ R(z,\mu) = R^*(\mu), \\ n(\mu - \widehat{\mu}_1)^\top \widehat{\Sigma}_1^{-1} (\mu - \widehat{\mu}_1) \le \tau_n^2 \end{array} \right\}.$$

In words, RDDR retains the excess-regret gradient of policy z on region $\mathcal{A}$ if z has the smallest maximum regret somewhere in $\mathcal{A}$ within τ_n estimated standard errors of the pilot estimate. Each check is a quadratic programme. It minimises the standardised distance to the pilot estimate over the points of $\mathcal{A}$ at which z minimises maximum regret, and these points satisfy linear constraints because $\mathcal{A}$ is polyhedral and excess regret is affine on it. Let $\widehat{r}_n := \dim(\widehat{\mathcal{S}}_{\ell,n})$, and let $\widehat{\Psi}_n$ be a matrix with orthonormal rows spanning $\widehat{\mathcal{S}}_{\ell,n}$, for example from a singular value decomposition of the retained gradients.

This construction parallels inference on directionally differentiable functions. To estimate the directional derivative of such a function at an unknown point, Fang and Santos (2019) and Hong and Li (2018) use a tuning parameter that tends to zero more slowly than $n^{-1/2}$, so that estimation error is negligible relative to it. Generalized moment selection (Andrews and Soares, 2010) selects nearly binding moment inequalities with a threshold of the same kind. The neighbourhood in RDDR plays the same role, with a radius that tends to zero in units of the reduced form and to infinity in estimated standard errors. The purpose differs. These papers estimate the derivative or the binding inequalities in order to conduct inference, whereas RDDR needs only the span of the relevant gradients, in order to reduce the dimension of the integrals in the design loss.

Given the estimated subspace, RDDR computes the same nested integrals as the full design loss, but with draws restricted to the subspace through the pilot estimate.

Definition 12 (RDDR design loss)**.** For $u \in \mathbb{R}^{\widehat{r}_n}$, define the scaled excess regret in the reduced coordinates centred at the pilot estimate by

$$\widehat{\ell}_n(z,u) := \sqrt{n}\left[R\left(z, \widehat{\mu}_1 + \widehat{\Psi}_n^\top u/\sqrt{n}\right) - R^*\left(\widehat{\mu}_1 + \widehat{\Psi}_n^\top u/\sqrt{n}\right)\right].$$

The RDDR design loss is

$$\widehat{C}_n(S_1, g) := \int \left[\min_{z \in \mathcal{Z}^T} \int \widehat{\ell}_n(z, u)\, N\left(\overline{u}, \widehat{\Psi}_n \widehat{\Sigma}_{+,g} \widehat{\Psi}_n^\top\right)(du) \right] N\left(0, \widehat{\Psi}_n(\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})\widehat{\Psi}_n^\top\right)(d\overline{u}).$$

The RDDR rule selects a candidate minimising this design loss, with ties broken by the smallest candidate index.

Here $\overline{u} = \sqrt{n}\,\widehat{\Psi}_n(\overline{\mu} - \widehat{\mu}_1)$ and $u = \sqrt{n}\,\widehat{\Psi}_n(\mu - \widehat{\mu}_1)$ are the posterior mean $\overline{\mu}$ and the draw μ in (18), projected onto the estimated subspace. The RDDR design loss therefore has the same nested structure as C_n, with both integrations in $\widehat{r}_n$ dimensions. The next proposition shows that RDDR recovers the regret-directed subspace with probability approaching one, and that the design it selects is as good as the one selected by the full design loss, up to an error that is negligible on the scale on which candidates differ.

Proposition 4 (Pilot recovery and first-order optimality of the RDDR rule)**.** *Suppose Assumptions 2– 4 and Assumption 5*(i) *hold, and use the regret-directed dimension reduction above with $\tau_n \to \infty$ and $\tau_n/\sqrt{n} \to 0$. Then, under the pilot law $Q_{n,0}^O$,*

$$\Pr\left\{\widehat{\mathcal{S}}_{\ell,n} = \mathcal{S}_\ell(P_0),\ \widehat{r}_n = r_\ell\right\} \longrightarrow 1.$$

Let $\widehat{\iota}_{\ell,n}$ be the smallest index $j \in \{1, \ldots, J\}$ minimising $\widehat{C}_n(S_1, g_{j,n}(S_1))$. Then

$$\sqrt{n}\left[C_n(S_1, g_{\widehat{\iota}_{\ell,n},n}(S_1)) - \min_{1 \le j \le J} C_n(S_1, g_{j,n}(S_1))\right] \xrightarrow{p} 0.$$

The second claim says that the full design loss of the candidate selected by RDDR exceeds the smallest full design loss by $o_p(n^{-1/2})$. This gap is negligible compared with the differences between candidates, which are of order $n^{-1/2}$ (Theorem 3(i)). Asymptotically, decisions based on RDDR are therefore equivalent to those based on the full design loss, and, as noted after Theorem 3, the RDDR rule also attains the minimum diffuse Gaussian design loss up to $o_p(1)$. In a given sample, a larger threshold τ_n retains more directions. It costs more computation but no extra design loss. In the extreme, RDDR keeps the excess-regret gradients of every policy on every region, regardless of the pilot estimate. We call this the *global reduction*. The resulting design loss then equals $\sqrt{n}\,C_n$ exactly and is computed in the span of these gradients. In Section 5, we use this global reduction to compute the full design loss exactly, over four combinations of μ instead of all nine coordinates, and compare it with the RDDR design losses for thresholds $\tau_n = a\sqrt{\log\log n}$ with $a = 0.5$, 1 and 2.

Running example. Return to $\ell(z_b, \cdot)$ and $\ell(z^\star, \cdot)$ in (12). Below the kink in $\underline{\theta}_{z^\star}$, where $[\Lambda_{\mathrm{SNAP}}(z_b)h]_2 \le 0$, they are the positive parts of $\pm[\Lambda_{\mathrm{SNAP}}(z_b)h]_1$. Above it, they are the positive parts of $\pm\big([\Lambda_{\mathrm{SNAP}}(z_b)h]_1 + [\Lambda_{\mathrm{SNAP}}(z_b)h]_2\big)$. By (11), these two linear functions are

$$\begin{aligned}[\Lambda_{\mathrm{SNAP}}(z_b)h]_1 &= 2(h_2^{Y1} - h_{z_b}^{Y1} - h_{z_b}^{Y0}) - h_2^D,\\ [\Lambda_{\mathrm{SNAP}}(z_b)h]_1 + [\Lambda_{\mathrm{SNAP}}(z_b)h]_2 &= 2(h_2^{Y1} - h_{z_b}^{Y1} - h_{z_b}^{Y0}) + h_2^{Y0}.\end{aligned}$$

Up to sign, the nonzero gradients of $\ell(z_b, \cdot)$ and $\ell(z^\star, \cdot)$ are therefore the coefficient vectors of these two functions. The two vectors are linearly independent, because h_2^D enters only the first function and h_2^{Y0} only the second. Excess regret is affine on each region of $\mathscr{A}$ and zero at μ_0 for $z \in \mathcal{Z}_0$, so these are the gradients in Definition 11. Hence

$$\mathcal{S}_\ell(P_0) = \mathrm{row}\big(\Lambda_{\mathrm{SNAP}}(z_b)\big), \qquad r_\ell = 2.$$

By Proposition 4, $\Pr\{\widehat{\mathcal{S}}_{\ell,n} = \mathcal{S}_\ell(P_0),\ \widehat{r}_n = 2\} \to 1$, and on this event the two integrations in $\widehat{C}_n$ are over $\mathbb{R}^2$ instead of $\mathbb{R}^9$.

Figure 1 shows the pilot neighbourhood in the coordinates $\Lambda_{\mathrm{SNAP}}(z_b)h$, in which the two gradients are $(1, 0)$ and $(1, 1)$. The tie boundary $\ell(z_b, h) = \ell(z^\star, h)$ is $[\Lambda_{\mathrm{SNAP}}(z_b)h]_1 = 0$ below the kink and $[\Lambda_{\mathrm{SNAP}}(z_b)h]_1 + [\Lambda_{\mathrm{SNAP}}(z_b)h]_2 = 0$ above it. A policy with a nonzero gradient on a region has the smallest maximum regret there only on this boundary. RDDR therefore retains the nonzero gradients from one side of the kink exactly when the neighbourhood meets the tie boundary on that side. This happens on neither side in panel (a), only below the kink in panel (b), and on both sides in panel (c), so $\widehat{r}_n = 0$, 1 and 2, respectively. If the neighbourhood also reaches the kink of $\overline{\theta}_{z^\star}$ or points where policy $1 - z_b$ has the smallest maximum regret, $\widehat{r}_n$ can exceed two. ◁

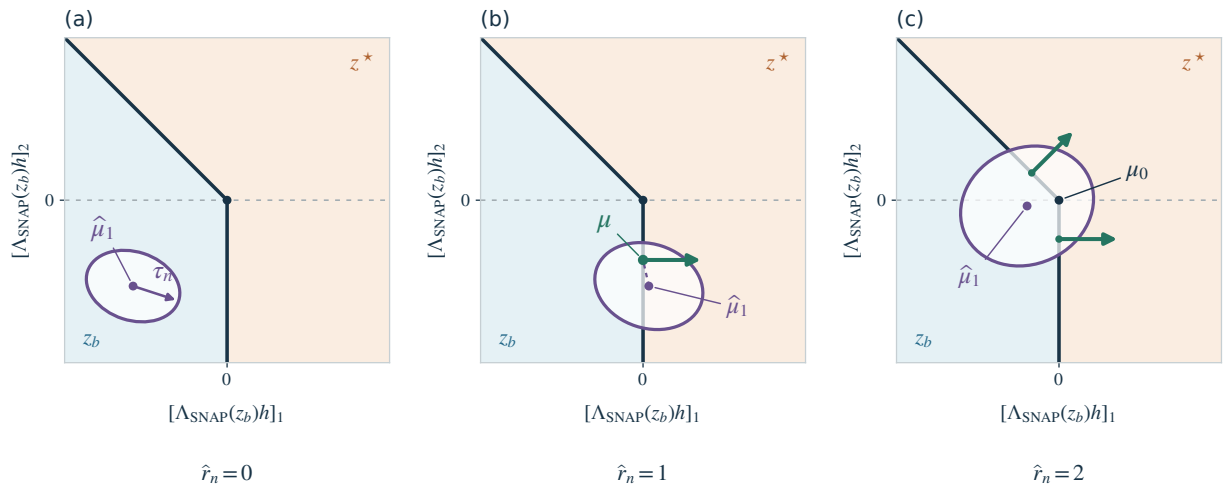


Figure 1: RDDR in the SNAP running example

Notes: In each panel, the purple ellipse is the image of the set defined by $n(\mu - \widehat{\mu}_1)^\top \widehat{\Sigma}_1^{-1}(\mu - \widehat{\mu}_1) \leq \tau_n^2$ in the coordinates $\Lambda_{\text{SNAP}}(z_b)h$, with $h = \sqrt{n}(\mu - \mu_0)$. Point labels denote images under this map, and τ_n is the radius in estimated standard-error units. The solid line separates the regions where z_b and $z^\star$ minimise maximum regret. The marked point μ in panel (b) satisfies $R(z_b, \mu) = R(z^\star, \mu) = R^*(\mu)$. The dashed line marks the kink in the lower bound $\underline{\theta}_{z^\star}$. Green arrows show representative excess-regret gradients in the displayed coordinates.

A further shortcut reduces the number of posterior integrations rather than their dimension. For each value $\overline{\mu}$ of the posterior mean in (18), it selects the policy that minimises maximum regret at $\overline{\mu}$, instead of the policy with the smallest expected excess regret under the posterior, and it can be combined with RDDR. Appendix S4.6 shows that this *policy plug-in* design loss can lose at first order in general, bounds how much it can lose in terms of differences in posterior covariance matrices across candidates, and finds that it is accurate in the SNAP calibration.

5 SNAP Experimental Design Numerical Exercises

This section applies the procedures of Section 4 to a calibration based on the SNAP outreach experiment of Finkelstein and Notowidigdo (2019). We measure how much choosing the design for the policy decision lowers expected excess regret relative to standard allocations, and how much computation RDDR saves. Subsection 5.1 describes the calibration and Subsection 5.2 the design rules. Subsections 5.3 and 5.4 compare the rules by expected excess regret and by the second-wave sample size each needs to match RDDR. Subsection 5.5 adds a binary covariate and measures the computation RDDR saves in this more complex problem.

5.1 Calibration

We simulate pilots and second waves from a law that matches published moments of the SNAP experiment, and we evaluate the chosen policies at this law. Finkelstein and Notowidigdo (2019) study roughly 30,000 Pennsylvania residents aged 60 and older who were not enrolled in SNAP. The three experimental arms $z = 0, 1, 2$ are the status quo, information outreach, and information plus application assistance. With $D(z)$ denoting SNAP enrolment within nine months, the published take-up probabilities are $\Pr\{D(0) = 1\} = 0.058$, $\Pr\{D(1) = 1\} = 0.105$, and $\Pr\{D(2) = 1\} = 0.176$. The outcome is food security, $Y(d) = \mathbf{1}\{\text{the household is food secure under enrolment state } d\}$. Ratcliffe and McKernan (2010) report $\Pr\{Y(0) = 1\} = 0.482$ and $\Pr\{Y(1) = 1\} = 0.644$. Kleinman (2026) reports a SNAP take-up rate of about 60 per cent among older Supplemental Security Income recipients under automatic enrolment. Using this participation level as an external calibration, we set $\Pr\{D(z^\star) = 1\} = 0.6$ in the data-generating law and use $\overline{p}^D_{z^\star} = 0.6$ as the upper bound on take-up in (7). Table 1 separates the published inputs from those we specify.

Table 1: Inputs to the SNAP calibration

Input	Value	Role in the exercise
Take-up probabilities by arm	$0.058, 0.105, 0.176$	Published estimates
Probabilities of being food secure	$0.482, 0.644$	External calibration
Proactive enrolment	$z^\star$, with take-up probability 0.6	External calibration
Implementation costs per person	$k_1 = 0.024$, $k_{z^\star} = 0.030$	Specified for this exercise
Type-specific outcome probabilities	See Table 2	Specified for this exercise

We impose the monotonicity restriction $D(0) \leq D(1) \leq D(2) \leq D(z^\star)$, as in the running example in Subsection 2.1. The four take-up probabilities then determine the shares of the five response types in (1). The shares of v_0 (never enrolled), v_1 (enrolled only under proactive enrolment), v_2 (enrolled with assistance), v_3 (enrolled with information), and v_4 (enrolled under the status quo) are 0.400, 0.424, 0.071, 0.047, and 0.058. Table 2 gives the probability of food security for each type with and without enrolment. These probabilities reproduce the two aggregate food-security probabilities above and satisfy $\Pr\{Y(1) = 1 \mid V = v\} \geq \Pr\{Y(0) = 1 \mid V = v\}$ for every type.

Table 2: Type-specific food-security probabilities

d	v_0: Never	v_1: Proactive	v_2: Assistance	v_3: Information	v_4: Baseline
0	0.933	0.195	0.069	0.379	0.058
1	0.935	0.545	0.081	0.379	0.267

Notes: Entries are $\Pr\{Y(d) = 1 \mid V = v\}$, rounded to three decimals. Calculations use the unrounded probabilities, which reproduce $\Pr\{Y(0) = 1\} = 0.482$ and $\Pr\{Y(1) = 1\} = 0.644$.

Using (3) and the bounds in (7), this calibration gives $\theta_1 = -0.023$ and $[\underline{\theta}_{z^\star}, \overline{\theta}_{z^\star}] = [-0.453, 0.339]$. The maximum regrets of 0, 1, and $z^\star$ are therefore 0.339, 0.362, and 0.453, so the status quo is the robust choice at the calibrated data generating process.

5.2 Candidate designs, design rules, and evaluation

We fix the pilot size at $n = 500$, with equal allocation across arms, and the second-wave size at $n_2 = 5{,}000$. An experiment $g = (\pi_0, \pi_1, \pi_2)$ allocates shares π_0, π_1, and π_2 of the second wave to the status quo ($z = 0$), information outreach ($z = 1$), and information plus application assistance ($z = 2$). The candidate list $\mathcal{G}_{13}$ consists of 13 fixed allocations. They are equal allocation $(1/3, 1/3, 1/3)$ and the six permutations each of $(0.1, 0.4, 0.5)$ and $(0.1, 0.3, 0.6)$.

We compare five design rules. Full and RDDR are the rules of Section 4, and the other three are benchmarks.

(1) *Full.* This rule minimises the full design loss C_n in (18) over $\mathcal{G}_{13}$, computed exactly with the global reduction of Subsection 4.3. By (7), maximum regret depends on μ only through four linearly independent combinations, namely θ_1, $\mu_2^{Y1} - \mu_0^{Y1} - \mu_0^{Y0}$, μ_2^D, and μ_2^{Y0}. The global reduction therefore integrates over these four combinations instead of all nine reduced form coordinates.

(2) *RDDR.* This rule minimises the RDDR design loss $\widehat{C}_n$ of Definition 12, with threshold $\tau_n = a\sqrt{\log\log n}$ for $a \in \{0.5, 1, 2\}$. With $n = 500$, these thresholds equal about 0.68, 1.35, and 2.70 estimated standard errors. If $\widehat{r}_n = 0$, all candidates have the same design loss to first order, and we let the rule selects equal allocation.

(3) LATE_{2b}. This rule adapts the design rule in Section 6.3 of Bugni and Gao (2023) to our three-arm experiment. Bugni and Gao consider a stratified experiment with two arms and imperfect compliance. Using a pilot experiment on the same population, they choose the share of units assigned to treatment in a later experiment. Their Theorem 6.3 gives the share that minimises the asymptotic variance of their estimator of the local average treatment effect, and they estimate this share from the pilot. With a single stratum, as here, this estimator is the Wald ratio of the two arms. In our three-arm setting, the rule compares arm 2 with a reference arm $b \in \{0, 1\}$ selected from the pilot. Write $\widehat{\mu}_z^D$, $\widehat{\mu}_z^{Y1}$, and $\widehat{\mu}_z^{Y0}$ for the arm-z coordinates of $\widehat{\mu}_1$. Set $b = 1$ if $\theta_1(\widehat{\mu}_1) > 0$ and $b = 0$ otherwise, and form the pilot Wald estimate

$$\widehat{\text{LATE}}_{2b} = \frac{(\widehat{\mu}_2^{Y1} + \widehat{\mu}_2^{Y0}) - (\widehat{\mu}_b^{Y1} + \widehat{\mu}_b^{Y0})}{\widehat{\mu}_2^D - \widehat{\mu}_b^D}.$$

Let $\mathrm{LATE}_{2b}(\mu)$ denote the corresponding ratio at μ, holding b fixed. After both waves, this ratio is estimated by $\mathrm{LATE}_{2b}(\widehat{\mu}_{+,g})$. Using $\widehat{\Sigma}_{+,g}(s_1)/n$ from Subsection 4.1 as the covariance of $\widehat{\mu}_{+,g}$, the delta method gives the variance of this estimate as

$$n^{-1}\nabla_\mu \mathrm{LATE}_{2b}(\widehat{\mu}_1)^\top \widehat{\Sigma}_{+,g}(s_1)\nabla_\mu \mathrm{LATE}_{2b}(\widehat{\mu}_1).$$

The rule selects the candidate in $\mathcal{G}_{13}$ that minimises this variance.

(4) *Equal allocation.* This rule always selects $g = (1/3, 1/3, 1/3)$.

(5) *SNAP original.* This rule always selects $(0.4, 0.2, 0.4)$, which approximates the arm shares in the main analysis sample of Finkelstein and Notowidigdo (2019, Table II). The original study was designed with a power calculation pre-registered in the AEA RCT Registry (Finkelstein and Notowidigdo, 2015). It randomly assigned 31,888 individuals in equal proportions to the three arms. Under this design, the minimum detectable effect on enrolment is 2 percentage points or less for a comparison between any two arms, at 80 per cent power and a 5 per cent significance level. The main analysis keeps half of the information arm, which gives the shares above.[26]

We evaluate each design rule by its expected excess regret at the calibrated law P. All five rules choose the terminal policy with the quasi-Bayesian rule $\delta^{\Pi}_{n,g}$ of Subsection 4.1, so that, with the sample sizes also fixed, differences in regret come only from how they allocate the second wave across arms. For a design g, let $F_{n,P,g}$ denote the Gaussian approximation to the sampling distribution of the second-wave estimate $\widehat{\mu}_{2,g}$, with mean and covariance computed from P and the second-wave sample sizes in each arm. We draw $M = 1{,}000$ independent pilots $S_1^{(1)}, \ldots, S_1^{(M)}$ from the calibrated law. For each pilot, the rule selects a design, and we average the excess regret of the policy chosen by $\delta^{\Pi}_{n,g}$ over $F_{n,P,g}$ at that design. Averaging over pilots gives

$$\widehat{\mathcal{R}}_n(P;\gamma_n) := \frac{1}{M}\sum_{m=1}^{M} \int \Big[R\Big(\delta^{\Pi}_{n,\gamma_n(S_1^{(m)})}(S_1^{(m)}, e), \mu(P)\Big) - R^*(\mu(P))\Big] F_{n,P,\gamma_n(S_1^{(m)})}(de).$$

We call this quantity *evaluation regret*.

[26] Table II in Finkelstein and Notowidigdo (2019) reports 10,630, 5,314, and 10,629 observations after restricting the information-only arm to its standard and marketing subgroups; see Section IV.C of the original study. The other half of the information arm received the standard letter without the reminder postcard or a letter that described the benefit amount differently.

5.3 Experiment selection, regret, and computation time

Table 3 reports, for each design rule, the average selected allocation, evaluation regret, and computation time. Design losses are computed by randomised quasi-Monte Carlo (RQMC) integration (see Appendix S5.1 for the number of nodes used for each integral).

Table 3: Experiment selection, regret, and computation time

Design rule	Mean (π_0, π_1, π_2)	Mean integration dimension	$10^3\times$ evaluation regret	Gain over equal allocation (per cent)	Computation time (minutes)
Full (C_n)	$(0.411, 0.362, 0.227)$	4.000	1.637	8.299	26.60
RDDR ($\widehat{C}_n$) $a = 0.5$	$(0.364, 0.371, 0.265)$	0.747	1.712	4.116	6.43
RDDR ($\widehat{C}_n$) $a = 1$	$(0.377, 0.379, 0.245)$	1.543	1.651	7.526	8.41
RDDR ($\widehat{C}_n$) $a = 2$	$(0.404, 0.370, 0.226)$	3.102	1.637	8.314	18.21
LATE_{2b}	$(0.340, 0.221, 0.440)$	—	3.198	−79.129	—
Equal allocation	$(0.333, 0.333, 0.333)$	—	1.785	0.000	—
SNAP original	$(0.400, 0.200, 0.400)$	—	2.245	−25.748	—

Notes: Gain is the percentage reduction in evaluation regret relative to equal allocation. For RDDR, integration dimension is the averaged estimated dimension $\widehat{r_n}$ over 1,000 pilots. Computation times are for computing the design losses of all 13 candidates for all 1,000 pilots on an Apple M3 Pro chip.

The Full rule lowers evaluation regret by 8.3 per cent relative to equal allocation. On average, it shifts the second wave towards arms 0 and 1, whose moments determine θ_1 and hence the choice between the status quo and information outreach. The RDDR rules with $a = 1$ and $a = 2$ lower evaluation regret by 7.5 and 8.3 per cent, with mean integration dimensions 1.54 and 3.10. With $a = 0.5$, the mean integration dimension falls to 0.75, but the gain falls to 4.1 per cent. The integration dimension is zero in 45.6, 16.6, and 0.5 per cent of pilots for $a = 0.5, 1, 2$, respectively, so a small threshold often discards the relevant directions and lead to poor performance. RDDR with $a = 1$ retains about 90 per cent of the gain of the Full rule at about one-third of its computation time, and $a = 2$ retains all of it at about two-thirds of its computation time.

Equal allocation performs well in this calibration, because it leaves no arm with a small sample and so preserves precision across the multiple policy comparisons relevant to the terminal policy decision. The two worst rules, SNAP original and the LATE_{2b} rule, place the least weight on arm 1. Relative to SNAP original, the Full rule lowers evaluation regret by 27.1 per cent, and RDDR with $a = 1$ and $a = 2$ lowers it by 26.5 and 27.1 per cent. The LATE_{2b} rule shifts the second wave towards arm 2, and its evaluation regret is 79 per cent higher than under equal allocation. Its criterion, the variance of a single Wald ratio, does not account for all the policy comparisons entering the terminal decision. It also does not depend

on the implementation costs, although these costs determine which policy comparisons are close.

5.4 Second-wave sample size

To make the differences in regret easier to interpret, we convert them into second-wave observations. That is, we ask how many second-wave observations a rule needs to match RDDR with $a = 1$ at $n_2 = 5{,}000$. Figure 2 repeats the calculation for $n_2 = 5{,}000, 5{,}500, \ldots, 7{,}500$, using the same pilots and candidate list at every sample size. For n_2 between 5,000 and 7,500, the Full rule lowers evaluation regret relative to SNAP original by 27.1 to 34.4 per cent, and the RDDR rule with $a = 1$ by 26.5 to 33.8 per cent.

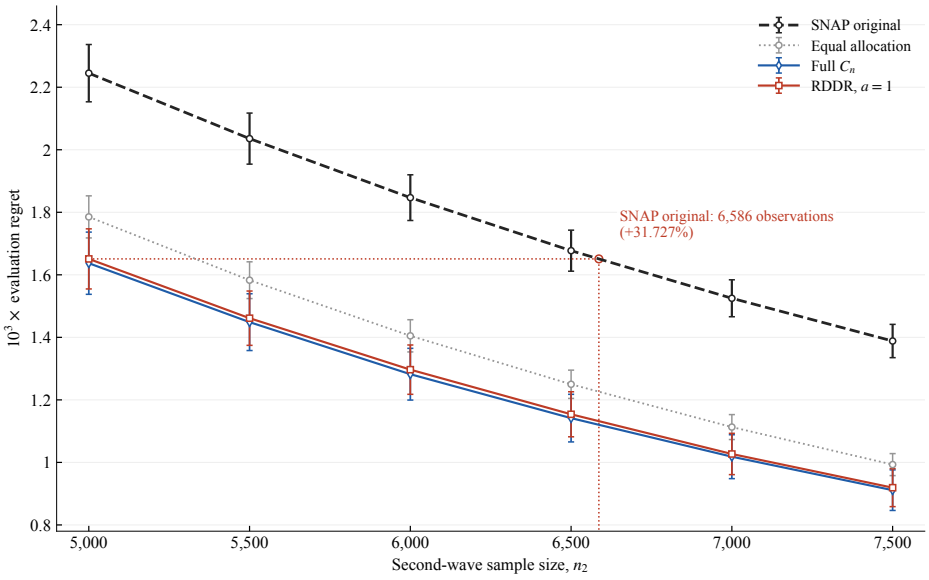


Figure 2: Evaluation regret and sample-size equivalence to SNAP original

Notes: Bars are the mean plus or minus 1.96 Monte Carlo standard errors across pilots. The dotted lines linearly interpolate the SNAP original curve to match the RDDR rule with $a = 1$ at $n_2 = 5{,}000$.

SNAP original needs about 6,586 second-wave observations, or 31.7 per cent more, to match the evaluation regret of RDDR with $a = 1$ at $n_2 = 5{,}000$.

Figure 3 adds the RDDR rules with $a = 0.5$ and $a = 2$ and the LATE_{2b} rule. For n_2 between 5,000 and 7,500, the RDDR rule with $a = 2$ lowers evaluation regret relative to SNAP original by 27.1 to 34.4 per cent, and the rule with $a = 0.5$, which retains fewer directions, by 23.8 to 30.7 per cent. The LATE_{2b} rule has higher evaluation regret than both SNAP original and equal allocation at every n_2 in this range.

Figure 4 extends the LATE_{2b} comparison to larger second-wave samples. The LATE_{2b} rule needs about 10,292 observations, or 105.8 per cent more, to match RDDR with $a = 1$

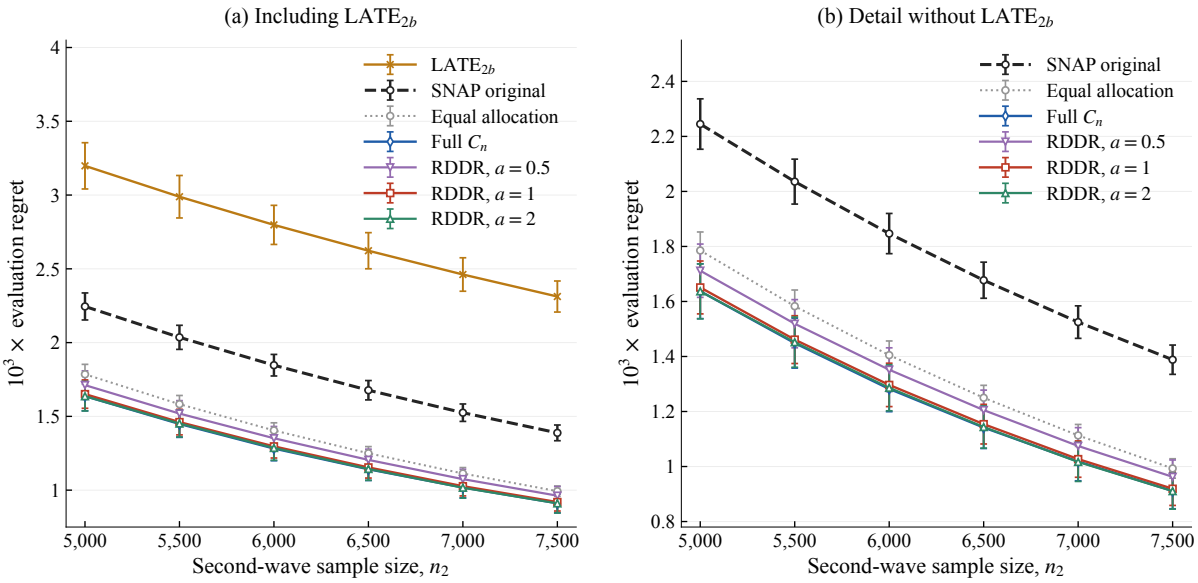


Figure 3: RDDR threshold sensitivity with SNAP original and the LATE$_{2b}$ rule

Notes: Panel (b) omits the LATE$_{2b}$ rule to show the remaining curves in greater detail. The three RDDR rules set $a \in \{0.5, 1, 2\}$. Bars are the mean plus or minus 1.96 Monte Carlo standard errors across pilots.

at 5,000 observations.

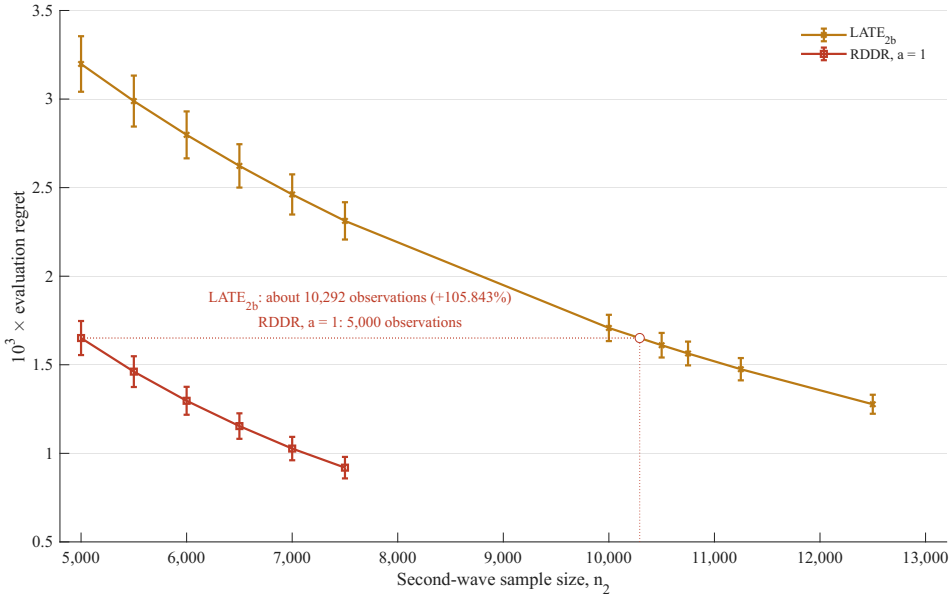


Figure 4: Second-wave sample size under the LATE$_{2b}$ and RDDR rules

Notes: The dotted lines match the evaluation regret of the LATE$_{2b}$ rule to the RDDR rule with $a = 1$ and 5,000 observations, interpolating linearly between 10,000 and 10,500 observations. Bars are the mean plus or minus 1.96 Monte Carlo standard errors across pilots. The RDDR curve is shown only over the computed range, 5,000–7,500.

5.5 Dimension reduction with a binary covariate

RDDR saves the most computation when the reduced form is large. To measure the saving in such a case, we add a binary covariate, which doubles the reduced form to 18 coordinates and lets the allocation differ across the two strata. Let $X = 1$ denote the 26 per cent of subjects with high predicted application barriers (see Appendix S5.3 for the complete calibration). The policy is still a single choice from $\{0, 1, z^\star\}$ for the whole population.

The pilot has $n = 500$ observations and uses equal allocation across arms within each stratum, and the follow-up experiment has $n_2 = 5{,}000$ observations. Within each stratum, the design chooses from equal allocation and the six permutations of $(0.2, 0.3, 0.5)$. Crossing the two lists gives 49 candidate experiments. Maximum regret depends on the 18 coordinates of the reduced form only through six combinations, and the Full rule integrates over 6 dimension using global reduction. Table 4 reports mean integration dimensions and computation times.

Table 4: Integration dimensions and design-loss computation times with a binary covariate

Design rule	Mean integration dimension	Estimated computation time 1,000 pilots
RDDR, $a = 0.5$	1.100	50.5 minutes
RDDR, $a = 1$	2.478	135.4 minutes
RDDR, $a = 2$	4.549	170.9 minutes
Full C_n	6	811.8 minutes

Notes: RDDR retains a pilot-dependent gradient span within the exact six-dimensional representation used by Full. Times are estimates based on RQMC integration (see Appendix S5.3 for the pilot counts, the numbers of nodes, and the timing extrapolation).

For 1,000 pilots, the Full rule takes about 13.5 hours and RDDR takes 0.84, 2.26, and 2.85 hours at $a = 0.5, 1, 2$. RDDR therefore saves substantial computation time.

Declaration of Generative AI Use

I used Claude and ChatGPT to assist with language editing, LaTeX formatting, programming, and checking mathematical arguments. I take full responsibility for the content of this paper.

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Appendix

The appendix is organised by the main-text section it supports. Sections S2–S5 correspond respectively to main-text Sections 2–5.

Additional Notation

We write $\mathcal{L}(X)$ and $\mathcal{L}(X \mid Y)$ for the law of X and its conditional law given Y, respectively. A subscript specifies the underlying probability law or parameter value when needed. For probability laws P, Q on the same measurable space, total variation is $\mathrm{TV}(P,Q) := \sup_A |P(A) - Q(A)|$, where the supremum is over measurable sets. We write $P \ll Q$ if P is absolutely continuous with respect to Q. For functions, $\|f\|_\infty = \sup_x |f(x)|$. We use $\|\cdot\|_F$ for the Frobenius norm of matrices. The operators $\mathrm{range}(A)$ and $\ker(A)$ denote the column and null spaces of A; for square A, $\mathrm{tr}(A) = \sum_i A_{ii}$. We write $A^\dagger$ for the Moore–Penrose inverse of a matrix A. The notation $X_n \xrightarrow{L^1} X$ means $\mathbb{E}\|X_n - X\| \to 0$.

S2 Proofs and Derivations for Section 2

S2.1 Proof of Theorem 1

Proof. By Definition 3 and (5),

$$P \in \mathcal{P}(\mu) \implies \alpha(P) \in \mathcal{F}(\mu), \qquad \theta_z(P) = \theta_z(\alpha(P)), \quad z \in \mathcal{Z}^T.$$

For sufficiency, fix $\alpha \in \mathcal{F}(\mu)$. Construct P_α by assigning type probabilities α^V_{xv} in each stratum and, conditional on $X = x, V = v$, drawing $Y(0)$ and $Y(1)$ independently with $Y(d) \sim \mathrm{Bernoulli}(m^d_{xv})$ for $d \in \{0,1\}$, where

$$m^d_{xv} = \begin{cases} \alpha^{Yd}_{xv}/\alpha^V_{xv}, & \alpha^V_{xv} > 0, \\ 0, & \alpha^V_{xv} = 0 \end{cases}.$$

Then $\alpha^V \in \mathcal{L}_V$, $m^d_{xv} \in [0,1]$, and

$$\mathbb{E}_{P_\alpha}[Y(d)\mathbf{1}\{V = v\} \mid X = x] = \alpha^V_{xv} m^d_{xv} = \alpha^{Yd}_{xv}, \qquad \alpha(P_\alpha) = \alpha.$$

Hence $P_\alpha \in \mathcal{P}(\mu)$ and $\theta_z(P_\alpha) = \theta_z(\alpha)$, proving the set equality.

For $\mu \in \mathcal{M}$, $\mathcal{F}(\mu)$ is a nonempty polyhedron with $0 \le \alpha^{Yd}_{xv} \le \alpha^V_{xv} \le 1$, hence compact and convex. Its affine image $\Theta_{\mathcal{Z}^T}(\mu)$ has the same properties. The set equality gives the

support-function formula, and compactness gives attainment. □

S2.2 Derivation of the SNAP payoff endpoints

Use the one-stratum SNAP specification in Subsection 2.5, suppressing the stratum index. Theorem 1 identifies the LP extrema below with the endpoints in (7).

The observed-moment constraints give

$$\begin{aligned}
\alpha^V_{v_4} &= \mu^D_0, & \alpha^V_{v_3} &= \mu^D_1 - \mu^D_0,\\
\alpha^V_{v_2} &= \mu^D_2 - \mu^D_1, & \alpha^V_{v_0} + \alpha^V_{v_1} &= 1 - \mu^D_2,\\
\alpha^{Y1}_{v_2} + \alpha^{Y1}_{v_3} &= \mu^{Y1}_2 - \mu^{Y1}_0, & \alpha^{Y0}_{v_2} + \alpha^{Y0}_{v_3} &= \mu^{Y0}_0 - \mu^{Y0}_2,\\
\alpha^{Y0}_{v_0} + \alpha^{Y0}_{v_1} &= \mu^{Y0}_2.
\end{aligned}$$

Only the split between v_0 and v_1 remains unresolved: neither type enrols under $0, 1, 2$, but v_1 enrols under $z^\star$. Substituting the identified v_2, v_3 components into (5) gives

$$\theta_{z^\star}(\alpha) = \mu^{Y1}_2 - \mu^{Y1}_0 - \mu^{Y0}_0 + \mu^{Y0}_2 + \alpha^{Y1}_{v_1} - \alpha^{Y0}_{v_1} - k_{z^\star}.$$

The unresolved components satisfy

$$\begin{aligned}
1 - \overline{p}^D_{z^\star} \le \alpha^V_{v_0} \le 1 - \underline{p}^D_{z^\star}, \qquad & \alpha^V_{v_1} = 1 - \mu^D_2 - \alpha^V_{v_0},\\
0 \le \alpha^{Y1}_{v_1} \le \alpha^V_{v_1}, \qquad & [\mu^{Y0}_2 - \alpha^V_{v_0}]_+ \le \alpha^{Y0}_{v_1} \le \min\{\mu^{Y0}_2, \alpha^V_{v_1}\}.
\end{aligned}$$

For the lower endpoint,

$$\begin{aligned}
\min_{\alpha \in \mathcal{F}(\mu)} \alpha^{Y1}_{v_1} &= 0,\\
\max_{\alpha \in \mathcal{F}(\mu)} \alpha^{Y0}_{v_1} &= \min\{\mu^{Y0}_2, \overline{p}^D_{z^\star} - \mu^D_2\},\\
\underline{\theta}_{z^\star}(\mu) &= \mu^{Y1}_2 - \mu^{Y1}_0 - \mu^{Y0}_0 + \mu^{Y0}_2\\
&\quad - \min\{\mu^{Y0}_2, \overline{p}^D_{z^\star} - \mu^D_2\} - k_{z^\star}.
\end{aligned}$$

Using $u - \min\{u, v\} = [u - v]_+$ gives the lower endpoint in (7).

For the upper endpoint, fix feasible $a = \alpha^V_{v_0}$ and set

$$\alpha^{Y1}_{v_1} = 1 - \mu^D_2 - a, \qquad \alpha^{Y0}_{v_1} = [\mu^{Y0}_2 - a]_+.$$

Then

$$\max_{\alpha\in\mathcal{F}(\mu):\,\alpha^V_{v_0}=a}\theta_{z^\star}(\alpha)=\mu_2^{Y1}-\mu_0^{Y1}-\mu_0^{Y0}+1-\mu_2^D$$
$$-k_{z^\star}-[a-\mu_2^{Y0}]_+.$$

This is weakly decreasing in a, so it is maximised at the smallest feasible split, $a=1-\overline{p}^D_{z^\star}$. Substitution gives (7).

S3 Proofs for Section 3

S3.1 Proof of Proposition 1

Proof. For $z',z\in\mathcal{Z}^T$, let $v_{z',z}(\mu):=\max_{\alpha\in\mathcal{F}(\mu)}\{\theta_{z'}(\alpha)-\theta_z(\alpha)\}$. By Appendix S2.1, $\mathcal{F}(\mu)$ is nonempty if and only if $\mu\in\mathcal{M}$, and it is then compact, so $v_{z',z}$ is finite on $\mathcal{M}$. Thus $\mathcal{M}$ is the projection of the polyhedron $\{(\mu,\alpha):\alpha\in\mathcal{F}(\mu)\}$, hence a polyhedron (Rockafellar, 1970, Theorem 19.3), and it is bounded because $\mathcal{M}\subseteq[0,1]^{d_\mu}$. The linear programme defining $v_{z',z}$ has an objective that does not depend on μ and a right-hand side that is affine in μ. By linear programming duality, $v_{z',z}$ is therefore the minimum of finitely many affine functions on $\mathcal{M}$ (Bertsimas and Tsitsiklis, 1997, Section 5.2).

By (6),

$$R(z,\cdot)=\max_{z'\in\mathcal{Z}^T}v_{z',z},\qquad R^*=\min_{z\in\mathcal{Z}^T}R(z,\cdot).$$

Finite maxima, minima, and differences preserve continuity and piecewise affinity. Thus finitely many closed polyhedra (cells) cover $\mathcal{M}$, and every $R(z,\cdot)$, R^*, and $R(z,\cdot)-R^*$ is affine on each cell. Convexity and boundedness of $\mathcal{M}$ give boundedness and a common Lipschitz constant L_R, the largest norm of the affine gradients.

(i) Assumption 2 gives $\mu_0\in\operatorname{int}\mathcal{M}$. Let $\mathcal{C}_0$ be the cells containing μ_0, and let T_C be the tangent cone of C at μ_0. For sufficiently small $\epsilon>0$, with $B_\epsilon:=\{\mu:\|\mu-\mu_0\|\le\epsilon\}$,

$$B_\epsilon\subseteq\mathcal{M},\qquad C\cap B_\epsilon=\varnothing\quad(C\notin\mathcal{C}_0),$$
$$C\cap B_\epsilon=(\mu_0+T_C)\cap B_\epsilon\quad(C\in\mathcal{C}_0),\qquad\bigcup_{C\in\mathcal{C}_0}T_C=\mathbb{R}^{d_\mu}.$$

Write $\beta_{z,C}$ for the gradient of $R(z,\cdot)$ on C, and set $\dot{R}_z(u):=\beta_{z,C}^\top u$ for $u\in T_C$. This is well defined, because for $u\in T_C\cap T_{C'}$ and small $s>0$,

$$\beta_{z,C}^\top u=\frac{R(z,\mu_0+su)-R(z,\mu_0)}{s}=\beta_{z,C'}^\top u.$$

Consequently, for every $z \in \mathcal{Z}^T$,

$$R(z, \mu_0 + u) = R(z, \mu_0) + \dot{R}_z(u), \qquad \|u\| \le \epsilon. \tag{19}$$

Hence $\dot{R}_z$ is positively homogeneous, piecewise linear, and L_R-Lipschitz, and $R(z, \cdot)$ is Hadamard directionally differentiable at μ_0 with derivative $\dot{R}_z$.

Since $\mathcal{Z}^T$ is finite and $R(z, \mu_0) > R^*(\mu_0)$ for $z \in \mathcal{Z}^T \setminus \mathcal{Z}_0$, shrink ϵ so that $B_\epsilon \subseteq \mathcal{N}_0$ and $R^* = \min_{z \in \mathcal{Z}_0} R(z, \cdot)$ on B_ϵ. Equation (19), positive homogeneity, and $R(z, \mu_0) = R^*(\mu_0)$ for $z \in \mathcal{Z}_0$ give

$$\ell_n(z, h) = \ell(z, h), \qquad z \in \mathcal{Z}_0, \quad \|h\| \le \epsilon\sqrt{n}. \tag{20}$$

(ii) For $z \in \mathcal{Z}_0$ and $h \in \mathcal{H}_n$,

$$\begin{aligned} 0 \le \ell_n(z, h) &= \sqrt{n}\left[R\left(z, \mu_0 + \frac{h}{\sqrt{n}}\right) - R(z, \mu_0) - R^*\left(\mu_0 + \frac{h}{\sqrt{n}}\right) + R^*(\mu_0)\right] \\ &\le 2L_R\|h\|. \end{aligned}$$

□

S3.2 Proof of Proposition 2

Proof. Fix a sequence of procedures in Definition 1 and an arbitrary subsequence. Under $h = 0$, let

$$\begin{aligned} \iota_n(s_1) &:= \min\{j \le J : \gamma_n(s_1) = g_{j,n}(s_1)\}, \\ B_n &:= \big(\mathsf{T}_{1,n}, g_{1,n}(S_1), \ldots, g_{J,n}(S_1), \iota_n(S_1)\big). \end{aligned}$$

Assumption 4, $\iota_n \in \{1, \ldots, J\}$, Prohorov's theorem (van der Vaart and Wellner, 1996, Theorem 1.3.9), and continuity of coordinate selection give a further subsequence such that

$$\begin{aligned} (B_n, G_n) &\Rightarrow (B, G), \\ B = \big(\mathsf{T}_1, g_1(\mathsf{T}_1), \ldots, g_J(\mathsf{T}_1), \iota\big), &\qquad G = g_\iota(\mathsf{T}_1). \end{aligned}$$

Lemma 3 and $|\mathcal{Z}^T| < \infty$ give, along a further subsequence,

$$\begin{aligned} W_n &:= \big(\mathsf{T}_{1,n}, G_n, \mathsf{T}_{2,n}(G_n), \delta_n(S_1, S_2(G_n))\big) \\ &\Rightarrow W := \big(\mathsf{T}_1, G, \mathsf{T}_{2,G}, Z_\infty\big), \\ \mathsf{T}_1 \sim N(0, I_1), &\qquad \mathsf{T}_{2,G} \mid (\mathsf{T}_1, G) \sim N(0, I_2(G)). \end{aligned}$$

We next construct a design rule and a policy rule that generate this limit. Write $Q_{\infty,0} := \mathcal{L}(W)$, and let $K_1(dg \mid t_1)$ and $K_2(dz \mid t_1, g, t_2)$ be regular conditional distributions of G given $\mathsf{T}_1 = t_1$ and of Z_∞ given $(\mathsf{T}_1, G, \mathsf{T}_{2,G}) = (t_1, g, t_2)$. Then

$$\begin{aligned} Q_{\infty,0}(dt_1, dg, dt_2, dz) = N(0, I_1)(dt_1)\, K_1(dg \mid t_1) \\ \times N(0, I_2(g))(dt_2)\, K_2(dz \mid t_1, g, t_2). \end{aligned} \tag{21}$$

Choose measurable conditional probabilities $p_j(t_1) := \Pr(\iota = j \mid \mathsf{T}_1 = t_1)$, assigning $(p_1, \ldots, p_J) = (1, 0, \ldots, 0)$ on any exceptional null set. Use

$$\begin{aligned} K_1(dg \mid t_1) &:= \sum_{j=1}^{J} p_j(t_1) \delta_{g_j(t_1)}(dg), \\ \gamma_\infty(t_1, u_1) &:= g_{\min\{j \le J : \sum_{k=1}^{j} p_k(t_1) \ge u_1\}}(t_1). \end{aligned}$$

Then $\gamma_\infty(t_1, u_1) \in \{g_1(t_1), \ldots, g_J(t_1)\}$ for every (t_1, u_1). The inverse transform of K_2 gives a measurable δ_∞ such that

$$\begin{aligned} \mathcal{L}\big(\gamma_\infty(t_1, U_1)\big) &= K_1(\cdot \mid t_1), \\ \mathcal{L}\big(\delta_\infty(t_1, t_2, g, U_2)\big) &= K_2(\cdot \mid t_1, g, t_2). \end{aligned}$$

On a new probability space with the independent variables specified in (15), these rules give

$$G = \gamma_\infty(\mathsf{T}_1, U_1), \qquad \mathcal{L}(\mathsf{T}_1, G, \mathsf{T}_{2,G}, Z_\infty) = Q_{\infty,0} \quad (h = 0).$$

It remains to show that the same two rules generate the limit of W_n under every local alternative. For fixed h, define

$$L_h(t_1, g, t_2, z) := \exp\Big(h^\top (t_1 + t_2) - \tfrac{1}{2} h^\top \{I_1 + I_2(g)\} h\Big),$$

which is continuous by Lemma 1. For every $A \succeq 0$, including singular A,

$$\frac{dN(Ah, A)}{dN(0, A)}(t) = e^{h^\top t - \frac{1}{2} h^\top A h}, \qquad t \in \operatorname{range}(A).$$

Multiplying (21) by L_h therefore replaces $N(0, I_1)$ and $N(0, I_2(g))$ by $N(I_1 h, I_1)$ and $N(I_2(g)h, I_2(g))$, and leaves K_1, K_2, and the rules that realise them unchanged. Thus

$Q_{\infty,h}(dw) := L_h(w) Q_{\infty,0}(dw)$ is a probability law and

$$\begin{aligned} Q_{\infty,h}(dt_1, dg, dt_2, dz) = N(I_1 h, I_1)(dt_1)\, K_1(dg \mid t_1) \\ \times N(I_2(g)h, I_2(g))(dt_2)\, K_2(dz \mid t_1, g, t_2). \end{aligned} \tag{22}$$

Lemma 4 gives

$$L_{n,h} := \frac{dQ_{n,h}^{\gamma_n}}{dQ_{n,0}^{\gamma_n}}, \qquad \log L_{n,h} = \log L_h(W_n) + o_{Q_{n,0}^{\gamma_n}}(1),$$
$$(W_n, L_{n,h}) \Rightarrow (W, L_h(W)).$$

Since $\mathbb{E}_{Q_{\infty,0}} L_h(W) = 1$, Le Cam's third lemma (van der Vaart, 1998, Theorem 6.6) yields

$$W_n \Rightarrow Q_{\infty,h} \quad \text{under } Q_{n,h}^{\gamma_n}. \tag{23}$$

The factorisation (22) is precisely the law generated by (15). Both the subsequence and $(\gamma_\infty, \delta_\infty)$ were chosen at $h = 0$, independently of h. □

S3.2.1 Auxiliary results used in the proof

The first lemma shows that the regularity of the second wave follows from that of the pilot, because a design only reweights the stratum–arm groups.

Lemma 1 (Second-wave regularity from the pilot)**.** *Suppose Assumptions 2 and 3*(i) *hold. For each stratum–arm group (x, z), let $p_{xz}^O := \Pr_{Q_{P_0}^O}(X = x, Z = z)$ and $p_{xz}(g) := q_x(g)\pi_{zx}(g)$, and let $C := 1/\min_{x,z} p_{xz}^O$. For $w = (y, d, z, x)$, let $r_g(w) := p_{xz}(g)/p_{xz}^O$. Then the following hold for every $g \in \mathcal{G}$ and $t \in \mathcal{N}_1$.*

(i) $Q_{t,g}^E(dw) = r_g(w)\, Q_t^O(dw)$ *and* $0 \le r_g \le C$. *In particular,* $Q_{t,g}^E \ll Q_{P_0,g}^E$.

(ii) The second-wave submodel $t \mapsto Q_{t,g}^E$ is DQM at μ_0 with score $\dot\ell_g^E = \dot\ell^O$, $Q_{P_0,g}^E$-almost surely. Its DQM remainder is at most C times that of the pilot submodel, so DQM holds uniformly over $g \in \mathcal{G}$.

(iii) $\sup_{g\in\mathcal{G}} \mathbb{E}_{Q_{P_0,g}^E} \|\dot\ell_g^E\|^{2+\delta} \le C\, \mathbb{E}_{Q_{P_0}^O} \|\dot\ell^O\|^{2+\delta} < \infty$.

(iv) $I_2(g) = \rho(g) \sum_{x,z} p_{xz}(g)\, \mathbb{E}_{Q_{P_0}^O}\big[\dot\ell^O \dot\ell^{O\top} \mid X = x, Z = z\big]$, *so $g \mapsto I_2(g)$ is continuous on $\mathcal{G}$.*

Proof. Let $K_{P,xz}$ denote the conditional law of (D, Y) in group (x, z). Assumption 1 gives, on that group,

$$Q_P^O = p_{xz}^O K_{P,xz}, \qquad Q_{P,g}^E = p_{xz}(g) K_{P,xz}, \qquad Q_{t,g}^E = r_g Q_t^O, \qquad 0 \le r_g \le C.$$

By Assumption 3(i),

$$Q^E_{P_0,g}(A) = 0 \implies Q^O_{P_0}(A \cap \{r_g > 0\}) = 0$$
$$\implies Q^O_t(A \cap \{r_g > 0\}) = 0 \implies Q^E_{t,g}(A) = 0.$$

For (ii), let $q_t := dQ^O_t / dQ^O_{P_0}$, so $q_{\mu_0} = 1$. Then

$$\int \Big[\sqrt{r_g q_t} - \sqrt{r_g} - \tfrac{1}{2}(t-\mu_0)^\top \dot\ell^O \sqrt{r_g}\Big]^2 dQ^O_{P_0}$$
$$= \int r_g \Big[\sqrt{q_t} - 1 - \tfrac{1}{2}(t-\mu_0)^\top \dot\ell^O\Big]^2 dQ^O_{P_0} \le C \int \Big[\sqrt{q_t} - 1 - \tfrac{1}{2}(t-\mu_0)^\top \dot\ell^O\Big]^2 dQ^O_{P_0},$$

with right-hand side $o(\|t-\mu_0\|^2)$, uniformly in g. For (iii)–(iv),

$$\mathbb{E}_{Q^E_{P_0,g}} \|\dot\ell^O\|^{2+\delta} = \int r_g \|\dot\ell^O\|^{2+\delta} dQ^O_{P_0} \le C \mathbb{E}_{Q^O_{P_0}} \|\dot\ell^O\|^{2+\delta},$$
$$I_2(g) = \rho(g) \int r_g \dot\ell^O \dot\ell^{O\top} dQ^O_{P_0}$$
$$= \rho(g) \sum_{x,z} p_{xz}(g)\, \mathbb{E}_{Q^O_{P_0}}\big[\dot\ell^O \dot\ell^{O\top} \mid X = x, Z = z\big].$$

Continuity of $\rho(g)$ and $p_{xz}(g) = q_x(g)\pi_{zx}(g)$ gives (iv). □

By Lemma 1(ii), we take the version $\dot\ell^E_g := \dot\ell^O$ of the second-wave score for every $g \in \mathcal{G}$. This version does not depend on g, so $(g, w) \mapsto \dot\ell^E_g(w)$ is jointly measurable. For every pilot-measurable design G_n, $\mathsf{T}_{2,n}(G_n) = n^{-1/2} \sum_{i=1}^{n_2(G_n)} \dot\ell^O(W^E_i)$ is therefore a well-defined random vector.

Let $I^{(1)}_2(g) := \mathbb{E}_{Q^E_{P_0,g}}\Big[\dot\ell^E_g \dot\ell^{E\top}_g\Big]$, so that $I_2(g) = \rho(g) I^{(1)}_2(g)$.

Lemma 2 (Pilot LAN and uniform second-wave LAN)**.** *Suppose Assumptions 2 and 3 hold. Then the following conclusions hold.*

(i) For every fixed $h \in \mathbb{R}^{d_\mu}$, $\log \frac{dQ^O_{n,h}}{dQ^O_{n,0}}(S_1) = h^\top \mathsf{T}_{1,n} - \frac{1}{2} h^\top I_1 h + o_{Q^O_{n,0}}(1)$, and $\mathsf{T}_{1,n} \Rightarrow N(I_1 h, I_1)$ under $Q^O_{n,h}$.

(ii) For a deterministic design $g \in \mathcal{G}$, define $r^E_n(h,g) := \log \frac{dQ^E_{n,h,g}}{dQ^E_{n,0,g}}\big(S_2(g)\big) - h^\top \mathsf{T}_{2,n}(g) + \frac{1}{2} h^\top I_2(g) h$. For every fixed $h \in \mathbb{R}^{d_\mu}$ and every $\epsilon > 0$,

$$\sup_{g \in \mathcal{G}} Q^E_{n,0,g}\big(|r^E_n(h,g)| > \epsilon\big) \longrightarrow 0. \tag{24}$$

Proof. Part (i) follows from DQM, the central limit theorem, and Le Cam's third lemma (van der Vaart, 1998, Theorem 7.2 and Example 6.7).

For (ii), we follow the proof of van der Vaart (1998, Theorem 7.2) and check that every remainder is uniform in g. Fix h and write $\mathbb{E}_g := \mathbb{E}_{Q^E_{P_{0,g}}}$. For $W^E \sim Q^E_{P_{0,g}}$, define

$$\xi_{n,g} := 2\left[\left\{\frac{dQ^E_{P_{n,h,g}}}{dQ^E_{P_{0,g}}}(W^E)\right\}^{1/2} - 1\right],$$
$$u_{n,g} := \frac{h^\top \dot{\ell}^E_g(W^E)}{\sqrt{n}}, \qquad a_{n,g} := \xi_{n,g} - u_{n,g}, \qquad \bar{\delta} := \min\{\delta, 2\}.$$

An additional index i denotes evaluation at W^E_i. Sums and maxima run over $1 \le i \le n_2(g)$ and are zero when $n_2(g) = 0$. All $o_p(1)$ and $O_p(1)$ statements below are uniform over $g \in \mathcal{G}$. Lemma 1 and $n_2(g) \le \bar{\rho} n$ give

$$\sup_{g\in\mathcal{G}} \mathbb{E}_g a^2_{n,g} = o(n^{-1}), \qquad \sup_{g\in\mathcal{G}} n_2(g)\mathbb{E}_g a^2_{n,g} = o(1), \tag{25}$$

$$\begin{aligned} \sup_{g\in\mathcal{G}} n_2(g)\mathbb{E}_g|u_{n,g}|^{2+\bar{\delta}} &= O(n^{-\bar{\delta}/2}) = o(1), \\ \sup_{g\in\mathcal{G}} n_2(g)\mathbb{E}_g u^2_{n,g} &= O(1). \end{aligned} \tag{26}$$

Union, Markov, and Chebyshev bounds yield

$$\max_i |\xi_{n,g,i}| = o_p(1), \qquad \sum_i a^2_{n,g,i} = o_p(1), \qquad \sum_i (a_{n,g,i} - \mathbb{E}_g a_{n,g}) = o_p(1),$$
$$\sum_i u^2_{n,g,i} = O_p(1).$$

Moreover, $\bar{\delta} \le 2$ gives

$$\begin{aligned} &n_2(g)\left\{\mathrm{Var}_g(u^2_{n,g}\mathbf{1}\{|u_{n,g}| \le 1\}) + \mathbb{E}_g[u^2_{n,g}\mathbf{1}\{|u_{n,g}| > 1\}]\right\} \\ &\quad \le 2n_2(g)\mathbb{E}_g|u_{n,g}|^{2+\bar{\delta}} = o(1). \end{aligned}$$

Truncation and Cauchy–Schwarz therefore give

$$\sum_i u_{n,g,i}^2 = n_2(g)\mathbb{E}_g u_{n,g}^2 + o_p(1),$$

$$\left|\sum_i \xi_{n,g,i}^2 - \sum_i u_{n,g,i}^2\right| \le 2\left(\sum_i u_{n,g,i}^2 \sum_i a_{n,g,i}^2\right)^{1/2} + \sum_i a_{n,g,i}^2 = o_p(1),$$

$$\sup_{g\in\mathcal{G}} n_2(g)\left|\mathbb{E}_g\xi_{n,g}^2 - \mathbb{E}_g u_{n,g}^2\right| = o(1).$$

The zero score mean and local absolute continuity imply

$$\mathbb{E}_g u_{n,g} = 0, \qquad \mathbb{E}_g(1+\xi_{n,g}/2)^2 = 1, \qquad \mathbb{E}_g a_{n,g} = \mathbb{E}_g\xi_{n,g} = -\tfrac{1}{4}\mathbb{E}_g\xi_{n,g}^2.$$

On $\{\max_i |\xi_{n,g,i}| \le 1\}$,

$$\left|\sum_i \left\{2\log(1+\xi_{n,g,i}/2) - \xi_{n,g,i} + \xi_{n,g,i}^2/4\right\}\right| = O\Big(\max_i |\xi_{n,g,i}|\Big)\sum_i \xi_{n,g,i}^2 = o_p(1).$$

Hence

$$\begin{aligned}\log\frac{dQ_{n,h,g}^E}{dQ_{n,0,g}^E}\big(S_2(g)\big) &= \sum_i \xi_{n,g,i} - \frac{1}{4}\sum_i \xi_{n,g,i}^2 + o_p(1)\\ &= \sum_i u_{n,g,i} + n_2(g)\mathbb{E}_g\xi_{n,g} - \frac{1}{4}n_2(g)\mathbb{E}_g u_{n,g}^2 + o_p(1) \qquad (27)\\ &= h^\top \mathsf{T}_{2,n}(g) - \frac{1}{2}\frac{n_2(g)}{n}h^\top I_2^{(1)}(g)h + o_p(1).\end{aligned}$$

Finally,

$$\sup_{g\in\mathcal{G}}\left\|\frac{n_2(g)}{n}I_2^{(1)}(g) - I_2(g)\right\| \le \frac{1}{n}\sup_{g\in\mathcal{G}}\|I_2^{(1)}(g)\| = O(n^{-1}), \qquad (28)$$

by $n_2(g) = \lfloor \rho(g)n \rfloor$ and Lemma 1(iii). Equations (27)–(28) give (24). □

Lemma 3 (Conditional Gaussian limit under the selected design)**.** *Suppose Assumptions 2 and 3 hold. Let $G_n = \gamma_n(S_1) \in \mathcal{G}$ be a pilot-measurable design and let B_n be a pilot-measurable random element taking values in a fixed Polish space. Along every subsequence for which $(B_n, G_n) \Rightarrow (B, G)$ under $Q_{n,0}^{\gamma_n}$, we have $\big(B_n, G_n, \mathsf{T}_{2,n}(G_n)\big) \Rightarrow \big(B, G, \mathsf{T}_{2,G}\big)$, where $\mathsf{T}_{2,G}$ is a random vector whose conditional distribution given (B, G) is $N(0, I_2(G))$.*

Proof. Fix $u \in \mathbb{R}^{d_\mu}$, let $\alpha = \min\{\delta, 1\}$, and define

$$\varepsilon_n(u) := \sup_{g \in \mathcal{G}} \left| \mathbb{E}_{Q^E_{n,0,g}} \left[e^{iu^\top \mathsf{T}_{2,n}(g)} \right] - \exp\left(-\frac{1}{2} u^\top I_2(g) u \right) \right|.$$

The zero score mean and the uniform $2+\delta$ moment bound in Lemma 1(iii) give

$$\sup_{g \in \mathcal{G}} \left| \mathbb{E}_{Q^E_{P_0,g}} \exp\left(\frac{iu^\top \dot{\ell}^E_g}{\sqrt{n}} \right) - 1 + \frac{u^\top I_2^{(1)}(g) u}{2n} \right| = O\big(n^{-1-\alpha/2}\big).$$

Taking $n_2(g)$-th powers by independence within the second-wave sample, and using $n_2(g) \le \bar{\rho} n$ and the rounding bound in (28), yields

$$\varepsilon_n(u) = O\big(n^{-\alpha/2}\big) \longrightarrow 0. \tag{29}$$

Since $\mathcal{L}(S_2(G_n) \mid S_1) = Q^E_{n,0,G_n}$,

$$\left| \mathbb{E}_{Q^{\gamma_n}_{n,0}} \left[e^{iu^\top \mathsf{T}_{2,n}(G_n)} \mid S_1 \right] - \exp\left(-\frac{1}{2} u^\top I_2(G_n) u \right) \right| \le \varepsilon_n(u) \longrightarrow 0.$$

Below, expectations of finite-sample variables are under $Q^{\gamma_n}_{n,0}$, and expectations of (B, G) are under their limiting law. For every bounded continuous f,

$$\begin{aligned} &\left| \mathbb{E}\left[f(B_n, G_n) e^{iu^\top \mathsf{T}_{2,n}(G_n)} \right] - \mathbb{E}\left[f(B_n, G_n) e^{-\frac{1}{2} u^\top I_2(G_n) u} \right] \right| \\ &\qquad \le \|f\|_\infty \varepsilon_n(u) \longrightarrow 0. \end{aligned}$$

Continuity of I_2 and $(B_n, G_n) \Rightarrow (B, G)$ imply

$$\mathbb{E}\left[f(B_n, G_n) e^{iu^\top \mathsf{T}_{2,n}(G_n)} \right] \longrightarrow \mathbb{E}\left[f(B, G) e^{-\frac{1}{2} u^\top I_2(G) u} \right].$$

Moreover,

$$\mathbb{E}\left[\|\mathsf{T}_{2,n}(G_n)\|^2 \mid S_1 \right] = \operatorname{tr}\left(\frac{n_2(G_n)}{n} I_2^{(1)}(G_n) \right) \le \bar{\rho} \sup_{g \in \mathcal{G}} \operatorname{tr} I_2^{(1)}(g) < \infty.$$

Thus $(B_n, G_n, \mathsf{T}_{2,n}(G_n))$ is tight, and every subsequential limit satisfies

$$\mathbb{E}\left[e^{iu^\top \mathsf{T}_{2,G}} \mid B, G \right] = e^{-\frac{1}{2} u^\top I_2(G) u}, \qquad u \in \mathbb{R}^{d_\mu}.$$

This identifies the joint limit, with the realisation

$$\mathsf{T}_{2,G} = I_2(G)^{1/2}\zeta_2, \qquad \zeta_2 \sim N(0, \mathrm{Id}_{d_\mu}), \qquad \zeta_2 \perp\!\!\!\perp (B, G).$$

□

Lemma 4 (Joint likelihood expansion and contiguity)**.** *Suppose Assumptions 2, 3, and 4 hold. Let $G_n = \gamma_n(S_1) \in \mathcal{G}_{n,J}(S_1)$ for a pilot-measurable design rule γ_n. For every fixed $h \in \mathbb{R}^{d_\mu}$, $\log \frac{dQ^{\gamma_n}_{n,h}}{dQ^{\gamma_n}_{n,0}} = h^\top \{\mathsf{T}_{1,n} + \mathsf{T}_{2,n}(G_n)\} - \frac{1}{2} h^\top \{I_1 + I_2(G_n)\} h + o_{Q^{\gamma_n}_{n,0}}(1)$. Moreover, $Q^{\gamma_n}_{n,h}$ and $Q^{\gamma_n}_{n,0}$ are mutually contiguous.*

Proof. Fix $h \in \mathbb{R}^{d_\mu}$. Since $\mathcal{L}_{Q^{\gamma_n}_{n,0}}(S_1) = Q^O_{n,0}$, Lemma 2(i) applies to the pilot. For every $\epsilon > 0$, (24) gives

$$\begin{aligned} &Q^{\gamma_n}_{n,0}\big(|r^E_n(h, G_n)| > \epsilon\big) \\ &\quad = \mathbb{E}_{Q^O_{n,0}}\left[Q^E_{n,0,G_n}\big(|r^E_n(h, G_n)| > \epsilon\big)\right] \\ &\quad \le \sup_{g \in \mathcal{G}} Q^E_{n,0,g}\big(|r^E_n(h, g)| > \epsilon\big) \longrightarrow 0. \end{aligned}$$

Since $G_n = \gamma_n(S_1)$ under both laws, choosing the design adds no factor to the likelihood ratio,

$$\frac{dQ^{\gamma_n}_{n,h}}{dQ^{\gamma_n}_{n,0}} = \frac{dQ^O_{n,h}}{dQ^O_{n,0}}(S_1)\frac{dQ^E_{n,h,G_n}}{dQ^E_{n,0,G_n}}\big(S_2(G_n)\big).$$

Taking logarithms proves the expansion.

For every subsequence, Assumption 4, $G_n \in \{g_{1,n}(S_1), \dots, g_{J,n}(S_1)\}$, continuity of coordinate selection, and Lemma 3 give a further subsequence such that, under $Q^{\gamma_n}_{n,0}$,

$$\begin{gathered} \big(\mathsf{T}_{1,n}, G_n, \mathsf{T}_{2,n}(G_n)\big) \Rightarrow \big(\mathsf{T}_1, G, \mathsf{T}_{2,G}\big), \\ \mathsf{T}_1 \sim N(0, I_1), \qquad \mathsf{T}_{2,G} \mid (\mathsf{T}_1, G) \sim N(0, I_2(G)), \\ \frac{dQ^{\gamma_n}_{n,h}}{dQ^{\gamma_n}_{n,0}} \Rightarrow L_h := e^{h^\top(\mathsf{T}_1 + \mathsf{T}_{2,G}) - \frac{1}{2}h^\top\{I_1 + I_2(G)\}h}. \end{gathered}$$

Consequently,

$$L_h > 0 \quad \text{a.s.}, \qquad \mathbb{E}[L_h \mid \mathsf{T}_1, G] = e^{h^\top \mathsf{T}_1 - \frac{1}{2}h^\top I_1 h}, \qquad \mathbb{E}L_h = 1.$$

Le Cam's first lemma (van der Vaart, 1998, Lemma 6.4(ii)–(iii)) gives mutual contiguity on this subsequence. The arbitrary-subsequence argument gives mutual contiguity of the full sequences. □

S3.3 Proof of Theorem 2

Proof. (i)

Fix a deterministic Gaussian decision procedure (γ, δ_∞) whose policy rule takes values in $\mathcal{Z}_0$, and write $G = \gamma(\mathsf{T}_1)$ and $Z_\infty = \delta_\infty(\mathsf{T}_1, \mathsf{T}_{2,G}, G)$. Tonelli's theorem ($\ell \geq 0$), conditioning on $(\mathsf{T}_1, \mathsf{T}_{2,G}, G)$, minimising over the terminal menu, and then conditioning on the pilot give

$$\begin{aligned}
\mathfrak{R}_{\infty,\sigma}(\gamma, \delta_\infty; P_0) &= \mathbb{E}_\sigma[\ell(Z_\infty, H)] \\
&= \mathbb{E}_\sigma[\mathbb{E}_\sigma[\ell(Z_\infty, H) \mid \mathsf{T}_1, \mathsf{T}_{2,G}, G]] \\
&= \mathbb{E}_\sigma\left[\int \ell(\delta_\infty(\mathsf{T}_1, \mathsf{T}_{2,G}, G), h)\, \pi_{\sigma,+}\left(dh \mid \mathsf{T}_1, \mathsf{T}_{2,G}, G\right)\right] \\
&\geq \mathbb{E}_\sigma\left[\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\sigma,+}(dh \mid \mathsf{T}_1, \mathsf{T}_{2,G}, G)\right] \\
&= \mathbb{E}_\sigma\left[\int \left[\min_{z \in \mathcal{Z}_0} \int \ell(z, h)\, \pi_{\sigma,+}(dh \mid \mathsf{T}_1, t_2, G)\right] \pi^{\mathrm{pred}}_{\sigma,G}(dt_2 \mid \mathsf{T}_1)\right] \\
&= \mathbb{E}_\sigma[C_\sigma(\mathsf{T}_1, G)] \\
&\geq \mathbb{E}_\sigma\left[\min_{1 \leq j \leq J} C_\sigma\big(\mathsf{T}_1, g_j(\mathsf{T}_1)\big)\right].
\end{aligned}$$

Both inequalities are equalities for $(\gamma^*_{\infty,\sigma}, \delta^*_{\infty,\sigma})$. For randomised rules, $(U_1, U_2) \perp (H, \zeta_1, \zeta_2)$, so conditioning on (U_1, U_2) gives the same bound. Thus

$$\mathfrak{R}^*_{\infty,\sigma}(P_0) = \mathfrak{R}_{\infty,\sigma}(\gamma^*_{\infty,\sigma}, \delta^*_{\infty,\sigma}; P_0) = \mathbb{E}_\sigma\left[\min_{1 \leq j \leq J} C_\sigma\big(\mathsf{T}_1, g_j(\mathsf{T}_1)\big)\right].$$

(ii) We first establish an implication for convergence of local-average excess regret that is also used in part (iii). Write $Z_n := \delta_n(S_1, S_2(G_n)) \in \mathcal{Z}_0$, and suppose that, for φ_σ-almost every h, $Z_n \Rightarrow Z_\infty$ under the corresponding fixed-h laws, where Z_∞ is generated by a Gaussian procedure $(\gamma_\infty, \delta_\infty)$. Since $\mathcal{Z}_0$ is finite, this means that the choice probabilities converge. With (20), it gives, for these h and all sufficiently large n,

$$\begin{aligned}
\mathbb{E}_{Q^{\gamma_n}_{n,h}}[\ell_n(Z_n, h)] &= \sum_{z \in \mathcal{Z}_0} \ell(z, h) Q^{\gamma_n}_{n,h}\{Z_n = z\} \\
&\longrightarrow \sum_{z \in \mathcal{Z}_0} \ell(z, h)\, \Pr_h\{Z_\infty = z\} = \mathbb{E}_h[\ell(Z_\infty, h)].
\end{aligned}$$

Extend the expectation by zero outside $\mathcal{H}_n$. Proposition 1(ii) and Subsection 3.1 give

$$0 \le \mathbf{1}\{h \in \mathcal{H}_n\}\mathbb{E}_{Q_{n,h}^{\gamma_n}}[\ell_n(Z_n, h)] \le 2L_R\|h\|, \qquad \int \|h\|\varphi_\sigma(h)\,dh < \infty,$$
$$\mathbf{1}\{h \in \mathcal{H}_n\} \longrightarrow 1, \qquad \int_{\mathcal{H}_n} \varphi_\sigma(h)\,dh \longrightarrow 1.$$

Dominated convergence yields

$$\mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) \longrightarrow \mathfrak{R}_{\infty,\sigma}(\gamma_\infty, \delta_\infty; P_0). \tag{30}$$

This also holds along subsequences.

For any admissible sequence $(\gamma_n, \delta_n) \in \mathbb{D}(P_0)$, set $L = \liminf_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0)$ and choose a subsequence $\{n_k\}$ along which $\mathfrak{R}_{n_k,\sigma}(\gamma_{n_k}, \delta_{n_k}; P_0) \to L$.

Fix $z_0 \in \mathcal{Z}_0$ and replace every terminal choice outside $\mathcal{Z}_0$ by z_0, writing the resulting policy rule as $\widetilde{\delta}_n$. Maximum regret is uniformly bounded by the argument in Appendix S3.1, say $\sup_{z,\mu} R(z, \mu) \le \bar{R} < \infty$, so

$$\left|\ell_n\left(\delta_n(S_1, S_2(G_n)), h\right) - \ell_n\left(\widetilde{\delta}_n(S_1, S_2(G_n)), h\right)\right| \le 2\bar{R}\sqrt{n}\,\mathbf{1}\{\delta_n(S_1, S_2(G_n)) \notin \mathcal{Z}_0\}.$$

Definition 6 therefore gives

$$\left|\mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) - \mathfrak{R}_{n,\sigma}(\gamma_n, \widetilde{\delta}_n; P_0)\right| = o(1),$$
$$\mathfrak{R}_{n_k,\sigma}(\gamma_{n_k}, \widetilde{\delta}_{n_k}; P_0) \to L.$$

By Proposition 2, there are a further subsequence $\{n_{k_\ell}\}$ and a Gaussian decision procedure $(\gamma_\infty, \delta_\infty)$ that generates the joint limiting distribution of the modified sequence under every fixed h. Since $\widetilde{\delta}_n$ takes values in $\mathcal{Z}_0$, the policy rule δ_∞ may also be chosen to take values in $\mathcal{Z}_0$. Equation (30) and part (i) give

$$\begin{aligned} L &= \lim_{\ell\to\infty} \mathfrak{R}_{n_{k_\ell},\sigma}(\gamma_{n_{k_\ell}}, \widetilde{\delta}_{n_{k_\ell}}; P_0) \\ &= \mathfrak{R}_{\infty,\sigma}(\gamma_\infty, \delta_\infty; P_0) \ge \mathfrak{R}^*_{\infty,\sigma}(P_0). \end{aligned}$$

(iii)

Fix $\sigma > 0$ and suppose there are no first-order ties at σ (Definition 10). Write $\gamma = \gamma^*_{\infty,\sigma}$ and $\delta_\infty = \delta^*_{\infty,\sigma}$, and suppress the σ subscript on the constructed finite-sample sequence.

Set

$$A_{\sigma,z}(t_1,t_2,g) := \int \ell(z,h)\,\pi_{\sigma,+}(dh \mid t_1,t_2,g),$$
$$J^*(t_1,a) := \operatorname*{arg\,min}_{1\le j\le J} C_\sigma(t_1,a_j), \qquad a=(a_1,\dots,a_J)\in\mathcal{G}^J,$$
$$\Phi(t_1,a) := a_{\min J^*(t_1,a)}.$$

The Gaussian posterior and predictive laws have means and covariances that are continuous, including in g (Lemma 1). Writing each law as its mean plus a covariance square root times a standard Gaussian vector, the Lipschitz bound on ℓ gives, with $x=(t_1,t_2,g)$,

$$x_k\to x \implies \max_{z\in\mathcal{Z}_0}|A_{\sigma,z}(x_k)-A_{\sigma,z}(x)|\to 0,$$
$$(t_{1,k},g_k)\to(t_1,g) \implies C_\sigma(t_{1,k},g_k)\to C_\sigma(t_1,g).$$

Thus δ_∞ is locally constant wherever $\arg\min_z A_{\sigma,z}$ is a singleton. For $(t_{1,k},a_k)\to(t_1,a)$, finiteness of the menu gives

$$J^*(t_{1,k},a_k)\subseteq J^*(t_1,a) \quad \text{eventually.}$$

If all $j\in J^*(t_1,a)$ give the same design g^*, then

$$\|\Phi(t_{1,k},a_k)-g^*\| \le \max_{j\in J^*(t_1,a)}\|a_{j,k}-g^*\| \longrightarrow 0.$$

Hence Φ is continuous at every point with a unique minimising design, even if that design appears at several indices.

Define

$$G_n=\gamma_n(S_1) := \Phi\big(\mathsf{T}_{1,n}, g_{1,n}(S_1),\dots,g_{J,n}(S_1)\big),$$
$$\delta_n(S_1,S_2(G_n)) := \delta_\infty(\mathsf{T}_{1,n},\mathsf{T}_{2,n}(G_n),G_n), \qquad G_\gamma := \gamma(\mathsf{T}_1).$$

Since $I_1\succ 0$, the pilot laws

$$\mathcal{L}_\sigma(\mathsf{T}_1)=N(0,I_1+\sigma^2 I_1^2), \qquad \mathcal{L}_{h=0}(\mathsf{T}_1)=N(0,I_1)$$

are mutually absolutely continuous. The condition of no first-order ties in the design choice therefore makes Φ continuous at $\big(\mathsf{T}_1,g_1(\mathsf{T}_1),\dots,g_J(\mathsf{T}_1)\big)$ almost surely under $h=0$. Assumption 4 and the continuous mapping theorem give

$$(\mathsf{T}_{1,n},G_n)\Rightarrow(\mathsf{T}_1,G_\gamma) \qquad \text{under } Q_{n,0}^{\gamma_n}.$$

Lemma 3, applied with $B_n=\mathsf{T}_{1,n}$, gives joint convergence with $\mathsf{T}_{2,n}(G_n)$ under $Q_{n,0}^{\gamma_n}$. The

likelihood expansion in Lemma 4 and the change-of-measure argument yielding (23) and (22) then give, for every fixed h,

$$\big(\mathsf{T}_{1,n}, \mathsf{T}_{2,n}(G_n), G_n\big) \Rightarrow \big(\mathsf{T}_1, \mathsf{T}_{2,G_\gamma}, G_\gamma\big) \qquad \text{under } Q_{n,h}^{\gamma_n},$$

where $\mathsf{T}_1 \sim N(I_1 h, I_1)$, $\mathsf{T}_{2,G_\gamma} = I_2(G_\gamma)h + I_2(G_\gamma)^{1/2}\zeta_2$, and $\zeta_2 \sim N(0, \mathrm{Id}_{d_\mu})$ is independent of T_1.

The condition of no first-order ties in the policy choice and Fubini's theorem give, for φ_σ-almost every h,

$$\Pr_h\left\{\left|\arg\min_{z\in\mathcal{Z}_0} A_{\sigma,z}(\mathsf{T}_1, \mathsf{T}_{2,G_\gamma}, G_\gamma)\right| = 1\right\} = 1.$$

Continuity of δ_∞ there and the continuous mapping theorem yield

$$\delta_n(S_1, S_2(G_n)) \Rightarrow \delta_\infty(\mathsf{T}_1, \mathsf{T}_{2,G_\gamma}, G_\gamma) \qquad \text{under } Q_{n,h}^{\gamma_n}.$$

The candidate-list constraint and $Z_n \in \mathcal{Z}_0$ give $(\gamma_n, \delta_n) \in \mathbb{D}(P_0)$. By (30) and part (i),

$$\lim_{n\to\infty} \mathfrak{R}_{n,\sigma}(\gamma_n, \delta_n; P_0) = \mathfrak{R}_{\infty,\sigma}(\gamma, \delta_\infty; P_0) = \mathfrak{R}^*_{\infty,\sigma}(P_0).$$

□

S3.4 Proof of Proposition 3

Proof. *(i)* Each $\dot{R}_z$ is continuous, positively homogeneous, and piecewise linear (Proposition 1 and its proof), hence Lipschitz. The functions $\ell(z,\cdot)$, $z \in \mathcal{Z}_0$, are therefore Lipschitz with a common constant L_ℓ. Fix t_1 and a design g, write $V_{\infty,1} := I_1^{-1}$ and $V_{\infty,+}(g) := I_+(g)^{-1}$, and let ϵ, ϵ_2 be independent $N(0, \mathrm{Id}_{d_\mu})$ vectors. For $\sigma \in (0,\infty]$ and $t_2 \in \mathbb{R}^{d_\mu}$, define

$$\begin{aligned} T_{2,\sigma} &:= I_2(g)V_{\sigma,1}t_1 + \{I_2(g) + I_2(g)V_{\sigma,1}I_2(g)\}^{1/2}\epsilon, \\ H_\sigma(t_2) &:= V_{\sigma,+}(g)(t_1 + t_2) + V_{\sigma,+}(g)^{1/2}\epsilon_2. \end{aligned}$$

Then $T_{2,\sigma} \sim \pi^{\mathrm{pred}}_{\sigma,g}(\cdot \mid t_1)$ and $H_\sigma(t_2) \sim \pi_{\sigma,+}(\cdot \mid t_1, t_2, g)$, so

$$C_\sigma(t_1, g) = \mathbb{E}_\epsilon\Big[\min_{z\in\mathcal{Z}_0} \mathbb{E}_{\epsilon_2}\ell(z, H_\sigma(T_{2,\sigma}))\Big].$$

Since $|\min_z a_z - \min_z b_z| \le \max_z |a_z - b_z|$,

$$|C_\sigma(t_1, g) - C_\infty(t_1, g)| \le L_\ell\, \mathbb{E}\|H_\sigma(T_{2,\sigma}) - H_\infty(T_{2,\infty})\| \longrightarrow 0 \qquad (\sigma \to \infty).$$

Here $H_\sigma(T_{2,\sigma})$ is linear in $(t_1, \epsilon, \epsilon_2)$, and its coefficient matrices converge because $V_{\sigma,1} \to I_1^{-1}$, $V_{\sigma,+}(g) \to I_+(g)^{-1}$, and the matrix square root is continuous on positive semidefinite matrices. Since $J < \infty$,

$$e_\sigma(t_1) := \max_{1 \le j \le J} |C_\sigma(t_1, g_j(t_1)) - C_\infty(t_1, g_j(t_1))| \longrightarrow 0.$$

If the minimising design $g^* := \gamma^*_\infty(t_1)$ is unique, set

$$\Delta_G(t_1) := \min_{j:\, g_j(t_1) \neq g^*} \{C_\infty(t_1, g_j(t_1)) - C_\infty(t_1, g^*)\} > 0,$$

with $\min \varnothing := \infty$. Then

$$2e_\sigma(t_1) < \Delta_G(t_1) \implies \gamma^*_{\infty,\sigma}(t_1) = \gamma^*_\infty(t_1).$$

Without first-order ties in the design choice at $\sigma = \infty$, this eventual equality holds at T_1 almost surely.

(ii) This part parallels the proof of CMS (2026, Theorem 3). Fix (t_1, t_2, g), define $H_\sigma(t_2)$ as in part (i), and set $a_\sigma(z) := \int \ell(z,u)\, \pi_{\sigma,+}(du \mid t_1, t_2, g)$. Then

$$e^Z_\sigma := \max_{z \in \mathcal{Z}_0} |a_\sigma(z) - a_\infty(z)| \le L_\ell\, \mathbb{E}\|H_\sigma(t_2) - H_\infty(t_2)\| \longrightarrow 0.$$

For a unique minimiser $z^* := \delta^*_\infty(t_1, t_2, g)$,

$$\Delta_Z := \min_{z \in \mathcal{Z}_0 \setminus \{z^*\}} \{a_\infty(z) - a_\infty(z^*)\} > 0,$$
$$2e^Z_\sigma < \Delta_Z \implies \delta^*_{\infty,\sigma}(t_1, t_2, g) = \delta^*_\infty(t_1, t_2, g).$$

Without first-order ties in the policy choice at $\sigma = \infty$, this eventual equality holds at $(\mathsf{T}_1, \mathsf{T}_{2,G}, G)$ almost surely.

□

S3.5 Verification for the SNAP running example

This subsection gives sufficient conditions for Assumptions 2 and 3 in the bounded-outcome SNAP model. The pilot and the second-wave experiments share the conditional laws of (D, Y) given (X, Z) and differ only in the probabilities of the stratum–arm groups. When these conditional laws are unrestricted, the efficient scores for the conditional means do not depend on these probabilities (Lemma 5). A path along which each conditional law moves

in its efficient-score direction is then least favourable for every experiment. Proposition 5 gives conditions under which such a path exists in $\mathcal{P}$. The common conditional laws are

$$\begin{aligned} Q_{xz,0}(B) &:= Q^O_{P_0}\big((D,Y)\in B \mid X=x, Z=z\big) \\ &= Q^E_{P_0,g}\big((D,Y)\in B \mid X=x, Z=z\big) \quad \text{if } Q^E_{P_0,g}(X=x,Z=z)>0, \end{aligned}$$

for $x\in\mathcal{X}$, $z\in\mathcal{Z}^E$, and Borel $B\subseteq\{0,1\}\times[0,1]$. For $t\in\mathbb{R}^{d_\mu}$, let $t_{xz}\in\mathbb{R}^3$ denote its block ordered as $(\mu^D_{xz},\mu^{Y1}_{xz},\mu^{Y0}_{xz})$. With $A:=(D, YD, Y(1-D))^\top$, write

$$\mu_{xz,0}:=\mathbb{E}_{Q_{xz,0}}A, \qquad \Sigma_{xz}:=\mathrm{Var}_{Q_{xz,0}}A.$$

Lemma 5 (Efficient scores and information for conditional means)**.** *Let Q be a law of $W=(Y,D,Z,X)$ under which (D,Y) given $(X,Z)=(x,z)$ has law $Q_{xz,0}$, and let $p_{xz}:=\Pr_Q(X=x,Z=z)$. Consider the model in which the law of (X,Z) is fixed and the conditional laws of (D,Y) given (X,Z) are unrestricted, and suppose $\Sigma_{xz}\succ 0$ whenever $p_{xz}>0$. Let $s_{xz}(D,Y):=\Sigma_{xz}^{-1}\{(D,YD,Y(1-D))^\top-\mu_{xz,0}\}$. For each (x,z) with $p_{xz}>0$, the efficient score for the (x,z)-block of μ is $\mathbf{1}\{X=x,Z=z\}s_{xz}(D,Y)$. The information matrix is* $\mathrm{blockdiag}_{x\in\mathcal{X},\,z\in\mathcal{Z}^E}(p_{xz}\Sigma_{xz}^{-1})$*, with zero blocks for the groups that Q does not sample.*

Proof. For $p_{xz}>0$, the efficient influence function is (van der Vaart, 1998, Sections 25.3–25.4)

$$\psi_{xz}:=p_{xz}^{-1}\mathbf{1}\{X=x,Z=z\}(A-\mu_{xz,0}), \qquad \mathbb{E}_Q[\psi_{xz}\psi_{xz}^\top]=\Sigma_{xz}/p_{xz}.$$

Hence the efficient score and its cross moments are

$$\begin{aligned} \dot\ell^{\mathrm{eff}}_{xz} &= (\Sigma_{xz}/p_{xz})^{-1}\psi_{xz}=\mathbf{1}\{X=x,Z=z\}s_{xz}, \\ \mathbb{E}_Q[\dot\ell^{\mathrm{eff}}_{xz}\dot\ell^{\mathrm{eff}\top}_{x'z'}] &= \mathbf{1}\{(x,z)=(x',z')\}\,p_{xz}\Sigma_{xz}^{-1}. \end{aligned}$$

□

For $d\in\{0,1\}$ and Borel $B\subseteq[0,1]$, Assumption 1 and monotonicity give

$$\begin{gathered} Q^d_{xz}(B):=Q_{xz,0}(\{d\}\times B)=\Pr_{P_0}\{D(z)=d, Y(d)\in B\mid X=x\}, \\ Q^0_{x,z+1}\le Q^0_{xz}, \qquad Q^1_{xz}\le Q^1_{x,z+1}, \\ Q^0_{x0}-Q^0_{x1}=\Pr_{P_0}\{V=v_3, Y(0)\in\cdot\mid X=x\}. \end{gathered}$$

Differences between adjacent arms are the outcome distributions of compliers, scaled by their shares, as in the last line.

Proposition 5 (Sufficient conditions for common least-favourable paths)**.** *Maintain Assumption 1 in the SNAP model of Section 2, where $\mathcal{L}_V$ imposes monotonicity and, possibly, bounds on take-up under proactive enrolment. Suppose that, for some $c > 0$ and every $x \in \mathcal{X}$, the following conditions hold.*

(i) For every $z \in \mathcal{Z}^E$, $\Sigma_{xz} \succ 0$. Equivalently, both take-up states occur in arm z, and the outcome is not almost surely constant given either state.

(ii) The monotonicity inequalities hold setwise with margin c, that is,

$$Q^0_{x,z+1} \le (1-c)\, Q^0_{xz} \quad \text{and} \quad Q^1_{xz} \le (1-c)\, Q^1_{x,z+1}, \qquad z \in \{0,1\}. \tag{31}$$

In addition, $\Pr_{P_0}(V = v_1 \mid X = x) > 0$.

Then there exists a path $t \mapsto P_t \in \mathcal{P}$ on a neighbourhood of μ_0 satisfying Assumptions 2 and 3, with $P_{\mu_0} = P_0$ and $\mu(P_t) = t$. The induced pilot and second-wave scores are those in Lemma 5, with information matrices

$$I_1 = \operatorname{blockdiag}_{x\in\mathcal{X},\ z\in\mathcal{Z}^E} \left(\Pr_{Q^Q_{P_0}}(X = x, Z = z) \Sigma^{-1}_{xz} \right),$$
$$I_2(g) = \rho(g) \operatorname{blockdiag}_{x\in\mathcal{X},\ z\in\mathcal{Z}^E} \left(\Pr_{Q^E_{P_0,g}}(X = x, Z = z) \Sigma^{-1}_{xz} \right).$$

Remark 4 (Interpretation of the conditions). Condition (i) is necessary for Assumptions 2 and 3, because

$$\begin{aligned} \Sigma_{xz} \text{ singular} &\implies \exists a \neq 0 :\ a^\top A = a^\top \mu_{xz,0} \quad Q_{xz,0}\text{-a.s.} \\ &\implies a^\top t_{xz} = a^\top \mu_{xz,0} \quad \text{for every } Q_{xz,t} \ll Q_{xz,0}, \end{aligned}$$

which precludes $\mu(P_t) = t$ on a neighbourhood of μ_0.

For $z = 0$, the first inequality in (31) reads

$$Q^0_{x1} \le (1-c) Q^0_{x0} \iff \Pr_{P_0}\{V = v_3 \mid D(0) = 0, Y(0) = y, X = x\} \ge c \quad \text{for } Q^0_{x0}\text{-almost every } y.$$

The other inequalities have the same interpretation for the corresponding complier groups. Each complier group must therefore keep a share of at least c at every outcome value. At $c = 0$, these inequalities extend the testable implications of Imbens and Rubin (1997) and Kitagawa (2015) to multiple arms. The condition $\Pr_{P_0}(V = v_1 \mid X = x) > 0$ permits the constructed path to keep proactive take-up fixed, and it is necessary when $\mathcal{L}_V$ fixes that take-up.

For binary Y, write $\alpha_{xv} := \Pr_{P_0}(V = v \mid X = x)$ and $q^d_{xv} := \Pr_{P_0}(Y(d) = 1 \mid V = v, X = x)$. Given condition (i),

$$\text{(ii)} \quad \iff \quad \begin{cases} \alpha_{xv} > 0, & x \in \mathcal{X},\ v \in \{v_1, v_2, v_3\}, \\ 0 < q^d_{xv} < 1, & x \in \mathcal{X},\ v \in \{v_2, v_3\},\ d \in \{0, 1\}. \end{cases}$$

The calibrations in Subsection 5.1 and Appendix S5.3 satisfy these conditions. If $\mathcal{L}_V$ fixes proactive take-up in every stratum, then

$$\text{(i) and (ii)} \quad \iff \quad \mu_0 \in \operatorname{int} \mathcal{M},$$

so both conditions are necessary for Assumption 2. ◁

Proof. Keep the pilot and design-specific sampling and assignment mechanisms fixed, and for t near μ_0 define $Q_{xz,t}$ by

$$\frac{dQ_{xz,t}}{dQ_{xz,0}} = 1 + (t_{xz} - \mu_{xz,0})^\top s_{xz}. \tag{32}$$

Bounded outcomes and condition (i) give $M := \max\{1, \max_{x,z} \operatorname{ess\,sup}_{Q_{xz,0}} \|s_{xz}\|\} < \infty$. Then

$$\begin{gathered} \mathbb{E}_{Q_{xz,0}} s_{xz} = 0, \qquad \mathbb{E}_{Q_{xz,0}}[(A - \mu_{xz,0}) s_{xz}^\top] = \mathrm{Id}_3, \\ \|t - \mu_0\| < 1/M \implies \frac{dQ_{xz,t}}{dQ_{xz,0}} > 0, \quad Q_{xz,t}(\{0,1\} \times [0,1]) = 1, \quad \mathbb{E}_{Q_{xz,t}} A = t_{xz}. \end{gathered}$$

Write $h := t - \mu_0$, let t^D_{xz} and h^D_{xz} be the first coordinates of t_{xz} and h_{xz}, and set $Q^d_{xz,t}(B) := Q_{xz,t}(\{d\} \times B)$. Then

$$\begin{aligned} Q^1_{xz,t}([0,1]) &= t^D_{xz}, & Q^0_{xz,t}([0,1]) &= 1 - t^D_{xz}, \\ Q^d_{xz,t}(B) - Q^d_{xz}(B) &= \mathbb{E}_{Q_{xz,0}}[\mathbf{1}\{D = d, Y \in B\}\, h_{xz}^\top s_{xz}]. \end{aligned}$$

Hence

$$\left|Q^d_{xz,t}(B) - Q^d_{xz}(B)\right| \le M\|h\|\, Q^d_{xz}(B) \tag{33}$$

for every $(x, z) \in \mathcal{X} \times \mathcal{Z}^E$, $d \in \{0, 1\}$, and Borel $B \subseteq [0, 1]$.

Set $P_{\mu_0} := P_0$ and $\alpha_{xv} := \Pr_{P_0}(V = v \mid X = x)$. Fix $t \neq \mu_0$ with $\|h\| < \min\{c/(2M), \min_{x \in \mathcal{X}} \alpha_{xv_1}\}$, a bound that is positive by condition (ii). Conditions (i)–

(ii) imply $c < 1$, hence $\|h\| < 1/M$. In each stratum x, set

$$\begin{aligned}
\alpha_{xv_0,t} &:= \alpha_{xv_0}, & \alpha_{xv_1,t} &:= 1 - t^D_{x2} - \alpha_{xv_0},\\
\alpha_{xv_2,t} &:= t^D_{x2} - t^D_{x1}, & \alpha_{xv_3,t} &:= t^D_{x1} - t^D_{x0},\\
\alpha_{xv_4,t} &:= t^D_{x0}, & \alpha_{xv,t} &:= 0 \quad (v \notin \{v_0, \dots, v_4\}).
\end{aligned}$$

Let G be a probability measure on $[0,1]$, and let $\bar{Q}^0_{x2,t} := Q^0_{x2,t}/(1 - t^D_{x2})$ be the outcome distribution of non-enrollees in arm 2. Define the $Y(d)$-margins $P^d_{xv,t}$ of the types by

v	$P^0_{xv,t}$	$P^1_{xv,t}$
v_0	$\alpha_{xv_0}\,\bar{Q}^0_{x2,t}$	$\alpha_{xv_0}\,G$
v_1	$\alpha_{xv_1,t}\,\bar{Q}^0_{x2,t}$	$\alpha_{xv_1,t}\,G$
v_2	$Q^0_{x1,t} - Q^0_{x2,t}$	$Q^1_{x2,t} - Q^1_{x1,t}$
v_3	$Q^0_{x0,t} - Q^0_{x1,t}$	$Q^1_{x1,t} - Q^1_{x0,t}$
v_4	$\alpha_{xv_4,t}\,G$	$Q^1_{x0,t}$

Margins that some arm reveals are differences between adjacent arms. Types v_0 and v_1, which no arm separates, both take the outcome distribution of arm-2 non-enrollees, and the remaining margins are multiples of G. The table gives

$$P^d_{xv,t}([0,1]) = \alpha_{xv,t}, \qquad \sum_v \alpha_{xv,t} = 1.$$

Nonnegativity follows from

$$\begin{gathered}
\alpha_{xv_1,t} = \alpha_{xv_1} - h^D_{x2} > 0, \qquad 1 - t^D_{x2} = \alpha_{xv_0} + \alpha_{xv_1,t} > 0,\\
\alpha_{xv_4,t} = Q^1_{x0,t}([0,1]) \geq 0,
\end{gathered}$$

and, by (33) and (31), for $z = 0, 1$,

$$\begin{aligned}
Q^0_{xz,t} - Q^0_{x,z+1,t} &\geq Q^0_{xz} - Q^0_{x,z+1} - M\|h\|(Q^0_{xz} + Q^0_{x,z+1})\\
&\geq (c - 2M\|h\|)Q^0_{xz} \geq 0,\\
Q^1_{x,z+1,t} - Q^1_{xz,t} &\geq Q^1_{x,z+1} - Q^1_{xz} - M\|h\|(Q^1_{x,z+1} + Q^1_{xz})\\
&\geq (c - 2M\|h\|)Q^1_{x,z+1} \geq 0.
\end{aligned}$$

Choose the product coupling

$$P_{xv,t} := \begin{cases} \alpha_{xv,t}^{-1} P^0_{xv,t} \otimes P^1_{xv,t}, & \alpha_{xv,t} > 0, \\ 0, & \alpha_{xv,t} = 0. \end{cases}$$

Since $\alpha_{xv_0,t} = \alpha_{xv_0}$, proactive take-up $1 - \alpha_{xv_0,t}$ is unchanged, so the nonnegative shares supported on $\{v_0, \dots, v_4\}$ satisfy $\mathcal{L}_V$. Together with the unchanged law of X, the measures $P_{xv,t}$ define $P_t \in \mathcal{P}$. Summing the table gives, for every x, z, d, B,

$$\begin{aligned} \Pr_{P_t}\{D(z) = d, Y(d) \in B \mid X = x\} &= \sum_{v:\, d_z(v) = d} P^d_{xv,t}(B) \\ &= Q^d_{xz,t}(B), \qquad \mu(P_t) = t. \end{aligned}$$

These identities also hold at $t = \mu_0$. The coupling of $Y(0)$ and $Y(1)$ under P_t need not converge to that under P_0. Assumptions 2 and 3 do not require this, because beyond $P_t \in \mathcal{P}$ and $P_{\mu_0} = P_0$ they depend on P_t only through $\mu(P_t)$ and its observed laws.

The fixed pilot sampling and assignment probabilities and (32) give, $Q^O_{P_0}$-almost surely,

$$\begin{aligned} \frac{dQ^O_t}{dQ^O_{P_0}}(W^O) &= 1 + h^\top \dot{\ell}^O(W^O) > 0, \\ \dot{\ell}^O_{xz}(W^O) &= \mathbf{1}\{X = x, Z = z\} s_{xz}(D, Y). \end{aligned}$$

Since $\|\dot{\ell}^O\| \le M$ and $|\sqrt{1+x} - 1 - x/2| \le x^2$ for $|x| \le 1/2$,

$$\int \left[\sqrt{\frac{dQ^O_t}{dQ^O_{P_0}}} - 1 - \frac{1}{2}(t - \mu_0)^\top \dot{\ell}^O \right]^2 dQ^O_{P_0} \le M^4 \|t - \mu_0\|^4$$

whenever $M\|t - \mu_0\| \le 1/2$. Thus the pilot path is DQM, with

$$\mathbb{E}_{Q^O_{P_0}} \|\dot{\ell}^O\|^{2+\delta} \le M^{2+\delta}, \qquad I_1 = \mathbb{E}_{Q^O_{P_0}}[\dot{\ell}^O \dot{\ell}^{O\top}] \succ 0,$$

where positive definiteness follows from positive pilot group probabilities and $\Sigma_{xz} \succ 0$. Lemma 1 gives DQM for each g, with $\dot{\ell}^E_g = \dot{\ell}^O$ on its sampled groups and $I_2(g) = \rho(g)\mathbb{E}_{Q^E_{P_0,g}}[\dot{\ell}^E_g \dot{\ell}^{E\top}_g]$.

For each experiment, the model of observed data induced by $\mathcal{P}$ contains the constructed path and is contained in the unrestricted conditional model. The path attains the information bound of the unrestricted model on the identified coordinates, so it is least favourable in the smaller model as well (van der Vaart, 1998, Sections 25.3–25.4). □

S4 Proofs and Constructions for Section 4

S4.1 Approximation by Gaussian design losses and proof of Theorem 3

The quasi-Bayesian procedure of Subsection 4.1 is one instance of a more general construction, which we call a *coherent inferential protocol*. We state the approximation result for this construction, so that it also covers the bootstrap implementation in Appendix S4.2. A protocol updates on a statistic of the second-wave sample, which keeps prediction and updating tractable. For each design g, let $E_{n,g} := E_{n,g}\big(S_2(g); S_1\big)$ denote this measurable statistic, with realisation e. In the quasi-Bayesian construction, $E_{n,g}$ is the second-wave estimate $\widehat{\mu}_{2,g}$ of μ.

Definition 13 (Coherent inferential protocol)**.** A coherent inferential protocol consists of a pilot law $\Pi_1(d\mu \mid S_1)$ on $\mathbb{R}^{d_\mu}$ with finite first moment and, for each $g \in \mathcal{G}$, a second-wave statistic $E_{n,g}$, its predictive law $\Pi^{\mathrm{pred}}_{n,g}(de \mid S_1)$, and the updated law $\Pi_{2,g}(d\mu \mid S_1, e, g)$. For every bounded measurable $f : \mathbb{R}^{d_\mu} \to \mathbb{R}$, these laws satisfy

$$\int \left[\int f(\mu) \Pi_{2,g}(d\mu \mid S_1, e, g) \right] \Pi^{\mathrm{pred}}_{n,g}(de \mid S_1) = \int f(\mu) \Pi_1(d\mu \mid S_1).$$

Suppose that, after observing the pilot, the policymaker commits to the same terminal policy under every candidate design, regardless of the second-wave evidence. Coherence requires the predictive average of this policy's posterior expected regret to equal its expected regret under the pilot law. Thus the protocol cannot assign value to an experiment whose evidence will not affect the terminal choice. For a coherent protocol, the policy rule and the full design loss are

$$\begin{aligned} \delta^{\Pi}_{n,g}(s_1, e) &:= \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} \int \{R(z,\mu) - R^*(\mu)\} \, \Pi_{2,g}(d\mu \mid s_1, e, g), \\ C_n(s_1, g) &:= \int \left[\min_{z \in \mathcal{Z}^T} \int \{R(z,\mu) - R^*(\mu)\} \, \Pi_{2,g}(d\mu \mid s_1, e, g) \right] \Pi^{\mathrm{pred}}_{n,g}(de \mid s_1). \end{aligned} \tag{34}$$

The quasi-Bayesian procedure of Subsection 4.1 is the protocol with $\Pi_1(\cdot \mid S_1) = N(\widehat{\mu}_1, \widehat{\Sigma}_1/n)$,

$$\begin{aligned} \Pi^{\mathrm{pred}}_{n,g}(\cdot \mid S_1) &= N\left(\widehat{\mu}_1, \{\widehat{\Sigma}_1 + \widehat{I}_2(g; S_1)^{-1}\}/n\right), \\ \Pi_{2,g}(\cdot \mid S_1, e, g) &= N\left(\widehat{\mu}_{+,g}(e; S_1), \widehat{\Sigma}_{+,g}/n\right), \end{aligned}$$

where $\widehat{\mu}_{+,g}(e; S_1) := \widehat{\Sigma}_{+,g}\{\widehat{I}_1\widehat{\mu}_1 + \widehat{I}_2(g; S_1)e\}$. The predictive law averages the Gaussian

approximation $N(\mu, \widehat{I}_2(g;S_1)^{-1}/n)$ to the sampling distribution of $\widehat{\mu}_{2,g}$ over Π_1, and $\Pi_{2,g}$ updates Π_1 with this approximation, so the quasi-Bayesian laws satisfy the coherence condition. If $E_{n,g}$ is drawn from the predictive law, $\widehat{\mu}_{+,g}(E_{n,g};S_1)$ is distributed as $N(\widehat{\mu}_1, (\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})/n)$, so (34) reduces to the policy rule and the full design loss (18) of Subsection 4.1. If $\widehat{I}_2(g;S_1)$ is singular, let $P_{n,g} := \widehat{I}_2(g;S_1)^{\dagger}\widehat{I}_2(g;S_1)$, where † denotes the Moore–Penrose inverse. The Gaussian approximation is then $N(P_{n,g}\mu, \widehat{I}_2(g;S_1)^{\dagger}/n)$ on range $\widehat{I}_2(g;S_1)$, the predictive law is the corresponding mixture over Π_1, and the formulas for $\Pi_{2,g}$ and $\widehat{\mu}_{+,g}(e;S_1)$ are unchanged. For the proofs, define

$$\widehat{\mathsf{T}}^{\Pi}_{1,n} := \widehat{I}_1\sqrt{n}(\widehat{\mu}_1 - \mu_0), \qquad \widehat{\mathsf{T}}^{\Pi}_{2,n}(g;e) := \widehat{I}_2(g;S_1)\sqrt{n}(e - \mu_0),$$

for a value e of $E_{n,g}$, and write $\widehat{\mathsf{T}}^{\Pi}_{2,n}(g) := \widehat{\mathsf{T}}^{\Pi}_{2,n}(g;\widehat{\mu}_{2,g})$. These statistics depend on the unknown μ_0 and are not needed to run the procedure. Since $\mathsf{T}_{1,n} = O_p(1)$, and given consistency of $\widehat{I}_1$, Assumption 5(i) is equivalent to $\widehat{\mathsf{T}}^{\Pi}_{1,n} = \mathsf{T}_{1,n} + o_p(1)$, and Assumption 5(ii) states that $\widehat{\mathsf{T}}^{\Pi}_{2,n}(G_n) = \mathsf{T}_{2,n}(G_n) + o_p(1)$.

The next lemma converts Gaussian approximation of a coherent protocol's predictive and updated laws into uniform approximation of its design losses. Let $e \mapsto \mathsf{T}^{\Pi}_{2,n}(g;e) \in \mathbb{R}^{d_\mu}$ be a measurable transformation of the second-wave statistic $E_{n,g}$.

For candidate $g_{j,n} = g_{j,n}(S_1)$, let $\widetilde{\Pi}^{T,\mathrm{pred}}_{n,j}(\cdot \mid S_1)$ be the predictive distribution of $\mathsf{T}^{\Pi}_{2,n}(g_{j,n};E_{n,g_{j,n}})$, where $E_{n,g_{j,n}} \sim \Pi^{\mathrm{pred}}_{n,g_{j,n}}(\cdot \mid S_1)$. Let $\Pi^{h,n}_{2,j}(\cdot \mid S_1, e)$ be the updated distribution of $\sqrt{n}(\mu - \mu_0)$ under $\Pi_{2,g_{j,n}}(\cdot \mid S_1, e, g_{j,n})$. Let W_1 denote the first Wasserstein distance induced by the Euclidean norm (Villani, 2009).

Lemma 6 (Approximation by Gaussian design losses)**.** *Suppose Assumptions 2–4 hold. Let* Π *be a coherent inferential protocol as in Definition 13. Under the pilot sampling distribution* $Q^{O}_{n,0}$*, assume:*

(i) The predictive and updated distributions defined above satisfy

$$\max_{1\le j\le J} W_1\Big(\widetilde{\Pi}^{T,\mathrm{pred}}_{n,j}(\cdot \mid S_1), \pi^{\mathrm{pred}}_{\infty,g_{j,n}}(\cdot \mid \mathsf{T}_{1,n})\Big) \xrightarrow{p} 0$$

and

$$\max_{1\le j\le J} \int W_1\Big(\Pi^{h,n}_{2,j}(\cdot \mid S_1, e), \pi_{\infty,+}\big(\cdot \mid \mathsf{T}_{1,n}, \mathsf{T}^{\Pi}_{2,n}(g_{j,n};e), g_{j,n}\big)\Big)\, \Pi^{\mathrm{pred}}_{n,g_{j,n}}(de \mid S_1) \xrightarrow{p} 0.$$

(ii) For some $\eta > 0$, $\int \|\sqrt{n}(\mu - \mu_0)\|^{1+\eta}\, \Pi_1(d\mu \mid S_1) = O_p(1)$.

Then

$$\max_{1\le j\le J} \big|\sqrt{n}C_n(S_1, g_{j,n}(S_1)) - C_\infty(\mathsf{T}_{1,n}, g_{j,n}(S_1))\big| \xrightarrow{p} 0. \tag{35}$$

Consequently, pairwise design-loss differences have the corresponding expansion, and every pilot-measurable $o_p(n^{-1/2})$-minimiser of C_n over the candidate list is an $o_p(1)$-minimiser of the corresponding Gaussian design losses.

Proof. By Definition 13, monotone convergence, and condition (ii), for every j,

$$\int\int \|h\|^{1+\eta}\,\Pi_{2,j}^{h,n}(dh \mid S_1, e)\,\Pi_{n,g_{j,n}}^{\mathrm{pred}}(de \mid S_1) = \int \|\sqrt{n}(\mu-\mu_0)\|^{1+\eta}\,\Pi_1(d\mu \mid S_1) =: M_n = O_p(1).$$

We first consider the policies in $\mathcal{Z}_0$.[27] For $z \in \mathcal{Z}_0$, let $f_{n,z}(h) := \sqrt{n}\{R(z, \mu_0 + h/\sqrt{n}) - R^*(\mu_0 + h/\sqrt{n})\}$. Let $L < \infty$ be a common Lipschitz constant for $f_{n,z}$ and $\ell(z,\cdot)$, $z \in \mathcal{Z}_0$. By (20),

$$|f_{n,z}(h) - \ell(z,h)| \le 2L\|h\|\,\mathbf{1}\{\|h\| > \epsilon\sqrt{n}\}, \qquad z \in \mathcal{Z}_0.$$

For probability laws P, Q with the indicated moments finite,

$$\begin{aligned}
&\max_{z\in\mathcal{Z}_0}\left|\int f_{n,z}\,dP - \int \ell(z,\cdot)\,dQ\right| \\
&\quad\le 2L\int_{\|h\|>\epsilon\sqrt{n}} \|h\|\,dP(h) + LW_1(P,Q) \\
&\quad\le \frac{2L}{(\epsilon\sqrt{n})^{\eta}}\int \|h\|^{1+\eta}\,dP(h) + LW_1(P,Q).
\end{aligned} \tag{36}$$

Write

$$\begin{aligned}
P_{j,e} &:= \Pi_{2,j}^{h,n}(\cdot \mid S_1, e), \\
Q_{j,e} &:= \pi_{\infty,+}(\cdot \mid \mathsf{T}_{1,n}, \mathsf{T}_{2,n}^{\Pi}(g_{j,n}; e), g_{j,n}), \\
\nu_j(de) &:= \Pi_{n,g_{j,n}}^{\mathrm{pred}}(de \mid S_1).
\end{aligned}$$

Coherence and condition (i) give

$$\begin{aligned}
&\max_j \int \max_{z\in\mathcal{Z}_0}\left|\int f_{n,z}\,dP_{j,e} - \int \ell(z,\cdot)\,dQ_{j,e}\right| \nu_j(de) \\
&\quad\le \frac{2LM_n}{(\epsilon\sqrt{n})^{\eta}} + L\max_j \int W_1(P_{j,e}, Q_{j,e})\,\nu_j(de) \\
&\quad= o_p(1).
\end{aligned}$$

With $F_{t_1,g}(t_2) := \min_{z\in\mathcal{Z}_0}\int \ell(z,h)\,\pi_{\infty,+}(dh \mid t_1, t_2, g)$, $|\min_z a_z - \min_z b_z| \le \max_z |a_z - b_z|$ yields

$$\max_j \left|\int \min_{z\in\mathcal{Z}_0}\int f_{n,z}\,dP_{j,e}\,\nu_j(de) - \int F_{\mathsf{T}_{1,n},g_{j,n}}\,d\widetilde{\Pi}_{n,j}^{T,\mathrm{pred}}(\cdot \mid S_1)\right| = o_p(1).$$

[27] This argument is related to CMS (2026, Appendix C).

If $\mathcal{Z}^T \neq \mathcal{Z}_0$, fix $z_0 \in \mathcal{Z}_0$. Finiteness and continuity give $r, \Delta_* > 0$, with $\mathcal{N}_* := B(\mu_0, r)$, such that

$$\overline{\mathcal{N}_*} \subset \mathcal{N}_0, \qquad \min_{z \notin \mathcal{Z}_0} \inf_{\mu \in \mathcal{N}_*} \{R(z,\mu) - R(z_0,\mu)\} \geq \Delta_*.$$

For $\Pi_2 := \Pi_{2,g}(\cdot \mid S_1, e, g)$,

$$\begin{aligned} 0 &\leq \min_{z \in \mathcal{Z}_0} \int R(z,\cdot)\, d\Pi_2 - \min_{z \in \mathcal{Z}^T} \int R(z,\cdot)\, d\Pi_2 \leq \max_{z \notin \mathcal{Z}_0} \int [R(z_0,\cdot) - R(z,\cdot)]_+ \, d\Pi_2 \\ &\leq 2 \int_{\mathcal{N}_*^c} \max_{z \in \mathcal{Z}^T} |R(z,\mu)|\, \Pi_2(d\mu). \end{aligned}$$

Coherence and linear growth give

$$\begin{aligned} &2\sqrt{n} \int \left[\int_{\mathcal{N}_*^c} \max_{z \in \mathcal{Z}^T} |R(z,\mu)|\, \Pi_{2,g}(d\mu \mid S_1, e, g) \right] \Pi_{n,g}^{\mathrm{pred}}(de \mid S_1) \\ &\quad = 2\sqrt{n} \int_{\mathcal{N}_*^c} \max_{z \in \mathcal{Z}^T} |R(z,\mu)|\, \Pi_1(d\mu \mid S_1) \leq C_r n^{-\eta/2} M_n \end{aligned}$$

for $C_r < \infty$. Thus, uniformly over candidates,

$$0 \leq \int \min_{z \in \mathcal{Z}_0} \int f_{n,z}\, dP_{j,e}\, \nu_j(de) - \sqrt{n} C_n(S_1, g_{j,n}) \leq C_r n^{-\eta/2} M_n = o_p(1).$$

By definition, $C_\infty(t_1, g) = \int F_{t_1,g}\, d\pi_{\infty,g}^{\mathrm{pred}}(\cdot \mid t_1)$. Posterior draws at t_2 and t_2' generated from the same centred Gaussian vector differ by $I_+(g)^{-1}(t_2 - t_2')$, so

$$|F_{t_1,g}(t_2) - F_{t_1,g}(t_2')| \leq L \|I_+(g)^{-1}(t_2 - t_2')\| \leq L \|I_1^{-1}\| \|t_2 - t_2'\|.$$

Consequently,

$$\begin{aligned} &\max_{1 \leq j \leq J} \left| \int F_{\mathsf{T}_{1,n}, g_{j,n}}\, d\widetilde{\Pi}_{n,j}^{T,\mathrm{pred}}(\cdot \mid S_1) - C_\infty(\mathsf{T}_{1,n}, g_{j,n}) \right| \\ &\quad \leq L \|I_1^{-1}\| \max_{1 \leq j \leq J} W_1\Big(\widetilde{\Pi}_{n,j}^{T,\mathrm{pred}}(\cdot \mid S_1), \pi_{\infty, g_{j,n}}^{\mathrm{pred}}(\cdot \mid \mathsf{T}_{1,n}) \Big), \end{aligned}$$

which is $o_p(1)$ by condition (i), proving (35). Write

$$C_{n,j} := C_n(S_1, g_{j,n}), \quad C_{\infty,j} := C_\infty(\mathsf{T}_{1,n}, g_{j,n}), \quad \varepsilon_n := \max_j |\sqrt{n} C_{n,j} - C_{\infty,j}| = o_p(1).$$

Then

$$\max_{j,k} |\sqrt{n}(C_{n,j} - C_{n,k}) - (C_{\infty,j} - C_{\infty,k})| \leq 2\varepsilon_n = o_p(1).$$

Hence a candidate whose finite-sample design loss is within $r_n = o_p(n^{-1/2})$ of the minimum has Gaussian design loss within $2\varepsilon_n + \sqrt{n} r_n = o_p(1)$ of the minimum. □

Proof of Theorem 3. For $h := \sqrt{n}(\mu - \mu_0)$, the quasi-Bayesian laws satisfy

$$\begin{aligned} h \mid S_1 &\sim N(\widehat{I}_1^{-1}\widehat{\mathsf{T}}^{\Pi}_{1,n}, \widehat{I}_1^{-1}), \\ \widehat{\mathsf{T}}^{\Pi}_{2,n}(g; E_{n,g}) \mid (h, S_1, g) &\sim N(\widehat{I}_2(g; S_1)h, \widehat{I}_2(g; S_1)). \end{aligned}$$

Thus

$$h \mid (S_1, e, g) \sim N\Big(\widehat{\Sigma}_{+,g}\left\{\widehat{\mathsf{T}}^{\Pi}_{1,n} + \widehat{\mathsf{T}}^{\Pi}_{2,n}(g; e)\right\}, \widehat{\Sigma}_{+,g}\Big). \tag{37}$$

The predictive parameters are

$$\begin{aligned} \widehat{m}_g &:= \widehat{I}_2(g; S_1)\widehat{I}_1^{-1}\widehat{\mathsf{T}}^{\Pi}_{1,n}, \\ \widehat{V}_g &:= \widehat{I}_2(g; S_1) + \widehat{I}_2(g; S_1)\widehat{I}_1^{-1}\widehat{I}_2(g; S_1), \\ \widehat{\mathsf{T}}^{\Pi}_{2,n}(g; E_{n,g}) \mid S_1 &\sim N(\widehat{m}_g, \widehat{V}_g). \end{aligned}$$

These are $\pi_{\infty,+}$ and $\pi^{\mathrm{pred}}_{\infty,g}$ of Subsection 3.4 with t_1, t_2, I_1, and $I_2(g)$ replaced by $\widehat{\mathsf{T}}^{\Pi}_{1,n}$, $\widehat{\mathsf{T}}^{\Pi}_{2,n}(g; e)$, $\widehat{I}_1$, and $\widehat{I}_2(g; S_1)$.

Part (i). Take $\mathsf{T}^{\Pi}_{2,n}(g; e) := \widehat{\mathsf{T}}^{\Pi}_{2,n}(g; e)$ in Lemma 6. Using the same standard normal vector for both laws gives

$$W_1(N(m, V), N(m', V')) \le \|m - m'\| + \|V^{1/2} - (V')^{1/2}\|_F.$$

Let (m_g, V_g) be the parameters of $\pi^{\mathrm{pred}}_{\infty,g}(\cdot \mid \mathsf{T}_{1,n})$, and let $(\widehat{m}_{+,g}(t_2), \widehat{V}_{+,g})$ and $(m_{+,g}(t_2), V_{+,g})$ be the parameters of (37) and $\pi_{\infty,+}(\cdot \mid \mathsf{T}_{1,n}, t_2, g)$. Assumption 5(i) and continuity of the matrix square root give, uniformly over candidates,

$$\begin{aligned} \|\widehat{m}_g - m_g\| + \|\widehat{V}_g^{1/2} - V_g^{1/2}\|_F &= o_p(1), \\ \|\widehat{m}_{+,g}(t_2) - m_{+,g}(t_2)\| &\le o_p(1)(1 + \|t_2\|), \\ \|\widehat{V}_{+,g}^{1/2} - V_{+,g}^{1/2}\|_F &= o_p(1), \\ \int \|t_2\|\, N(dt_2; \widehat{m}_g, \widehat{V}_g) &= O_p(1). \end{aligned}$$

These bounds verify condition (i) of the lemma. Moreover,

$$\int \left\|\sqrt{n}(\mu - \mu_0)\right\|^2 \Pi_1(d\mu \mid S_1) = \left\|\widehat{I}_1^{-1}\widehat{\mathsf{T}}^{\Pi}_{1,n}\right\|^2 + \mathrm{tr}(\widehat{I}_1^{-1}) = O_p(1),$$

verifying condition (ii) with $\eta = 1$. The lemma gives part (i).

Part (ii). Fix a local direction h. Since there are no first-order ties in the design choice at $\sigma = \infty$, the minimising design is almost surely unique under the null Gaussian law $N(0, I_1)$

of T_1.

Work first under $Q_{n,0}^{\gamma_n^C}$. Assumption 4 gives joint convergence of the $\mathsf{T}_{1,n}$ and the candidate list, with $\mathsf{T}_1 \sim N(0, I_1)$, and part (i) supplies the design-loss expansion. Continuity of I_2, the W_1 bound in part (i), and the Lipschitz bound on ℓ make $C_\infty(t_1, g)$ continuous in (t_1, g). After subtracting candidate 1, this continuity and the design-loss expansion give

$$\big(\sqrt{n}\,[C_n(S_1, g_{j,n}(S_1)) - C_n(S_1, g_{1,n}(S_1))]\big)_{j=1}^J \Rightarrow \big(C_\infty\big(\mathsf{T}_1, g_j(\mathsf{T}_1)\big) - C_\infty\big(\mathsf{T}_1, g_1(\mathsf{T}_1)\big)\big)_{j=1}^J .$$

The map from a finite candidate list and its design losses to the selected design is continuous whenever all minimising labels represent the same design. Joint convergence of the candidate list and design losses, uniqueness of the limiting minimising design, and continuity of I_2 therefore give $(\mathsf{T}_{1,n}, G_n, I_2(G_n)) \Rightarrow (\mathsf{T}_1, G_\infty, I_2(G_\infty))$, where $G_\infty = \gamma_\infty^*(\mathsf{T}_1)$. From this point, the argument in part (iii) of the proof of Theorem 2, with G_γ replaced by G_∞, gives $\big(\mathsf{T}_{1,n}, G_n, \mathsf{T}_{2,n}(G_n)\big) \Rightarrow \big(\mathsf{T}_1, G_\infty, \mathsf{T}_{2,G_\infty}\big)$ under $Q_{n,h}^{\gamma_n^C}$, where $\mathsf{T}_1 \sim N(I_1 h, I_1)$ and $\mathsf{T}_{2,G_\infty} \mid (\mathsf{T}_1, G_\infty) \sim N\big(I_2(G_\infty)h, I_2(G_\infty)\big)$. By contiguity (Lemma 4), Assumption 5(ii) also holds under $Q_{n,h}^{\gamma_n^C}$, so $\widehat{\mathsf{T}}_{2,n}^{\Pi}(G_n; E_{n,G_n})$ has the same limit as $\mathsf{T}_{2,n}(G_n)$.

It remains to identify the terminal policy. Let $\widehat{\pi}_n$ be the realised posterior law of $u = \sqrt{n}(\mu - \mu_0)$, and write $\pi_n := \pi_{\infty,+}(\cdot \mid \mathsf{T}_{1,n}, \mathsf{T}_{2,n}(G_n), G_n)$. The Gaussian parameters of $\widehat{\pi}_n$ are given by (37), evaluated at $g = G_n$ and $e = E_{n,G_n}$. Assumption 5, contiguity, the preceding W_1 bound, and Gaussian second moments give $W_1(\widehat{\pi}_n, \pi_n) = o_p(1)$ and $\int \|u\|^2\, \widehat{\pi}_n(du) = O_p(1)$. By (20) and (36) with $\eta = 1$,

$$\begin{aligned}
&\max_{z \in \mathcal{Z}_0} \left| \int f_{n,z}\, d\widehat{\pi}_n - \int \ell(z, \cdot)\, d\pi_n \right| \\
&\quad \le \frac{2L}{\epsilon\sqrt{n}} \int \|u\|^2\, \widehat{\pi}_n(du) + L W_1(\widehat{\pi}_n, \pi_n) = o_p(1).
\end{aligned}$$

Posterior expectations of ℓ are continuous in the Gaussian parameters. The continuous mapping theorem therefore gives, jointly with $\mathsf{T}_{1,n}$, $\mathsf{T}_{2,n}(G_n)$, and the selected design, and over $z \in \mathcal{Z}_0$,

$$\begin{aligned}
&\sqrt{n} \int \{R(z, \mu) - R^*(\mu)\} \Pi_{2,G_n}(d\mu \mid S_1, E_{n,G_n}, G_n) \\
&\qquad \Rightarrow \int \ell(z, u) \pi_{\infty,+}\left(du \mid \mathsf{T}_1, \mathsf{T}_{2,G_\infty}, G_\infty\right).
\end{aligned}$$

Fix $z_0 \in \mathcal{Z}_0$. Since $\int \|u\|\, \widehat{\pi}_n(du) = O_p(1)$, global Lipschitz continuity gives, for every $z \notin \mathcal{Z}_0$,

$$\begin{aligned}
&\int \left[R\left(z, \mu_0 + \frac{u}{\sqrt{n}}\right) - R\left(z_0, \mu_0 + \frac{u}{\sqrt{n}}\right) \right] \widehat{\pi}_n(du) \\
&\qquad = R(z, \mu_0) - R(z_0, \mu_0) + O_p(n^{-1/2}).
\end{aligned}$$

The leading term is strictly positive, so finiteness of the terminal menu excludes all policies outside $\mathcal{Z}_0$ with probability tending to one. Since there are no first-order ties in the policy choice at $\sigma = \infty$, continuity of the argmin map on the finite menu therefore yields the joint convergence $\big(G_n, \delta_n^C(S_1, S_2(G_n))\big) \Rightarrow (G_\infty, \delta_\infty^*(\mathsf{T}_1, \mathsf{T}_{2,G_\infty}, G_\infty))$. This proves part (ii). □

S4.2 Implementation with a bootstrap pilot distribution

This subsection uses a bootstrap distribution as the pilot law Π_1 in Definition 13. For each candidate, a Gaussian approximation specifies the conditional distribution of a second-wave statistic. Averaging over Π_1 gives its predictive distribution, and its likelihood updates Π_1. The resulting full design loss C_n has the same first-order approximation as in the quasi-Bayesian construction.

Construction. Let $\Pi_1(d\mu \mid S_1)$ be a bootstrap distribution on $\mathbb{R}^{d_\mu}$ with finite first moment, based on the efficient estimator $\widehat{\mu}_1$. Set $u_n(\mu) := \sqrt{n}(\mu - \widehat{\mu}_1)$. To compute candidate g's design loss, draw μ from Π_1 and generate $\mathsf{Y}_g = \widehat{I}_2(g; S_1)^{1/2} u_n(\mu) + \varepsilon$, with $\varepsilon \sim N(0, \mathrm{Id}_{d_\mu})$ independent of the bootstrap draw. Conditional on (μ, S_1, g), the density of Y_g is $q_{n,g}(y \mid \mu, S_1) := \varphi_1(y - \widehat{I}_2(g; S_1)^{1/2} u_n(\mu))$, where φ_1 is the $N(0, \mathrm{Id}_{d_\mu})$ density. Define prediction and updating by

$$\begin{aligned} \Pi_{n,g}^{\mathrm{pred}}(dy \mid S_1) &:= \left[\int q_{n,g}(y \mid \mu, S_1)\Pi_1(d\mu \mid S_1)\right] dy, \\ \Pi_{2,g}(d\mu \mid S_1, y, g) &:= \frac{q_{n,g}(y \mid \mu, S_1)\Pi_1(d\mu \mid S_1)}{\displaystyle\int q_{n,g}(y \mid \widetilde{\mu}, S_1)\Pi_1(d\widetilde{\mu} \mid S_1)}. \end{aligned} \tag{38}$$

Integrating the update against the predictive law recovers Π_1, as required by Definition 13. Substituting these laws into (34) gives C_n, with the same terminal tie-breaking convention.

The update depends on y only through $\widehat{I}_2(g; S_1)^{1/2} y$. For the observed second-wave sample, replace this statistic by $\widehat{I}_2(g; S_1)\sqrt{n}\{\widehat{\mu}_{2,g}(S_2(g); S_1) - \widehat{\mu}_1\}$. Equivalently, at the second-wave estimate $e = \widehat{\mu}_{2,g}(S_2(g); S_1)$, weight the pilot distribution by

$$\exp\Big(-\frac{n}{2}(e - \mu)^\top \widehat{I}_2(g; S_1)(e - \mu)\Big).$$

Proposition 6 (Bootstrap approximation of diffuse Gaussian design losses)**.** *Suppose Assumptions 2–4 hold. Let $\Pi_1^{u,n}$ be the conditional law of $u_n(\mu)$ under Π_1. Under $Q_{n,0}^O$, assume*

$$\begin{aligned} \sqrt{n}(\widehat{\mu}_1 - \mu_0) &= I_1^{-1}\mathsf{T}_{1,n} + o_p(1), \\ W_1\big(\Pi_1^{u,n}, N(0, I_1^{-1})\big) \xrightarrow{p} 0, &\qquad \int \|u\|^{1+\eta}\Pi_1^{u,n}(du) = O_p(1) \end{aligned} \tag{39}$$

for some $\eta > 0$, and

$$\max_{1\le j\le J} \left\| \widehat{I}_2(g_{j,n}(S_1); S_1) - I_2(g_{j,n}(S_1)) \right\| \xrightarrow{p} 0. \tag{40}$$

Then

$$\max_{1\le j\le J} \left| \sqrt{n} C_n(S_1, g_{j,n}(S_1)) - C_\infty(\mathsf{T}_{1,n}, g_{j,n}(S_1)) \right| \xrightarrow{p} 0. \tag{41}$$

Consequently, every pilot-measurable $o_p(n^{-1/2})$-minimiser of C_n over the candidate list is an $o_p(1)$-minimiser of the corresponding diffuse Gaussian design losses.

Condition (40) is the last part of Assumption 5(i). Convergence of realised terminal choices additionally requires the analogue of Assumption 5(ii) for the design selected by the bootstrap protocol, together with no first-order ties at $\sigma = \infty$.

Proof. Write $a_n := \sqrt{n}(\widehat{\mu}_1 - \mu_0)$ and $c_n := I_1^{-1}\mathsf{T}_{1,n}$. By (39),

$$\begin{aligned} M_n &:= \int \|\sqrt{n}(\mu - \mu_0)\|^{1+\eta}\, \Pi_1(d\mu \mid S_1) \\ &\le 2^\eta \left\{ \|a_n\|^{1+\eta} + \int \|u\|^{1+\eta} \Pi_1^{u,n}(du) \right\} = O_p(1). \end{aligned}$$

Cancelling the predictive density in (38) gives

$$\begin{aligned} C_n(S_1, g) = \int_{\mathbb{R}^{d_\mu}} \min_{z\in\mathcal{Z}^T} \Bigg\{ \int \{R(z,\mu) - R^*(\mu)\} \\ \times q_{n,g}(y \mid \mu, S_1) \Pi_1(d\mu \mid S_1) \Bigg\} dy. \end{aligned} \tag{42}$$

Let C_n^0 denote (42) with $\mathcal{Z}^T$ replaced by $\mathcal{Z}_0$. The coherence argument in Lemma 6 gives, uniformly over candidates,

$$0 \le \sqrt{n}(C_n^0 - C_n) \le C_r n^{-\eta/2} M_n = o_p(1).$$

Since $\int q_{n,g}(y \mid \mu, S_1)\,dy = 1$, (20) gives

$$\begin{aligned}
&\int \max_{z\in\mathcal{Z}_0} \int \bigl|f_{n,z}(\sqrt{n}(\mu-\mu_0)) - \ell(z,\sqrt{n}(\mu-\mu_0))\bigr| \\
&\qquad\qquad \times q_{n,g}(y \mid \mu, S_1)\,\Pi_1(d\mu \mid S_1)\,dy \\
&\quad \le \frac{2LM_n}{(\epsilon\sqrt{n})^{\eta}} = o_p(1), \\
&\max_{z\in\mathcal{Z}_0} \int |\ell(z,a_n+u) - \ell(z,c_n+u)|\,\Pi_1^{u,n}(du) \le L\|a_n - c_n\| = o_p(1).
\end{aligned}$$

Consequently,

$$\begin{aligned}
\sqrt{n}C_n(S_1,g) = \int \min_{z\in\mathcal{Z}_0} \Bigg\{ \int \ell\bigl(z, I_1^{-1}\mathsf{T}_{1,n} + u\bigr) \\
\times \varphi_1(y - \widehat{I}_2(g;S_1)^{1/2}u)\Pi_1^{u,n}(du) \Bigg\} dy + o_p(1).
\end{aligned}$$

By (39), choose a conditional coupling

$$U_n \sim \Pi_1^{u,n}, \qquad V \sim N(0, I_1^{-1}), \qquad d_n := \mathbb{E}(\|U_n - V\| \mid S_1) = o_p(1).$$

Write $\widehat{B}_g := \widehat{I}_2(g;S_1)^{1/2}$ and $B_g := I_2(g)^{1/2}$. By (40),

$$b_n := \max_j \|\widehat{B}_{g_{j,n}} - B_{g_{j,n}}\| = o_p(1), \qquad \max_j \|\widehat{B}_{g_{j,n}}\| = O_p(1).$$

Since $\int |\varphi_1(y-a) - \varphi_1(y-b)|\,dy \le \min\{2, \|a-b\|\}$ and $|\ell(z,h)| \le L\|h\|$, for every $z \in \mathcal{Z}_0$,

$$\begin{aligned}
&\int \Bigl|\mathbb{E}\bigl\{\ell(z,c_n+U_n)\varphi_1(y - \widehat{B}_g U_n) - \ell(z,c_n+V)\varphi_1(y - B_g V) \bigm| S_1\bigr\}\Bigr|\,dy \\
&\quad \le L\,\mathbb{E}(\|U_n - V\| \mid S_1) + L\,\mathbb{E}\Bigl[\|c_n + V\| \min\bigl\{2, \|\widehat{B}_g\|\|U_n - V\| + \|\widehat{B}_g - B_g\|\|V\|\bigr\} \Bigm| S_1\Bigr].
\end{aligned}$$

By Cauchy–Schwarz and $\min\{2,x\}^2 \le 2x$, this is bounded by

$$Ld_n + L\left[2\mathbb{E}(\|c_n+V\|^2 \mid S_1)\{\|\widehat{B}_g\|d_n + b_n\mathbb{E}\|V\|\}\right]^{1/2} = o_p(1),$$

uniformly over candidates. Thus $|\min_z b_z - \min_z c_z| \le \max_z |b_z - c_z|$ allows replacement of $(\Pi_1^{u,n}, \widehat{B}_g)$ by $(N(0, I_1^{-1}), B_g)$.

Under the resulting Gaussian law, let $\varepsilon \sim N(0, \mathrm{Id}_{d_\mu})$ be independent of V. Then

$$\begin{aligned} H &:= c_n + V \sim \pi_{\infty,1}(\cdot \mid \mathsf{T}_{1,n}), \\ Y &:= B_g V + \varepsilon, \qquad \mathsf{T}_{2,g} := B_g Y + I_2(g) c_n, \\ \mathsf{T}_{2,g} \mid (H, S_1, g) &\sim N(I_2(g) H, I_2(g)), \\ \mathcal{L}(H \mid Y, S_1, g) &= \mathcal{L}(H \mid \mathsf{T}_{2,g}, S_1, g) = \pi_{\infty,+}(\cdot \mid \mathsf{T}_{1,n}, \mathsf{T}_{2,g}, g). \end{aligned}$$

The first equality in the last line holds because the posterior depends on Y only through $I_2(g)^{1/2} Y$, so $\mathsf{T}_{2,g}$ is sufficient for H. Since the marginal law of $\mathsf{T}_{2,g}$ is $\pi^{\mathrm{pred}}_{\infty,g}(\cdot \mid \mathsf{T}_{1,n})$, the joint integral equals $C_\infty(\mathsf{T}_{1,n}, g)$, proving (41). The conclusion about approximate minimisers follows from the same bound as in Lemma 6. □

Computation. Generate bootstrap draws $\mu^{(1)}, \ldots, \mu^{(B)}$ from $\Pi_1(\cdot \mid S_1)$, and precompute $R(z, \mu^{(b)})$ for each policy and draw. For each candidate g, select a draw index b_0 uniformly from $\{1, \ldots, B\}$ and generate $y = \widehat{I}_2(g; S_1)^{1/2} u_n(\mu^{(b_0)}) + \varepsilon$. Assign the bootstrap draws normalised weights proportional to $\exp(-\|y - \widehat{I}_2(g; S_1)^{1/2} u_n(\mu^{(b)})\|^2/2)$, and average the minimum weighted excess regret over the generated values of y. Reusing the same bootstrap draws for prediction and updating gives empirical laws that satisfy Definition 13 and avoids new regret evaluations for each y. Proposition 6 still applies if the error from using finitely many draws and from approximating the outer integral is $o_p(n^{-1/2})$, uniformly over candidates. Alternatively, one may take Π_1 to be the empirical distribution of the draws, if it satisfies the conditions of the proposition; then only the error from the outer integral needs to be $o_p(n^{-1/2})$.

S4.3 Stratum–arm mean implementation

S4.3.1 Construction

Write $A_i := (D_i, Y_i D_i, Y_i(1 - D_i))^\top$, and mark second-wave observations with a superscript E. For $(x, z) \in \mathcal{X} \times \mathcal{Z}^E$, let $N_{xz} := \sum_{i=1}^n \mathbf{1}\{X_i = x, Z_i = z\}$ be the pilot count and, for a design g, let $N_{xz,g} := \sum_{i=1}^{n_2(g)} \mathbf{1}\{X_i^E = x, Z_i^E = z\}$ be the second-wave count.

Pilot wave. The pilot conditional-mean estimate for (x, z) is

$$\widehat{m}^O_{xz} := \frac{1}{N_{xz}} \sum_{i=1}^n \mathbf{1}\{X_i = x, Z_i = z\} A_i, \tag{43}$$

and $\widehat{\Sigma}_{xz}$ is the sample covariance matrix of the A_i in the same group, with divisor N_{xz}. Stack the $\widehat{m}^O_{xz}$ to obtain $\widehat{\mu}_1$, and set $\widehat{\Sigma}_1(S_1) := \text{blockdiag}_{(x,z)\in\mathcal{X}\times\mathcal{Z}^E}(n\widehat{\Sigma}_{xz}/N_{xz})$.[28]

Second wave. For a design g, recall that $q_x(g)$ and $\pi_{zx}(g)$ are its stratum sampling and within-stratum assignment probabilities. Assumption 1(iii) supplies randomised assignment. Let $\nu_{xz}(g) := n_2(g)q_x(g)\pi_{zx}(g)$ be the expected count of group (x,z). If $N_{xz,g} \geq 1$, set

$$\widehat{m}_{xz,g} := \frac{1}{N_{xz,g}} \sum_{i=1}^{n_2(g)} \mathbf{1}\{X_i^E = x, Z_i^E = z\} A_i^E; \tag{44}$$

otherwise set $\widehat{m}_{xz,g} := \widehat{m}^O_{xz}$. Stack these means to obtain $\widehat{\mu}_{2,g}$. The (x,z)-block of $\widehat{I}_2(g;S_1)$ is $\widehat{I}_{2,xz}(g;S_1) := n^{-1}\nu_{xz}(g)\widehat{\Sigma}_{xz}^{-1}$.

S4.3.2 Assumption 5 for the stratum–arm means

Proposition 7 (Quasi-Bayesian decision procedure with stratum–arm means)**.** *Suppose Assumptions 2–4 hold and that, in both waves, the sample mean of $(D, YD, Y(1-D))$ in each stratum–arm group is an efficient estimator of the group's moments. If the quasi-Bayesian procedure uses the stratum–arm means and covariances constructed above, then Assumption 5 holds and $(\gamma_n^C, \delta_n^C) \in \mathbb{D}(P_0)$.*

Proof. Efficiency identifies the scores with those in Lemma 5. In particular,

$$p_{xz} := \Pr_{Q^O_{P_0}}(X = x, Z = z),$$
$$I_1 = \text{blockdiag}_{(x,z)}(p_{xz}\Sigma_{xz}^{-1}) \succ 0 \quad \Longrightarrow \quad \min_{x,z} p_{xz} > 0.$$

Assumption 5(i). On $\mathcal{E}_{1,n}$, (43) and Lemma 5 give, for each (x,z),

$$\widehat{I}_{1,xz} = \frac{N_{xz}}{n}\widehat{\Sigma}_{xz}^{-1}, \qquad \widehat{\mathsf{T}}^{\Pi}_{1,n,xz} = \widehat{\Sigma}_{xz}^{-1}\Sigma_{xz}\mathsf{T}_{1,n,xz}.$$

The law of large numbers and Lemma 2(i) give

$$N_{xz}/n \xrightarrow{p} p_{xz}, \qquad \widehat{\Sigma}_{xz} \xrightarrow{p} \Sigma_{xz} \succ 0, \qquad \Pr(\mathcal{E}_{1,n}) \to 1, \qquad \mathsf{T}_{1,n} = O_p(1),$$
$$\widehat{\mathsf{T}}^{\Pi}_{1,n} - \mathsf{T}_{1,n} = o_p(1), \qquad \widehat{I}_1 - I_1 = o_p(1),$$
$$\sqrt{n}(\widehat{\mu}_1 - \mu_0) = I_1^{-1}\mathsf{T}_{1,n} + o_p(1), \qquad \widehat{\Sigma}_1 \xrightarrow{p} I_1^{-1}.$$

[28] Let $\mathcal{E}_{1,n}$ be the event that every N_{xz} is positive and every $\widehat{\Sigma}_{xz}$ is positive definite. Off $\mathcal{E}_{1,n}$, set $\widehat{m}^O_{xz} = 0_3$ and $\widehat{\Sigma}_{xz} = \text{Id}_3$ for every (x,z), and $\widehat{\Sigma}_1 = \text{Id}_{d_\mu}$. The second-wave calculations use these same pilot estimates.

Contiguity extends the expansion to every fixed local alternative, and Lemma 5 then gives regularity and efficiency.

For the second wave, uniformly over g,

$$\|\widehat{I}_{2,xz}(g;S_1) - I_{2,xz}(g)\| \le \bar{\rho}\|\widehat{\Sigma}_{xz}^{-1} - \Sigma_{xz}^{-1}\| + n^{-1}\|\widehat{\Sigma}_{xz}^{-1}\| = o_p(1),$$
$$\max_{j\le J}\|\widehat{I}_2(g_{j,n}(S_1);S_1) - I_2(g_{j,n}(S_1))\| = o_p(1),$$

using $|n_2(g)/n - \rho(g)| \le n^{-1}$ and $\rho(g) \le \bar{\rho}$.

Assumption 5(ii). If $N_{xz,g} \ge 1$, (44) and Lemma 5 give

$$\widehat{\mathsf{T}}^{\Pi}_{2,n,xz}(g;E_{n,g}) = \frac{\nu_{xz}(g)}{N_{xz,g}}\,\widehat{\Sigma}_{xz}^{-1}\Sigma_{xz}\,\mathsf{T}_{2,n,xz}(g).$$

Choose $M_n \to \infty$ with $M_n = o(\sqrt{n})$. Uniformly over the finite candidate list and groups, Chebyshev's inequality and boundedness give, under $Q_{n,0}^{\gamma_n^C}$,

$$\begin{aligned}
\nu_{xz}(g) = 0 \quad &\Longrightarrow \quad \widehat{\mathsf{T}}^{\Pi}_{2,n,xz}(g;E_{n,g}) = \mathsf{T}_{2,n,xz}(g) = 0,\\
\nu_{xz}(g) \ge M_n \quad &\Longrightarrow \quad N_{xz,g}/\nu_{xz}(g) = 1 + o_p(1),\\
0 < \nu_{xz}(g) < M_n \quad &\Longrightarrow \quad \begin{cases} \|\widehat{\mathsf{T}}^{\Pi}_{2,n,xz}(g;E_{n,g})\| \lesssim M_n/\sqrt{n} = o(1),\\ \mathbb{E}[\|\mathsf{T}_{2,n,xz}(g)\|^2 \mid S_1, g] \lesssim M_n/n = o(1).\end{cases}
\end{aligned}$$

The last bound for $\widehat{\mathsf{T}}^{\Pi}_{2,n,xz}$ also holds when $N_{xz,g} = 0$. Together with

$$\max_{x,z}\|\widehat{\Sigma}_{xz}^{-1}\Sigma_{xz} - \mathrm{Id}_3\| = o_p(1), \qquad \mathsf{T}_{2,n}(G_n) = O_p(1)$$

from Assumption 3 and Lemma 1, these bounds imply

$$\widehat{\mathsf{T}}^{\Pi}_{2,n}(G_n;E_{n,G_n}) - \mathsf{T}_{2,n}(G_n) = o_p(1).$$

Probability of selecting outside $\mathcal{Z}_0$. If $\mathcal{Z}^T = \mathcal{Z}_0$, admissibility is immediate. Otherwise, adapt CMS (2026, Appendix D, Lemmas 8 and 12) as follows. Fix $z_0 \in \mathcal{Z}_0$, and choose $L_D \ge 1$ and $\epsilon_* > 0$ such that

$$\Delta_* := \min_{z\notin\mathcal{Z}_0}\{R(z,\mu_0) - R(z_0,\mu_0)\} > 0, \qquad 4L_D\epsilon_* < \Delta_*,$$
$$|\{R(z,\mu) - R(z_0,\mu)\} - \{R(z,\mu_0) - R(z_0,\mu_0)\}| \le L_D\|\mu - \mu_0\|.$$

On $\mathcal{E}_{1,n}$, cancellation of $\widehat{\Sigma}_{xz}^{-1}$ gives

$$\widehat{\mu}_{+,g,xz} = \frac{N_{xz}\widehat{m}_{xz}^{O} + \nu_{xz}(g)\widehat{m}_{xz,g}}{N_{xz} + \nu_{xz}(g)}. \tag{45}$$

Define

$$\mathcal{B}_{1,n} := \mathcal{E}_{1,n}^{c} \cup \{\|\widehat{\mu}_1 - \mu_0\| > \epsilon_*\} \cup \bigcup_{x,z} \{N_{xz} < np_{xz}/2\}.$$

Hoeffding's inequality (Hoeffding, 1963), applied to the frequencies and first two sample moments, gives $r, C, c > 0$ such that

$$\sup_{h \in \mathcal{H}_n,\ \|h\| \le r\sqrt{n}} Q_{n,h}^{O}(\mathcal{B}_{1,n}) \le Ce^{-cn}. \tag{46}$$

This uses $p_{xz} > 0$ and the fact that, by Assumption 3 and boundedness, the conditional covariance matrices along the path converge to $\Sigma_{xz} \succ 0$ as $t \to \mu_0$.

Fix sufficiently small $\eta, \epsilon \in (0,1)$. Off $\mathcal{B}_{1,n}$, (45) and boundedness give, uniformly over designs,

$$\begin{aligned} \nu_{xz}(g) < \eta n \quad &\Longrightarrow \quad \|\widehat{\mu}_{+,g,xz} - \widehat{m}_{xz}^{O}\| \le \frac{2\sqrt{3}\eta}{p_{xz}}, \\ \nu_{xz}(g) \ge \eta n \quad &\Longrightarrow \quad \Pr\{N_{xz,g} < \nu_{xz}(g)/2 \mid S_1, g\} \le e^{-\eta n/8}. \end{aligned}$$

Conditional on $(S_1, g, N_{xz,g} = m)$, the m observations have coordinates in $[0,1]$ and mean $\mu_{xz,0} + h_{xz}/\sqrt{n}$. Hoeffding's inequality and a union bound give

$$\Pr\Big\{ \big\|\widehat{m}_{xz,g} - \mu_{xz,0} - h_{xz}/\sqrt{n}\big\| > \epsilon \;\Big|\; S_1, g, N_{xz,g} = m \Big\} \le 6e^{-2m\epsilon^2/3},$$

which is at most $6e^{-\eta\epsilon^2 n/3}$ for $m \ge \nu_{xz}(g)/2$. For fixed $\sigma > 0$ and all large n,

$$\varphi_{n,\sigma} \le 2\varphi_\sigma \text{ on } \mathcal{H}_n, \qquad \int_{\mathcal{H}_n \cap \{\|h\| > r\sqrt{n}\}} \varphi_{n,\sigma}(h)\, dh \le 2^{1+d_\mu/2} e^{-r^2 n/(4\sigma^2)}.$$

Combining these bounds with (46) and a union bound over (x,z), and taking r smaller if necessary, gives $C_1, c_1 > 0$ such that

$$\int_{\mathcal{H}_n} Q_{n,h}^{\gamma_n^C}\Big(\mathcal{B}_{1,n} \cup \big\{\|\widehat{\mu}_{+,G_n}(E_{n,G_n}; S_1) - \mu_0\| > 2\epsilon_*\big\}\Big)\, \varphi_{n,\sigma}(h)\, dh \le C_1 e^{-c_1 n}. \tag{47}$$

Off the event in (47), write

$$\widehat{\Pi}_2 := \Pi_{2,G_n}(\cdot \mid S_1, E_{n,G_n}, G_n) = N(m_n, V_n).$$

Boundedness and $N_{xz} \geq np_{xz}/2$ give

$$\|m_n - \mu_0\| \leq 2\epsilon_*, \qquad V_n \preceq Cn^{-1}\mathrm{Id}_{d_\mu}, \qquad \int \|\mu - \mu_0\| \, \widehat{\Pi}_2(d\mu) \leq 2\epsilon_* + \sqrt{Cd_\mu/n}.$$

Hence, for every $z \notin \mathcal{Z}_0$,

$$\int \{R(z,\mu) - R(z_0,\mu)\} \, \widehat{\Pi}_2(d\mu) \geq \Delta_* - L_D\{2\epsilon_* + \sqrt{Cd_\mu/n}\} > 0$$

for all large n, so no policy outside $\mathcal{Z}_0$ is selected. By (47), for every fixed $\sigma > 0$,

$$\sqrt{n} \int_{\mathcal{H}_n} \Pr_{Q_{n,h}^{\gamma_n^C}} \{\delta_n^C(S_1, S_2(G_n)) \notin \mathcal{Z}_0\} \, \varphi_{n,\sigma}(h) \, dh \leq C_1 \sqrt{n} e^{-c_1 n} \longrightarrow 0.$$

Since $\gamma_n^C(s_1) \in \mathcal{G}_{n,J}(s_1)$ by construction, $(\gamma_n^C, \delta_n^C) \in \mathbb{D}(P_0)$. □

S4.4 Proof of Theorem 4

Proof. By Proposition 1 and its proof, $\ell(z,\cdot)$, $z \in \mathcal{Z}_0$, is continuous, positively homogeneous, and piecewise linear. Define

$$\begin{aligned}
\mathscr{A}_0 &:= \{\mathcal{A} \in \mathscr{A} : \mu_0 \in \mathcal{A}\}, \\
\beta_{z,\mathcal{A}} &:= \nabla_\mu\{R(z,\mu) - R^*(\mu)\}, \qquad \mu \in \mathrm{int}(\mathcal{A}), \\
T_{\mathcal{A}} &:= \{h : \exists r > 0, \ \mu_0 + th \in \mathcal{A} \ \forall t \in [0,r]\}, \qquad \mathcal{A} \in \mathscr{A}_0.
\end{aligned}$$

For $z \in \mathcal{Z}_0$, affinity and zero excess regret at μ_0 give

$$\begin{aligned}
R(z,\mu) - R^*(\mu) &= \beta_{z,\mathcal{A}}^\top(\mu - \mu_0), \quad \mu \in \mathcal{A}, \quad \mathcal{A} \in \mathscr{A}_0, \\
\ell(z,h) &= \beta_{z,\mathcal{A}}^\top h, \qquad\qquad h \in T_{\mathcal{A}}.
\end{aligned}$$

Since $\mu_0 \in \mathrm{int}(\mathcal{M})$, $\bigcup_{\mathcal{A} \in \mathscr{A}_0} T_{\mathcal{A}} = \mathbb{R}^{d_\mu}$, and each $T_{\mathcal{A}}$ is full-dimensional. At differentiable boundary points, adjacent gradients coincide. Hence

$$\mathcal{S}_\ell(P_0) = \mathrm{span}\{\nabla_h \ell(z,h) : z \in \mathcal{Z}_0, \ \ell(z,\cdot) \text{ is differentiable at } h\},$$

which is independent of $\mathscr{A}$.

For $v \in \ker \Psi$ and $z \in \mathcal{Z}_0$, continuity and piecewise linearity give

$$\beta_{z,\mathcal{A}}^\top v = 0 \quad (\mathcal{A} \in \mathscr{A}_0) \quad \Longrightarrow \quad \ell(z, h + tv) = \ell(z,h) \quad (t \in \mathbb{R}).$$

Taking $v = \Psi^\top \Psi h - h \in \ker \Psi$ gives

$$\ell(z, h) = \ell(z, \Psi^\top \Psi h). \tag{48}$$

If $r_\ell = 0$, then $\ell(z, h) = \ell(z, 0) = 0$ and $C_\sigma(t_1, g) = 0$. Assume $r_\ell > 0$.

Fix (σ, t_1, g). The posterior mean $\bar{H}_g$ below is affine in $\mathsf{T}_{2,g}$, and the posterior covariance $V_{\sigma,+}(g)$ does not depend on $\mathsf{T}_{2,g}$. Gaussian updating and total variance therefore give

$$\begin{aligned} \bar{H}_g &:= V_{\sigma,+}(g)(t_1 + \mathsf{T}_{2,g}), \\ \bar{H}_g \mid (\mathsf{T}_1 = t_1, g) &\sim N(V_{\sigma,1} t_1, V_{\sigma,1} - V_{\sigma,+}(g)), \\ H \mid (\bar{H}_g, t_1, g) &\sim N(\bar{H}_g, V_{\sigma,+}(g)). \end{aligned}$$

Projecting by Ψ and using (48) yields

$$\begin{aligned} C_\sigma(t_1, g) = \int \Big[\min_{z \in \mathcal{Z}_0} \int \ell(z, \Psi^\top u)\, N_{r_\ell}(du; m, \Sigma_{\ell,\sigma}(g)) \Big] \\ \times N_{r_\ell}\big(dm; \Psi V_{\sigma,1} t_1, \Psi V_{\sigma,1} \Psi^\top - \Sigma_{\ell,\sigma}(g)\big). \end{aligned} \tag{49}$$

This representation uses only r_ℓ-dimensional Gaussian laws, and all dependence on g is through $\Sigma_{\ell,\sigma}(g)$.

For the comparison, write

$$\Sigma_k := \Sigma_{\ell,\sigma}(g_k), \qquad A := \Psi V_{\sigma,1} \Psi^\top, \qquad m_0 := \Psi V_{\sigma,1} t_1,$$

and suppose $\Sigma_2 \succeq \Sigma_1$. Define the Gaussian coupling

$$\begin{aligned} M_2 &\sim N_{r_\ell}(m_0, A - \Sigma_2), \\ M_1 \mid M_2 &\sim N_{r_\ell}(M_2, \Sigma_2 - \Sigma_1), \\ U \mid (M_1, M_2) &\sim N_{r_\ell}(M_1, \Sigma_1). \end{aligned}$$

Then

$$M_k \sim N_{r_\ell}(m_0, A - \Sigma_k), \qquad U \mid M_k \sim N_{r_\ell}(M_k, \Sigma_k), \qquad k = 1, 2.$$

Thus $N_{r_\ell}(M_2, \Sigma_2)$ is a mixture of $N_{r_\ell}(M_1, \Sigma_1)$ over the conditional law of M_1 given M_2. With $F_k(m) := \min_{z \in \mathcal{Z}_0} \int \ell(z, \Psi^\top u)\, N_{r_\ell}(du; m, \Sigma_k)$,

$$F_2(M_2) = \min_{z \in \mathcal{Z}_0} \mathbb{E}\Big[\int \ell(z, \Psi^\top u)\, N_{r_\ell}(du; M_1, \Sigma_1) \;\Big|\; M_2 \Big] \geq \mathbb{E}\{F_1(M_1) \mid M_2\},$$

hence

$$C_\sigma(t_1, g_2) = \mathbb{E}F_2(M_2) \geq \mathbb{E}F_1(M_1) = C_\sigma(t_1, g_1).$$

This is the Gaussian case of Blackwell's comparison (Blackwell, 1951, 1953). The substitutions $V_{\sigma,1} = I_1^{-1}$ and $V_{\sigma,+}(g) = I_+(g)^{-1}$ give the conclusions for C_∞. □

S4.5 Proof of Proposition 4

Proof. Use $\beta_{z,\mathcal{A}}$ from the proof of Theorem 4. Define

$$\begin{aligned}
a_n &:= \sqrt{n}(\widehat{\mu}_1 - \mu_0), \\
E_n &:= \{\mu : n(\mu - \widehat{\mu}_1)^\top \widehat{\Sigma}_1^{-1}(\mu - \widehat{\mu}_1) \leq \tau_n^2\}, \\
F_{z,\mathcal{A}} &:= \{\mu \in \mathcal{A} : R(z, \mu) = R^*(\mu)\}, \\
\mathcal{J}_0 &:= \{(z, \mathcal{A}) : z \in \mathcal{Z}_0,\ \mathcal{A} \in \mathscr{A},\ \mu_0 \in \mathcal{A}\}, \\
\widehat{\mathcal{J}}_n &:= \{(z, \mathcal{A}) \in \mathcal{Z}^T \times \mathscr{A} : E_n \cap F_{z,\mathcal{A}} \neq \varnothing\}.
\end{aligned}$$

Thus $\widehat{\mathcal{J}}_n$ is the set of policy–region pairs that RDDR retains in Subsection 4.3. The pilot central limit theorem and Assumption 5(i) give

$$\begin{aligned}
a_n = O_p(1), \qquad \widehat{\Sigma}_1 \xrightarrow{p} I_1^{-1} \succ 0, \\
\Pr(\mu_0 \in E_n) = \Pr(a_n^\top \widehat{\Sigma}_1^{-1} a_n \leq \tau_n^2) \longrightarrow 1, \\
\sup_{\mu \in E_n} \|\mu - \mu_0\| \leq \|\widehat{\mu}_1 - \mu_0\| + \frac{\tau_n \sqrt{\lambda_{\max}(\widehat{\Sigma}_1)}}{\sqrt{n}} = o_p(1).
\end{aligned}$$

Each $F_{z,\mathcal{A}}$ is closed, and

$$\begin{aligned}
(z, \mathcal{A}) \in \mathcal{J}_0 &\iff \mu_0 \in F_{z,\mathcal{A}}, \\
(z, \mathcal{A}) \notin \mathcal{J}_0 &\implies \operatorname{dist}(\mu_0, F_{z,\mathcal{A}}) > 0,
\end{aligned}$$

with $\operatorname{dist}(\mu_0, \varnothing) = \infty$. Since $\mathcal{Z}^T \times \mathscr{A}$ is finite,

$$\begin{aligned}
\Pr\{\widehat{\mathcal{J}}_n = \mathcal{J}_0\} \longrightarrow 1, \\
\Pr\{\widehat{\mathcal{S}}_{\ell,n} = \mathcal{S}_\ell(P_0),\ \widehat{r}_n = r_\ell\} \longrightarrow 1.
\end{aligned}$$

On $\mathcal{R}_n := \{\widehat{\mathcal{S}}_{\ell,n} = \mathcal{S}_\ell(P_0)\}$, define $O_n := \widehat{\Psi}_n \Psi^\top$. Then $O_n^\top O_n = \mathrm{Id}_{r_\ell}$ and $\widehat{\Psi}_n = O_n \Psi$. By (20) and global Lipschitz continuity,

$$\begin{aligned}
\widehat{\ell}_n(z,u) &= f_{n,z}(a_n + \widehat{\Psi}_n^\top u), \\
0 \le \widehat{\ell}_n(z,u) &\le L(\|a_n\| + \|u\|), \\
|\widehat{\ell}_n(z,u) - \ell(z, a_n + \Psi^\top O_n^\top u)| &\le \frac{2L}{\epsilon\sqrt{n}} \|a_n + \widehat{\Psi}_n^\top u\|^2, \qquad z \in \mathcal{Z}_0.
\end{aligned}$$

If $\mathcal{Z}^T \setminus \mathcal{Z}_0 \neq \varnothing$, fix $z_0 \in \mathcal{Z}_0$ and let $\Delta := \min_{z \in \mathcal{Z}^T \setminus \mathcal{Z}_0} \{R(z,\mu_0) - R(z_0,\mu_0)\} > 0$. All expectations below are conditional on S_1, with $(\overline{U}_g, U_g)$ drawn as in Definition 12. Define

$$\begin{aligned}
B_{n,g} &:= \|a_n\| + \mathbb{E}(\|U_g\| \mid \overline{U}_g), \\
\mathbb{E}B_{n,g}^2 &\le 2\|a_n\|^2 + 2\,\mathrm{tr}(\widehat{\Sigma}_1) = O_p(1).
\end{aligned}$$

Increasing L if necessary, global Lipschitz continuity gives

$$\begin{aligned}
\mathbb{E}\{\widehat{\ell}_n(z,U_g) - \widehat{\ell}_n(z_0,U_g) \mid \overline{U}_g\} &\ge \sqrt{n}\Delta - LB_{n,g}, \quad z \notin \mathcal{Z}_0, \\
0 \le \mathbb{E}\{\widehat{\ell}_n(z_0,U_g) \mid \overline{U}_g\} &\le LB_{n,g}.
\end{aligned}$$

Let $\widehat{C}_n^0$ denote the RDDR design loss with the minimum restricted to $\mathcal{Z}_0$. Then, uniformly over candidates,

$$\begin{aligned}
0 \le \widehat{C}_n^0(S_1,g) - \widehat{C}_n(S_1,g) &\le L\mathbb{E}[B_{n,g}\mathbf{1}\{LB_{n,g} \ge \sqrt{n}\Delta\}] \\
&\le \frac{L^2 \mathbb{E}B_{n,g}^2}{\sqrt{n}\Delta} = O_p(n^{-1/2}).
\end{aligned}$$

If $\mathcal{Z}^T = \mathcal{Z}_0$, set $\widehat{C}_n^0 = \widehat{C}_n$. On $\mathcal{R}_n$, the approximation of $\widehat{\ell}_n$ by ℓ above gives

$$\begin{aligned}
&\mathbb{E}\max_{z \in \mathcal{Z}_0} |\widehat{\ell}_n(z,U_g) - \ell(z, a_n + \Psi^\top O_n^\top U_g)| \\
&\quad \le \frac{4L}{\epsilon\sqrt{n}} \{\|a_n\|^2 + \mathrm{tr}(\widehat{\Sigma}_1)\} = O_p(n^{-1/2}).
\end{aligned}$$

For the Gaussian comparison, define

$$\begin{aligned}
\widehat{A}_g &:= \Psi(\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g})\Psi^\top, & A_g &:= \Psi(I_1^{-1} - I_+(g)^{-1})\Psi^\top, \\
\widehat{B}_g &:= \Psi\widehat{\Sigma}_{+,g}\Psi^\top, & B_g &:= \Psi I_+(g)^{-1}\Psi^\top.
\end{aligned}$$

For independent $Z_1, Z_2 \sim N(0, \mathrm{Id}_{r_\ell})$, couple

$$\overline{V}_{n,g} := \widehat{A}_g^{1/2} Z_1, \quad V_{n,g} := \overline{V}_{n,g} + \widehat{B}_g^{1/2} Z_2,$$
$$\overline{V}_g := A_g^{1/2} Z_1, \quad V_g := \overline{V}_g + B_g^{1/2} Z_2.$$

On $\mathcal{R}_n$, $(\overline{V}_{n,g}, V_{n,g}) \overset{d}{=} (O_n^\top \overline{U}_g, O_n^\top U_g)$. Assumption 5(i) and continuity of the matrix square root give

$$c_n := I_1^{-1} \mathsf{T}_{1,n}, \qquad \|a_n - c_n\| = o_p(1),$$
$$\max_j \mathbb{E}\|V_{n,g_{j,n}} - V_{g_{j,n}}\| \le \max_j \big\{ \|\widehat{A}^{1/2}_{g_{j,n}} - A^{1/2}_{g_{j,n}}\|_F + \|\widehat{B}^{1/2}_{g_{j,n}} - B^{1/2}_{g_{j,n}}\|_F \big\} = o_p(1).$$

By (49), projection invariance, and $|\min_z b_z - \min_z c_z| \le \max_z |b_z - c_z|$,

$$\max_j |\widehat{C}_n^0(S_1, g_{j,n}) - C_\infty(\mathsf{T}_{1,n}, g_{j,n})| \le O_p(n^{-1/2}) + L\|a_n - c_n\| + L \max_j \mathbb{E}\|V_{n,g_{j,n}} - V_{g_{j,n}}\| = o_p(1).$$

Since $\Pr(\mathcal{R}_n) \to 1$,

$$\max_{1 \le j,k \le J} \Big| \widehat{C}_n(S_1, g_{j,n}(S_1)) - \widehat{C}_n(S_1, g_{k,n}(S_1)) - [C_\infty(\mathsf{T}_{1,n}, g_{j,n}(S_1)) - C_\infty(\mathsf{T}_{1,n}, g_{k,n}(S_1))] \Big| \xrightarrow{p} 0.$$

Theorem 3(i) and the triangle inequality therefore yield

$$\max_{1 \le j,k \le J} \Big| \widehat{C}_n(S_1, g_{j,n}(S_1)) - \widehat{C}_n(S_1, g_{k,n}(S_1)) - \sqrt{n}\,[C_n(S_1, g_{j,n}(S_1)) - C_n(S_1, g_{k,n}(S_1))] \Big| \xrightarrow{p} 0. \tag{50}$$

Let Δ_n be the left-hand side of (50). Since $\widehat{C}_n(S_1, g_{\widehat{\iota}_{\ell,n},n}(S_1)) \le \widehat{C}_n(S_1, g_{\iota_n^C(S_1),n}(S_1))$ and $\iota_n^C(S_1)$ minimises C_n,

$$0 \le \sqrt{n}\Big[C_n(S_1, g_{\widehat{\iota}_{\ell,n},n}(S_1)) - \min_{1 \le j \le J} C_n(S_1, g_{j,n}(S_1))\Big] \le \Delta_n \xrightarrow{p} 0. \qquad \square$$

S4.6 Policy plug-in design losses

This subsection develops the policy plug-in shortcut mentioned at the end of Subsection 4.3. Regret-directed dimension reduction reduces the dimension of the posterior and predictive integrations in C_n. A further computational saving is possible by reducing the number of policy-specific posterior integrations. For each possible realisation of second-wave evidence, we select the policy that minimises regret evaluated at the updated posterior mean, rather than regret averaged over the updated posterior. To obtain the resulting design loss, we compute posterior expected excess regret only for the selected policy and average over the predictive distribution of second-wave evidence. We call this the policy plug-in design loss $K_n(s_1, g)$.

CMS (2026) show that plug-in policy rules can be asymptotically suboptimal in a static partially identified decision problem. Here the plug-in step is used only to rank experiments, and the terminal policy is still chosen by $\delta^{\Pi}_{n,g}$. Any shortfall therefore comes from selecting a different experiment. We measure it by the full design loss of the selected experiment relative to the minimum, bound it by the cross-candidate variation in the plug-in penalty $K_n - C_n$, and give sufficient conditions for first-order optimality.

More specifically, for a realisation e of $E_{n,g}$ under candidate g, let

$$\overline{\mu}_{n,+}(s_1, e, g) = \int_{\mathbb{R}^{d_\mu}} \mu \, \Pi_{2,g}(d\mu \mid s_1, e, g)$$

be the updated mean.

Definition 14 (Policy plug-in design loss)**.** Define the tie-broken posterior-mean policy

$$z^K_n(s_1, e, g) = \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} R(z, \overline{\mu}_{n,+}(s_1, e, g)).$$

The resulting policy plug-in design loss is

$$K_n(s_1, g) = \int\int_{\mathbb{R}^{d_\mu}} \left[R(z^K_n(s_1, e, g), \mu) - R^*(\mu)\right] \Pi_{2,g}(d\mu \mid s_1, e, g) \Pi^{\text{pred}}_{n,g}(de \mid s_1).$$

Let $\gamma^K_n(s_1)$ minimise $K_n(s_1, \cdot)$ over the candidate list $\mathcal{G}_{n,J}(s_1)$, with ties broken by the smallest candidate index.

At each e, C_n computes posterior expected excess regret for every terminal policy before minimising; K_n selects a policy at the posterior mean and computes that policy's posterior expected excess regret. The two integrands therefore agree at e whenever the selected policy also minimises posterior expected regret. Because policy comparisons depend on

$R(z,\cdot) - R(z',\cdot)$, a nonlinear term common to all policies cancels. If all these differences are affine on the convex hull of the posterior support, the two policy comparisons agree exactly. With nonlinear regret differences, posterior averaging can change the minimising policy. The SNAP comparison in (17) illustrates how uncertainty across the lower-bound kink can produce such a discrepancy in the policy used to compute the design loss.

For each candidate, define $d_n(s_1, g) := K_n(s_1, g) - C_n(s_1, g) \geq 0$.

Suppressing s_1,

$$\begin{aligned} K_n(\gamma_n^K) \leq K_n(\gamma_n^C) &\iff C_n(\gamma_n^K) + d_n(\gamma_n^K) \leq C_n(\gamma_n^C) + d_n(\gamma_n^C), \\ 0 \leq C_n(\gamma_n^K) - \min_g C_n(g) &\leq d_n(\gamma_n^C) - d_n(\gamma_n^K) \\ &\leq \max_g d_n(g) - \min_g d_n(g), \end{aligned}$$

with extrema over $\mathcal{G}_{n,J}(s_1)$.

Only cross-candidate variation in the penalty affects experiment ranking. If this range is $o_p(n^{-1/2})$ under $Q_{n,0}^O$, the K_n-rule is a first-order approximate minimiser of C_n. Under the assumptions of Theorem 3, it then also attains the minimum diffuse Gaussian design loss up to $o_p(1)$, as noted after that theorem.

We next bound this cross-candidate variation in terms of the differences in the posterior covariances $\widehat{\Sigma}_{+,g}$ across candidates.

Proposition 8 (Covariance bound on plug-in penalties)**.** *Fix a pilot realisation s_1, which we suppress in the covariance matrices, and use the quasi-Bayesian implementation in Subsection 4.1. Suppose $\|\widehat{\Sigma}_1\| \leq M$ and $\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g} \succeq \kappa \mathrm{Id}_{d_\mu}$ for every candidate g, with fixed constants $M, \kappa > 0$. If each regret difference $R(z,\cdot) - R(z',\cdot)$ is the sum of an affine function and a L_Δ-Lipschitz function, with a common $L_\Delta < \infty$, then, for any $g, g' \in \mathcal{G}_{n,J}(s_1)$,*

$$|d_n(s_1, g) - d_n(s_1, g')| \leq \frac{cL_\Delta}{\sqrt{n}} \|\widehat{\Sigma}_{+,g}^{1/2} - \widehat{\Sigma}_{+,g'}^{1/2}\|_F, \tag{51}$$

where c depends only on d_μ, M, κ.

The restrictions on $\widehat{\Sigma}_1$ and $\widehat{\Sigma}_{+,g}$ require the pilot to be informative and every candidate design to provide nondegenerate additional information in all directions. Proposition 8.1 in Appendix S4.6.1 below gives a general bound allowing singular cases. The bound combines the sensitivity of the non-affine parts of regret differences between policies, captured by L_Δ, with differences in the posterior uncertainty left by candidate experiments. Candidate designs with similar posterior covariance matrices have similar plug-in penalties, even when the penalties themselves are not small.

Under the stated uniform bounds, the right-hand side of (51) is $O_p(n^{-1/2})$, but need not be $o_p(n^{-1/2})$. Appendix S4.6.2 gives a two-policy Gaussian example using the SNAP expressions for ℓ in (12). Selecting experiments by K_n incurs a positive first-order design-loss gap even though the terminal policy remains posterior optimal.

Application to RDDR. The same plug-in step applies to the RDDR design loss in Definition 12. For $m \in \mathbb{R}^{\widehat{r}_n}$, define

$$\widehat{z}^K_{\ell,n}(m) := \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} \widehat{\ell}_n(z, m).$$

Using the joint Gaussian law in that definition, conditional on the pilot, set

$$\widehat{K}_n(S_1, g) := \mathbb{E}\left[\widehat{\ell}_n\big(\widehat{z}^K_{\ell,n}(\overline{U}_g), U_g\big)\right]. \tag{52}$$

Minimising $\widehat{K}_n$ over the candidate list combines RDDR with the policy plug-in step. The same penalty argument applies in the reduced coordinates; Appendix S4.6.1 gives the corresponding bound on the design-loss gap and conditions for first-order optimality.

Finally, we summarise the computational savings from RDDR and the plug-in step. Consider a direct implementation using nested numerical integration. Let N_e and N_μ denote the total numbers of nodes used for the predictive and posterior integrations, respectively. Let $N_e^{(\widehat{r}_n)}, N_\mu^{(\widehat{r}_n)}$ be their reduced-dimension counterparts. Up to common linear-algebra costs, the numbers of maximum-regret evaluations per candidate are[29]

design loss	maximum-regret evaluations
C_n	$N_e N_\mu \lvert\mathcal{Z}^T\rvert$
K_n	$N_e\{N_\mu + \lvert\mathcal{Z}^T\rvert\}$
$\widehat{C}_n$	$N_e^{(\widehat{r}_n)} N_\mu^{(\widehat{r}_n)} \lvert\mathcal{Z}^T\rvert$
$\widehat{K}_n$	$N_e^{(\widehat{r}_n)}\{N_\mu^{(\widehat{r}_n)} + \lvert\mathcal{Z}^T\rvert\}$.

The plug-in saving is largest when the terminal menu is large or each regret evaluation requires a costly linear programme. In practice, C_n can be computed directly when d_μ is small. When d_μ is large, RDDR provides the main saving, and its threshold should not be too small, since Section 5 shows that small thresholds often discard relevant directions. The plug-in step can then be added, and Proposition 8 indicates that it is most reliable when the candidate designs leave similar posterior covariance matrices.

[29] The counts omit the common R^* term, which does not affect experiment ranking.

S4.6.1 Bounds on policy plug-in penalties and the RDDR design-loss gap

Proposition 8.1 (General Gaussian bound on plug-in penalties)**.** *Fix a pilot realisation s_1, and use the quasi-Bayesian implementation in Subsection 4.1. Suppose each regret difference $R(z,\cdot)-R(z',\cdot)$ is the sum of an affine function and a L_Δ-Lipschitz function, with a common $L_\Delta < \infty$. Then, for any candidates g, g',*

$$\begin{aligned} &|d_n(s_1,g) - d_n(s_1,g')| \\ &\quad \le \frac{L_\Delta}{\sqrt{n}}\Big[\|\widehat{\Sigma}_{+,g}^{1/2} - \widehat{\Sigma}_{+,g'}^{1/2}\|_F + \min\{\sqrt{\operatorname{tr}\widehat{\Sigma}_{+,g}}, \sqrt{\operatorname{tr}\widehat{\Sigma}_{+,g'}}\} \\ &\qquad\qquad \times \mathrm{TV}(N(0,\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g}), N(0,\widehat{\Sigma}_1 - \widehat{\Sigma}_{+,g'}))\Big]. \end{aligned} \tag{53}$$

Covariance arguments s_1 are suppressed. Let $B_n(g,g')$ denote the right-hand side, and set $k = \gamma_n^K(s_1)$. Suppressing s_1,

$$\begin{aligned} 0 &\le C_n(k) - \min_g C_n(g) \\ &\le \max_g \left[B_n(k,g) - \{K_n(g) - K_n(k)\}\right]_+, \end{aligned} \tag{54}$$

where the extrema are over $\mathcal{G}_{n,J}(s_1)$.

For RDDR, let $\widehat{d}_n(s_1,g) := \widehat{K}_n(s_1,g) - \widehat{C}_n(s_1,g) \ge 0$, and let $\widehat{\gamma}_n^K(s_1)$ minimise $\widehat{K}_n(s_1,\cdot)$ over the same candidate list, with fixed tie-breaking. Under the conditions of Proposition 4,

$$\begin{aligned} 0 &\le \sqrt{n}\left[C_n(S_1, \widehat{\gamma}_n^K(S_1)) - \min_{g\in\mathcal{G}_{n,J}(S_1)} C_n(S_1,g)\right] \\ &\le \max_{g\in\mathcal{G}_{n,J}(S_1)} \widehat{d}_n(S_1,g) - \min_{g\in\mathcal{G}_{n,J}(S_1)} \widehat{d}_n(S_1,g) + o_p(1). \end{aligned} \tag{55}$$

Convergence is under the pilot law $Q_{n,0}^O$.

By (50), suppressing S_1,

$$\begin{aligned} 0 &\le \sqrt{n}\{C_n(\widehat{\gamma}_n^K) - \min_g C_n(g)\} \\ &= \widehat{K}_n(\widehat{\gamma}_n^K) - \widehat{K}_n(\gamma_n^C) + \widehat{d}_n(\gamma_n^C) - \widehat{d}_n(\widehat{\gamma}_n^K) + o_p(1) \\ &\le \max_g \widehat{d}_n(g) - \min_g \widehat{d}_n(g) + o_p(1), \end{aligned}$$

which proves (55).

Under the condition on regret differences in Proposition 8.1, the bound (53) holds for $\widehat{d}_n$, with prefactor L_Δ in place of $L_\Delta/\sqrt{n}$ and covariance substitutions $\widehat{\Sigma}_1 \mapsto \widehat{\Psi}_n\widehat{\Sigma}_1\widehat{\Psi}_n^\top$ and $\widehat{\Sigma}_{+,g} \mapsto \widehat{\Psi}_n\widehat{\Sigma}_{+,g}\widehat{\Psi}_n^\top$. The rule that minimises $\widehat{K}_n$ achieves first-order optimality for the full

design loss if the range of $\widehat{d}_n$ over candidate designs is $o_p(1)$.

Proof of Proposition 8.1. For continuous piecewise-affine losses f_z with finitely many pieces, write

$$c_{a,z} := f_a - f_z = A_{a,z} + r_{a,z}, \qquad A_{a,z} \text{ affine}, \quad \mathrm{Lip}(r_{a,z}) \le L_\Delta.$$

Let $Z \sim N(0, \mathrm{Id}_d)$ and

$$0 \preceq \Sigma_g \preceq S, \qquad P_g := N(0, S - \Sigma_g), \qquad \overline{U}_g \sim P_g, \quad Z \perp \overline{U}_g,$$
$$U_g := \overline{U}_g + \Sigma_g^{1/2} Z, \qquad z^K(m) := \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} f_z(m).$$

Let $C(g)$ and $K(g)$ be the full and policy plug-in design losses for these losses, with the same terminal tie-breaking for every candidate. Then

$$D_g(m) := \max_{z \in \mathcal{Z}^T} \mathbb{E} c_{z^K(m),z}(m + \Sigma_g^{1/2} Z) \ge 0,$$
$$d(g) := K(g) - C(g) = \int D_g \, dP_g.$$

For $a = z^K(m)$,

$$c_{a,z}(m) \le 0, \qquad \mathbb{E} A_{a,z}(m + \Sigma_g^{1/2} Z) = A_{a,z}(m),$$
$$\begin{aligned} \mathbb{E} c_{a,z}(m + \Sigma_g^{1/2} Z) &= c_{a,z}(m) + \mathbb{E}\{r_{a,z}(m + \Sigma_g^{1/2} Z) - r_{a,z}(m)\} \\ &\le L_\Delta \mathbb{E} \|\Sigma_g^{1/2} Z\|. \end{aligned}$$

Thus

$$\begin{aligned} 0 \le D_g(m) &\le L_\Delta \, \mathbb{E} \|\Sigma_g^{1/2} Z\| \\ &\le L_\Delta \sqrt{\mathrm{tr}\, \Sigma_g}. \end{aligned} \tag{56}$$

At the same mean m, the plug-in policy $z^K(m)$ is common to all candidates. Using the same Z for g and g' therefore gives

$$\begin{aligned} |D_g(m) - D_{g'}(m)| &\le \max_z \left| \mathbb{E}\{r_{a,z}(m + \Sigma_g^{1/2} Z) - r_{a,z}(m + \Sigma_{g'}^{1/2} Z)\} \right| \\ &\le L_\Delta \, \mathbb{E} \|(\Sigma_g^{1/2} - \Sigma_{g'}^{1/2}) Z\| \\ &\le L_\Delta \|\Sigma_g^{1/2} - \Sigma_{g'}^{1/2}\|_F. \end{aligned} \tag{57}$$

By (56)–(57),

$$|d(g) - d(g')| \le \int |D_g - D_{g'}|\, dP_g + \left| \int D_{g'}\, d(P_g - P_{g'}) \right|$$
$$\le L_\Delta \|\Sigma_g^{1/2} - \Sigma_{g'}^{1/2}\|_F + L_\Delta \sqrt{\operatorname{tr} \Sigma_{g'}}\ \mathrm{TV}(P_g, P_{g'}).$$

Exchange g, g' to replace the trace factor by $\min\{\sqrt{\operatorname{tr} \Sigma_g}, \sqrt{\operatorname{tr} \Sigma_{g'}}\}$.

Full design losses. Take

$$f_z(u) := \sqrt{n}\{R(z, \widehat{\mu}_1 + u/\sqrt{n}) - R^*(\widehat{\mu}_1 + u/\sqrt{n})\},$$
$$(S, \Sigma_g, d(g)) = (\widehat{\Sigma}_1, \widehat{\Sigma}_{+,g}, \sqrt{n}\, d_n(s_1, g)).$$

The common R^* cancels in $f_a - f_z$, and the scaling preserves L_Δ, giving (53).

For $k = \gamma_n^K(s_1)$, suppressing s_1,

$$0 \le C_n(k) - \min_g C_n(g) = \max_g \{d_n(g) - d_n(k) - [K_n(g) - K_n(k)]\}$$
$$\le \max_g [B_n(k, g) - \{K_n(g) - K_n(k)\}]_+,$$

proving (54).

RDDR design losses. Take

$$f_z = \widehat{\ell}_n(z, \cdot), \qquad d(g) = \widehat{d}_n(s_1, g),$$
$$S = \widehat{\Psi}_n \widehat{\Sigma}_1 \widehat{\Psi}_n^\top, \qquad \Sigma_g = \widehat{\Psi}_n \widehat{\Sigma}_{+,g} \widehat{\Psi}_n^\top.$$

For $R(a, \cdot) - R(z, \cdot) = A_{a,z} + r_{a,z}$,

$$f_a(u) - f_z(u) = \sqrt{n}\,(A_{a,z} + r_{a,z})(\widehat{\mu}_1 + \widehat{\Psi}_n^\top u/\sqrt{n}),$$
$$\mathrm{Lip}\Big(u \mapsto \sqrt{n}\, r_{a,z}(\widehat{\mu}_1 + \widehat{\Psi}_n^\top u/\sqrt{n})\Big) \le L_\Delta \|\widehat{\Psi}_n^\top\| \le L_\Delta.$$

The general bound therefore applies with prefactor L_Δ. □

Proof of Proposition 8. Suppress the pilot realisation and set

$$S := \widehat{\Sigma}_1, \qquad \Sigma_g := \widehat{\Sigma}_{+,g}, \qquad A := S - \Sigma_g, \quad B := S - \Sigma_{g'},$$
$$A, B \succeq \kappa \mathrm{Id}_{d_\mu}, \qquad P := N(0, A), \quad Q := N(0, B).$$

Pinsker's inequality (Tsybakov, 2009, Lemma 2.5) and the symmetrized Gaussian KL formula

give

$$\begin{aligned}\mathrm{TV}(P,Q)^2 &\le \tfrac{1}{4}\{\mathrm{KL}(P\|Q)+\mathrm{KL}(Q\|P)\}\\ &= \tfrac{1}{8}\operatorname{tr}(A^{-1}(A-B)B^{-1}(A-B))\\ &= \tfrac{1}{8}\|A^{-1/2}(A-B)B^{-1/2}\|_F^2 \le \frac{\|A-B\|_F^2}{8\kappa^2}.\end{aligned}$$

For $D := \Sigma_g^{1/2} - \Sigma_{g'}^{1/2}$,

$$\begin{aligned}\Sigma_g - \Sigma_{g'} &= \Sigma_g^{1/2} D + D\Sigma_{g'}^{1/2},\\ \|A-B\|_F &\le 2\sqrt{\|S\|}\,\|D\|_F, \qquad \operatorname{tr}\Sigma_g \le \operatorname{tr} S \le d_\mu M.\end{aligned}$$

Substitution in (53) yields

$$\begin{aligned}|d_n(g)-d_n(g')| &\le \frac{L_\Delta}{\sqrt{n}}\left(1+\frac{\sqrt{\|S\|\operatorname{tr} S}}{\sqrt{2}\,\kappa}\right)\|D\|_F\\ &\le \frac{L_\Delta}{\sqrt{n}}\left(1+\frac{M\sqrt{d_\mu}}{\sqrt{2}\,\kappa}\right)\|D\|_F,\end{aligned}$$

proving (51). □

S4.6.2 A Gaussian counterexample

In the coordinates $\Lambda_{\mathrm{SNAP}}(z_b)h = (u_1 - 2u_2, 4u_2)^\top$, write policies $z_b, z^\star$ as $0, 1$. Then (12) becomes

$$\begin{aligned}\ell(0,u) &= [u_1 + 2|u_2|]_+, \qquad \ell(1,u) = [-u_1 - 2|u_2|]_+,\\ \ell(0,u) - \ell(1,u) &= u_1 + 2|u_2|, \qquad L_\Delta = 2.\end{aligned}$$

Consider an auxiliary Gaussian design problem with these functions and with information matrices unrestricted by the SNAP allocation model. Let $U \sim N(0, \mathrm{Id}_2)$ be the pilot posterior, and let idealised experiments g_1, g_2 reveal U_1, U_2, respectively, and let $a := 2\sqrt{2/\pi}$. Under g_1, the posterior comparison selects policy 0 when $U_1 < -a$, and the plug-in comparison at $(U_1, 0)$ selects it when $U_1 < 0$. Under g_2, both select policy 1 almost surely. Gaussian integration gives

$$C(g_1) - C(g_2) = -\phi(a) + a\Phi(-a) < 0, \qquad K(g_1) - K(g_2) = \phi(0) > 0.$$

For finite positive-definite information, take $0 < \varepsilon < 1/2$ and

$$\Sigma_{g_1} := \begin{pmatrix} \varepsilon & 0 \\ 0 & 1-\varepsilon \end{pmatrix}, \qquad \Sigma_{g_2} := \begin{pmatrix} 1-\varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix},$$
$$\mathrm{Id}_2 - \Sigma_e \succ 0, \qquad I_2(e) := \Sigma_e^{-1} - \mathrm{Id}_2 \succ 0, \qquad e \in \{g_1, g_2\}.$$

For pilot posterior $N(m, \mathrm{Id}_2)$, use the coupling

$$\overline{U}_e = m + (\mathrm{Id}_2 - \Sigma_e)^{1/2} Z, \qquad U_e = \overline{U}_e + \Sigma_e^{1/2} Z', \qquad Z, Z' \overset{\text{i.i.d.}}{\sim} N(0, \mathrm{Id}_2),$$

and denote the design losses by $C_\varepsilon(m, e), K_\varepsilon(m, e)$. Since ℓ is Lipschitz and limiting plug-in ties require $Z_1 = 0$ or $Z_2 = 0$, which are null events, dominated convergence gives

$$(C_\varepsilon(m, e), K_\varepsilon(m, e)) \longrightarrow (C(e), K(e)) \quad \text{as } (m, \varepsilon) \to (0, 0),$$
$$\exists\, \varepsilon \in (0, 1/2),\ r, c_0 > 0 : \quad \inf_{\|m\| \le r} \{C_\varepsilon(m, g_2) - C_\varepsilon(m, g_1)\} \ge c_0,$$
$$\inf_{\|m\| \le r} \{K_\varepsilon(m, g_1) - K_\varepsilon(m, g_2)\} > 0.$$

Fix this ε. With $\widehat{\mu}_1 \sim N(0, \mathrm{Id}_2/n)$, a flat prior, and independent second-wave information $nI_2(e)$, positive homogeneity gives

$$\sqrt{n}\, C_n(m/\sqrt{n}, e) = C_\varepsilon(m, e), \qquad \sqrt{n}\, K_n(m/\sqrt{n}, e) = K_\varepsilon(m, e),$$
$$\|\sqrt{n}\widehat{\mu}_1\| \le r \implies \gamma_n^K = g_2, \quad \sqrt{n}\{C_n(\widehat{\mu}_1, g_2) - C_n(\widehat{\mu}_1, g_1)\} \ge c_0.$$

Consequently,

$$\liminf_{n\to\infty} \Pr\left\{ \sqrt{n} \left[C_n(\widehat{\mu}_1, \gamma_n^K) - \min_{e \in \{g_1, g_2\}} C_n(\widehat{\mu}_1, e) \right] \ge c_0 \right\} \ge \Pr(\|Z\| \le r) > 0.$$

Hence minimising the policy plug-in design loss need not achieve first-order optimality for the full design loss.

S4.6.3 Policy plug-in in the SNAP calibration

This subsection evaluates the policy plug-in shortcut in the SNAP calibration of Section 5, in the setting without covariates. We first compare three design rules. For pilot realisation s_1 and second-wave evidence e, let $\overline{\mu}_{n,+}(s_1, e, g)$ denote the posterior mean, as in Subsec-

tion S4.6, and recall

$$z_n^K(s_1, e, g) = \min_{\preceq_T} \arg\min_{z \in \mathcal{Z}^T} R(z, \overline{\mu}_{n,+}(s_1, e, g)).$$

We display the full design loss C_n and the policy plug-in design loss K_n alongside a regret plug-in design loss L_n:

$$\begin{aligned}
C_n(s_1, g) &= \int \left[\min_{z \in \mathcal{Z}^T} \int \{R(z, \mu) - R^*(\mu)\} \, \Pi_{2,g}(d\mu \mid s_1, e, g) \right] \Pi_{n,g}^{\text{pred}}(de \mid s_1), \\
K_n(s_1, g) &= \int \left[\int \left[R(z_n^K(s_1, e, g), \mu) - R^*(\mu) \right] \, \Pi_{2,g}(d\mu \mid s_1, e, g) \right] \Pi_{n,g}^{\text{pred}}(de \mid s_1), \\
L_n(s_1, g) &= \int \min_{z \in \mathcal{Z}^T} R(z, \overline{\mu}_{n,+}(s_1, e, g)) \, \Pi_{n,g}^{\text{pred}}(de \mid s_1).
\end{aligned} \tag{58}$$

The regret plug-in design loss L_n evaluates minimum regret at the updated posterior mean and averages it over the predictive distribution of second-wave evidence. It uses raw regret because subtracting R^* at the posterior mean would make the integrand identically zero. This comparison distinguishes two shortcuts. The policy plug-in design loss K_n selects a policy at the updated mean but still averages its regret over the posterior, while the regret plug-in design loss L_n replaces this posterior averaging by evaluation at the updated mean.

For the rules without RDDR, the K_n-rule selects a different experiment from C_n in only 16 pilots and loses 0.012 per cent of the C_n-minimising rule's gain in C_n over equal allocation. In contrast, L_n selects a different experiment in 574 pilots and loses more than that entire gain. Its evaluation regret is also higher than under equal allocation. The small difference in evaluation regret between K_n and C_n is less than one paired Monte Carlo standard error.

Table 5 compares RDDR with and without the policy plug-in step (52). We denote this design rule by RDDR + policy plug-in ($\widehat{K}_n$); the posterior policy rule remains $\delta_{n,g}^{\Pi}$.

Table 5: RDDR with and without policy plug-in

Design rule	Different from Full (C_n) (%)	C_n gain lost (%)	Evaluation regret (10^{-3})	Regret gain over equal allocation (%)	Computation time (minutes)
RDDR ($\widehat{C}_n$) $a = 0.5$	53.800	69.889	1.712	4.116	6.43
RDDR + policy plug-in ($\widehat{K}_n$), $a = 0.5$	53.800	69.889	1.712	4.116	5.09
RDDR ($\widehat{C}_n$) $a = 1$	38.700	14.944	1.651	7.526	8.41
RDDR + policy plug-in ($\widehat{K}_n$), $a = 1$	38.700	14.893	1.649	7.617	7.30
RDDR ($\widehat{C}_n$) $a = 2$	15.400	1.638	1.637	8.314	18.21
RDDR + policy plug-in ($\widehat{K}_n$), $a = 2$	15.700	1.626	1.637	8.334	14.35

Notes: The gain from minimising the full design loss C_n is the reduction in the mean of C_n relative to equal allocation. Gain lost is the percentage of this improvement that each rule gives up. Evaluation regret is defined in Subsection 5.2; regret gain is its percentage reduction relative to equal allocation. Computation times use the same RQMC node budgets and timing convention as the corresponding RDDR rules in Table 3.

Adding the plug-in step reduces RDDR computation time by 20.8, 13.2, and 21.2 per cent for $a = 0.5, 1, 2$, respectively. The selected experiments differ from the corresponding RDDR choices in 0.000, 0.600, and 1.900 per cent of pilots for $a = 0.5, 1, 2$, respectively. Evaluation regret is nearly unchanged. In this SNAP calibration, the plug-in step is therefore a useful computational shortcut that does not materially affect the selected experiment or evaluation regret. It need not be accurate in general, as shown by the counterexample in Appendix S4.6.2. Appendix S5.4 examines how the increase in design loss from the plug-in step changes with the candidate list and the covariance structure in a simplified model.

S5 Details for the SNAP Exercises

S5.1 Calibration and numerical implementation

The calibration without covariates uses the inputs and type-specific outcome probabilities in Tables 1 and 2. The stratified exercise below uses the same aggregate inputs and policy costs. Reported calibration entries are rounded to three decimals; calculations use unrounded values. For the proactive-only type, the food-security probabilities are $\Pr\{Y(0) = 1 \mid V = v_1\} \approx 0.195$ and $\Pr\{Y(1) = 1 \mid V = v_1\} \approx 0.545$. Together with the type masses $(0.400, 0.424, 0.071, 0.047, 0.058)$, the unrounded probabilities preserve the

aggregate food-security calibration.

The data-generating distribution for Table 3 was fixed before the 1,000 evaluation pilots were drawn. The calibration held aggregate enrolment probabilities and food-security probabilities fixed, imposed $\Pr\{Y(1) = 1 \mid V = v\} \geq \Pr\{Y(0) = 1 \mid V = v\}$ for every response type v, and required all four joint (D, Y) categories to have positive probability within every arm.

For each arm z, index the four joint (D, Y) categories $(1,1), (1,0), (0,1), (0,0)$ by $c = 1, \ldots, 4$. Let N_{zc} count pilot observations in arm z and category c, and let n_{1z} denote the arm-specific sample size. The pilot estimator uses the smoothed category frequencies $\widehat{p}_{zc} = \frac{N_{zc}+1/2}{n_{1z}+2}$. The resulting estimate of the mean of $(D, YD, Y(1-D))$ is $\widehat{\mu}_1$.

The full and RDDR design losses and the policy rule all use the quasi-Bayesian implementation in Subsection 4.1. Regret is computed from five affine functions of μ: the payoff $\theta_1(\mu)$ in (3), and the two affine functions entering each endpoint $\underline{\theta}_{z^\star}(\mu)$ and $\overline{\theta}_{z^\star}(\mu)$ in (7). Their covariance matrix has rank four. RDDR selects gradients on this four-dimensional affine support as described in Subsection 4.3, using only regions with nonempty interior relative to that support. Reported integration dimensions are computed as the rank of the retained excess-regret gradients in the original five-coordinate representation. The covariance matrix used to define the neighbourhood is estimated from the smoothed category frequencies; posterior updating and evaluation retain the calibrated covariance matrices.

For the timing exercises, the predictive (outer) and posterior (inner) integrals are evaluated using randomized quasi-Monte Carlo (RQMC). In the setting without covariates, Full uses 4,096 nodes for each integral. RDDR uses 2,048 nodes for each integral at $a = 0.5, 1$ and 3,072 at $a = 2$. Each policy plug-in rule uses the same inner and outer node counts as its corresponding RDDR rule. These are total node counts for each integral. Computation times were measured in separate runs from those used for the reported experiment selection and evaluation-regret results.

S5.2 Sample-size comparisons

The LATE_{2b} comparison in Figure 4 adds second-wave sample sizes of 10,000, 10,500, 10,750, 11,250, and 12,500. Sample-size equivalents use linear interpolation within the computed grids.

Figure 5 repeats the comparison of RDDR with RDDR + policy plug-in ($\widehat{K}_n$) in Table 5 at each second-wave sample size from 5,000 to 7,500. In each panel, the RDDR and RDDR + policy plug-in curves nearly coincide.

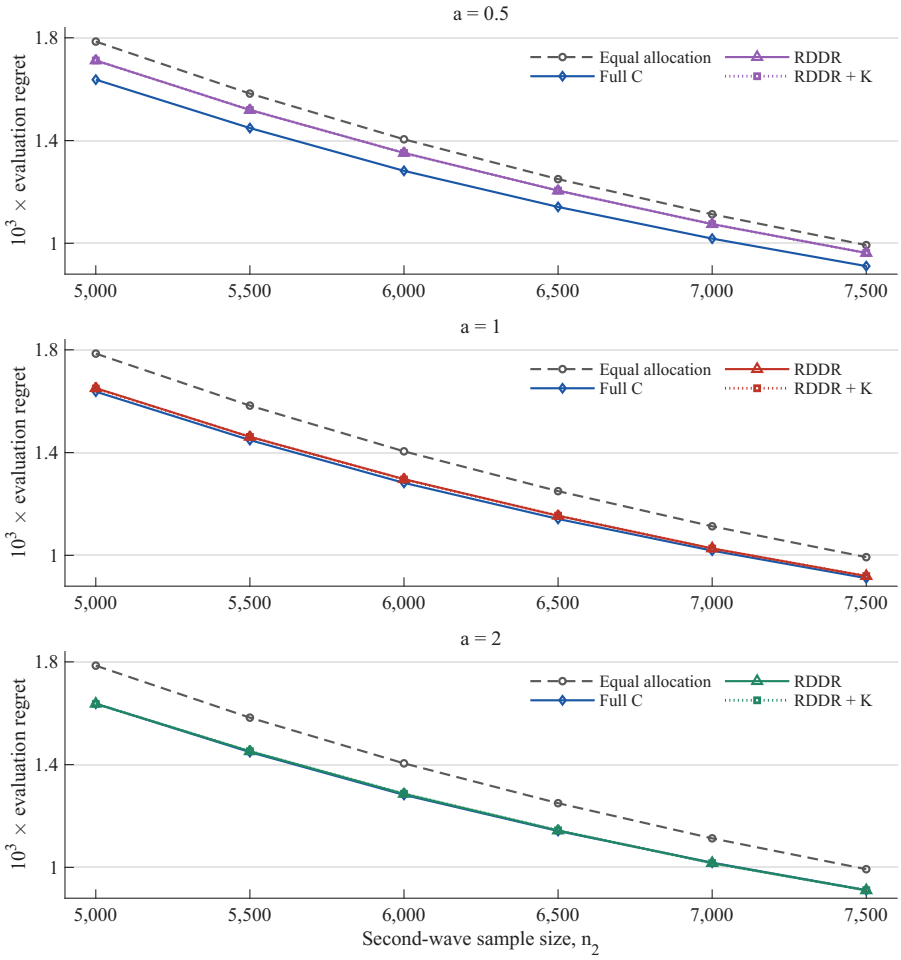


Figure 5: RDDR and RDDR + policy plug-in ($\widehat{K}_n$) as the second-wave sample size varies

Notes: Each panel fixes a and uses $n_2 = 5{,}000, 5{,}500, \ldots, 7{,}500$. Solid triangles denote RDDR and dotted squares denote RDDR + policy plug-in ($\widehat{K}_n$). Points are mean evaluation regret.

S5.3 Stratified calibration and numerical implementation

The lower- and higher-barrier strata have population weights 0.740 and 0.260. Table 6 gives their conditional take-up probabilities. Under the monotone support, these probabilities determine the conditional type-mass vectors α_0, α_1; the unrounded values satisfy $0.740\alpha_0 + 0.260\alpha_1 = \alpha$, where α is the aggregate type-mass vector in Subsection 5.1. Proactive take-up is 0.620 in the higher-barrier stratum and $(0.600 - 0.260 \times 0.620)/0.740 \approx 0.593$ in the lower-barrier stratum, giving population take-up 0.600.

Table 6: Take-up calibration with a binary covariate

Stratum	$D(0)$	$D(1)$	$D(2)$	$D(z^\star)$
Lower predicted barriers, $X = 0$	0.066	0.119	0.168	0.593
Higher predicted barriers, $X = 1$	0.035	0.065	0.200	0.620
Population	0.058	0.105	0.176	0.600

Table 7: Type-specific food-security probabilities with a binary covariate

Stratum	d	Never	Proactive	Assistance	Information	Baseline
Lower barriers, $X = 0$	0	0.715	0.346	0.714	0.221	0.757
Higher barriers, $X = 1$	0	0.465	0.062	0.714	0.221	0.757
Lower barriers, $X = 0$	1	0.758	0.551	0.886	0.975	0.906
Higher barriers, $X = 1$	1	0.508	0.281	0.886	0.975	0.906

Notes: Entries are conditional on response type and stratum. Population weighting reproduces $\Pr\{Y(0) = 1\} = 0.482$ and $\Pr\{Y(1) = 1\} = 0.644$.

For the proactive-only type in the lower-barrier stratum, the food-security probabilities are approximately 0.346 for $Y(0)$ and 0.551 for $Y(1)$.

The identified set imposes the separate restrictions

$$\Pr_P\{D(z^\star) = 1 \mid X = x\} \leq \overline{p}^D_{x,z^\star}, \qquad x \in \{0, 1\},$$

where each upper bound equals the unrounded entry in the $D(z^\star)$ column of Table 6. With zero per-recipient costs and a common terminal policy, the population payoff endpoints are the weighted sums of the stratum-specific endpoints in (7). The policy regrets therefore depend only on the six affine coordinates

$$\big(\theta_1(\mu), A, d_0, b_0, d_1, b_1\big), \qquad A = \sum_{x=0}^{1} \omega_x\big(\mu^{Y1}_{x2} - \mu^{Y1}_{x0} - \mu^{Y0}_{x0}\big) - k_{z^\star},$$

where $d_x = \mu^D_{x2}$ and $b_x = \mu^{Y0}_{x2}$. Full integrates over the induced six-dimensional Gaussian law. RDDR screens the affine regions in this representation and uses the rank of the retained

excess-regret gradients as its integration dimension. Dimension statistics use 1,000 stratified pilots. For $a = 0.5, 1, 2$, these give mean integration dimensions 1.100, 2.478, and 4.549. The integration dimension is zero in 39.800, 12.400, and 0.400 per cent of pilots, respectively.

All computation times in Table 4 are extrapolated linearly to 1,000 pilots from measurements on 10 pilots for each rule. Each timed pilot evaluates all 49 candidate experiments specified in Subsection 5.5. The inner and outer RQMC node counts are both 8,192 for Full and both 2,048, 4,096, and 4,096 for RDDR at $a = 0.5, 1, 2$, respectively.

S5.4 Gaussian stress-test specification and computation

We consider a Gaussian decision model with state $H = (H_T, H_F, H_U)^\top \in \mathbb{R}^3$ and terminal actions $\{0, 1\}$. Fix $d_0 \approx 1.641$, $f_0 \approx -0.817$, and reference $h_0 = (-f_0, 0, f_0)^\top$. Writing $D(H) = d_0 - H_F + H_U$, define the raw regret functions by

$$\begin{aligned} R(0, H) &= \left[\frac{H_T + D(H)}{2} + \min\{H_U, 0\}\right]_+, \\ R(1, H) &= \left[-\frac{H_T - D(H)}{2} - [H_F]_+\right]_+. \end{aligned} \tag{59}$$

The regret functions are defined on all of $\mathbb{R}^3$; near h_0, their difference $R(0, H) - R(1, H)$ is $H_T + [H_F]_+ + H_U$, placing the reference at both a kink and a policy tie.

Covariances and candidate experiments. The pilot has 500 observations, with arm counts $(167, 167, 166)$, and the follow-up experiment has 5,000. The seven original allocations are

$$\begin{aligned} (g_1, \ldots, g_7) = \big((0.2, 0.2, 0.6), (0.2, 0.4, 0.4), (0.2, 0.6, 0.2), \\ (0.4, 0.2, 0.4), (0.4, 0.4, 0.2), (0.6, 0.2, 0.2), (1/3, 1/3, 1/3)\big). \end{aligned}$$

The expanded list adds, in order,

$$\begin{aligned} (g_8, \ldots, g_{13}) = \big((0.1, 0.1, 0.8), (0.1, 0.8, 0.1), (0.8, 0.1, 0.1), \\ (0.1, 0.45, 0.45), (0.45, 0.1, 0.45), (0.45, 0.45, 0.1)\big). \end{aligned}$$

Equal allocation remains g_7, with second-wave counts $(1667, 1667, 1666)$. All other second-wave counts equal 5,000 times the allocation fractions.

Table 8 directly specifies covariance structures I and II. For counts $N = (N_1, N_2, N_3)$, set $V(N) = \sum_{a=1}^{3} \frac{\Omega_a^H}{N_a}$. With $\Sigma^{\text{pil}} = V(167, 167, 166)$ and $\Sigma_g^{\text{obs}} = V(N_g)$ for second-wave counts N_g, the model conditional on pilot mean m is $H \mid m \sim N(m, \Sigma^{\text{pil}})$ and $Y_g \mid H, m \sim$

$N(H, \Sigma_g^{\text{obs}})$. The posterior covariance and mean after observing Y_g are $\Sigma_g = \{(\Sigma^{\text{pil}})^{-1} + (\Sigma_g^{\text{obs}})^{-1}\}^{-1}$ and $\bar{H}_g = \Sigma_g\{(\Sigma^{\text{pil}})^{-1}m + (\Sigma_g^{\text{obs}})^{-1}Y_g\}$.

Table 8: Two covariance structures for the Gaussian model

Arm	$10^3\Omega_{11}^H$	$10^3\Omega_{12}^H$	$10^3\Omega_{13}^H$	$10^3\Omega_{22}^H$	$10^3\Omega_{23}^H$	$10^3\Omega_{33}^H$
Covariance structure I						
1	52.463	0	0	0	0	0
2	0	0	0	0	0	0
3	166.219	−81.122	0.372	147.449	3.088	3.751
Covariance structure II						
1	26.231	22.008	0	18.465	0	0
2	0	0	0	0	0	0
3	83.128	−21.570	0.372	274.002	3.088	3.751

Notes: Upper triangles are reported in coordinates (H_T, H_F, H_U), multiplied by 10^3. Numbers throughout this subsection are rounded for display; calculations use unrounded values. Each unrounded contribution is positive semidefinite, and $V(N)$ is positive definite for all count vectors used here.

Writing $T = H_T + H_U$, the pilot ratio $\text{SD}(T \mid H_U)/\,\text{SD}(H_F \mid H_U)$ is approximately 1.228 under I and 0.614 under II, with conditional correlations approximately -0.458 and zero. Both structures use the same regret functions, reference state, and candidate lists.

Design losses and pilot means. Conditional on pilot mean m, the future posterior mean $\bar{H}_g$ and parameter have the joint Gaussian representation $\bar{H}_g \sim N(m, \Sigma^{\text{pil}} - \Sigma_g)$ and $H \mid \bar{H}_g \sim N(\bar{H}_g, \Sigma_g)$. We evaluate $C(m,g), K(m,g), L(m,g)$ from (58) under this law and the regret functions in (59), suppressing fixed sample sizes. Policy ties select 0. In C and K, computation omits the common oracle term $\mathbb{E}R^*(H)$, which cancels in design comparisons. Write $g^q(u) = g_{\min \arg\min_j q(m(u),g_j)}$, $q = C, K, L$, and abbreviate $q(m(u), g)$ by $q(u, g)$.

The deterministic 41×41 grid is $(u_T, u_F) \in [-3.25, 3.25]^2$, with unstandardised means $m(u_T, u_F) = (\sigma_T u_T - f_0,\ \sigma_F u_F,\ f_0)^\top$, $\sigma_T^2 = \text{Var}(T)$, and $\sigma_F^2 = \text{Var}(H_F)$, where variances use the relevant pilot covariance. The scales are $(\sigma_T, \sigma_F) \approx (0.037, 0.030)$ under structure I and $(\sigma_T, \sigma_F) \approx (0.026, 0.042)$ under structure II. Weights come from $m \sim N(h_0, \Sigma^{\text{pil}})$, conditional on its third coordinate equalling f_0. The distributions of the standardised pilot means have mean zero, with approximate covariances under I and II:

$$\begin{pmatrix} 0.980 & -0.449 \\ -0.449 & 0.983 \end{pmatrix}, \qquad \begin{pmatrix} 0.960 & 0 \\ 0 & 0.991 \end{pmatrix}.$$

The square contains approximately 0.998 of the probability under either distribution; integrals use this square without renormalisation. The covariance comparison thus also changes the distribution of unstandardised pilot means.

Gaussian expectations use conditional Gauss–Legendre quadrature, split at policy thresholds and kinks of the regret functions, with Gaussian tails truncated at nine standard deviations. The 13-candidate grids use 24 nodes per interval in each of $(\bar{H}_F, \bar{H}_U)$ and 48 per conditional $\bar{H}_T$ interval under structure I, and 20 and 48 under structure II; fixed-mean comparisons use 24 and 48. Table 9 uses bilinear interpolation of design-loss differences and the trapezoidal rule with Gaussian weights at 1,501 points per axis.

Design-loss gaps. Table 9 reports all four combinations, evaluating selected experiments with C and a policy rule that minimises posterior expected regret. The percentage of the gain lost is

$$100\,\frac{\int [C(u, g^q(u)) - C(u, g^C(u))]\, w(u)\, du}{\int [C(u, g_7) - C(u, g^C(u))]\, w(u)\, du}, \qquad q \in \{K, L\},$$

where w is the specified Gaussian density and both integrals are over the displayed square.

Table 9: Covariance, candidate list, and plug-in design-loss gap

Covariance	Candidates	C/K differ (%)	C gain (10^{-5})	Design-loss gap of K (10^{-6})	Gain lost: K (%)	Gain lost: L (%)
I	7	52.632	3.812	0.362	0.949	276.876
I	13	21.747	5.767	1.179	2.045	512.709
II	7	0.517	3.127	1.33×10^{-5}	4.25×10^{-5}	292.937
II	13	47.668	4.419	5.010	11.338	574.524

Notes: Gaussian weights are specific to each covariance, with no renormalisation on the square. The gain is relative to equal allocation. With 13 candidates, C selects g_{12} at all 1,681 grid points under both covariances, so the reported gain is achieved by a fixed allocation. The very small seven-candidate design-loss gap under structure II is reported as an approximate numerical value.

Expanding the candidate list from 7 to 13 increases the plug-in design-loss gap under both covariances; changing covariance has opposite effects for the two lists. Figure 6 shows the selected experiments and design-loss gaps with 13 candidates.

The bound on the design-loss gap. Subtracting $(H_T + H_F + H_U)/2$ from the difference $R(0, H) - R(1, H)$ in (59) leaves a globally $\sqrt{2}$-Lipschitz function, as the gradients on its affine pieces show. We evaluate (54) with $L_\Delta = \sqrt{2}$ and the positive semidefinite square roots of the covariance matrices. We bound total variation by the minimum of one and the two bounds obtained from Pinsker's inequality, one for each direction of the Gaussian KL divergence. Posterior-mean covariances are positive definite. Using 1,001 weighting points per axis gives approximate integrated upper bounds $(8.96, 17.5, 9.83, 18.9)\times10^{-3}$ in the order of Table 9, well above the computed design-loss gaps.

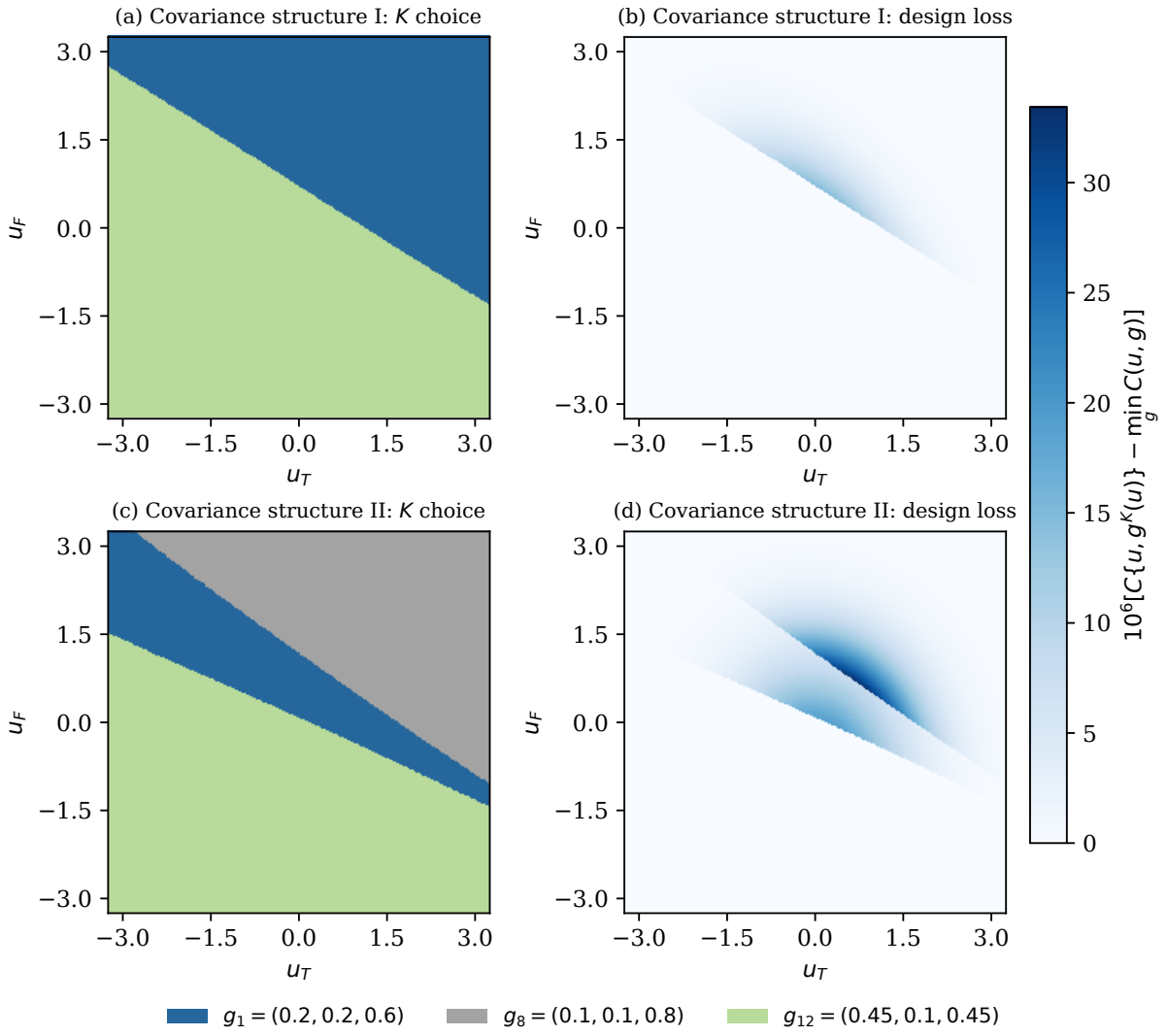


Figure 6: Policy plug-in choices and design-loss gaps with 13 candidates

Notes: The left panels report the K-selected experiment; C selects g_{12} throughout both panels. The right panels report the resulting design-loss gap on a common colour scale. Candidate design-loss differences are interpolated bilinearly before selecting experiments. The third unstandardised pilot coordinate is fixed at f_0. The axes use each covariance's own standard deviations, so identical plotted coordinates need not represent identical unstandardised means.

Comparison at identical unstandardised means. Table 10 holds the unstandardised mean fixed across covariances, using structure I's scales at $u_T, u_F \in \{-1.95, 0, 1.95\}$, the grid coordinates nearest $\{-2, 0, 2\}$. All nine comparisons use the same regret functions and 13-candidate list, without weighting or normalisation.

Table 10: Policy plug-in comparison at identical unstandardised pilot means

		K-selected experiment		Design-loss gap (10^{-6})	
u_T	u_F	I	II	I	II
−1.95	−1.95	g_{12}	g_{12}	0	0
0	−1.95	g_{12}	g_{12}	0	0
1.95	−1.95	g_{12}	g_{12}	0	0
−1.95	0	g_{12}	g_{12}	0	0
0	0	g_{12}	g_{12}	0	0
1.95	0	g_1	g_8	1.191	1.313
−1.95	1.95	g_1	g_1	1.376	0.112
0	1.95	g_1	g_8	1.121	16.404
1.95	1.95	g_1	g_8	0.006	0.224

Notes: Coordinates in both columns use structure I's scales and the same fixed third coordinate f_0. The C-selected experiment is g_{12} in every row under both covariances. The design-loss gap is $C(m, g^K) - \min_g C(m, g)$, with a policy rule that minimises posterior expected regret.